Improving and Quantifying Utility under Local Differential Privacy

Ye Zheng

Advisor: Dr. Yidan Hu

Committee: Dr. Sumita Mishra, Dr. Haibo Yang, Dr. Weijie Zhao

Chairperson: Dr. Anurag Agrawal

RIT | Rochester Institute of Technology

PDF & slides 👉 https://zhengyeah.com

1st-Author Publications

RIT

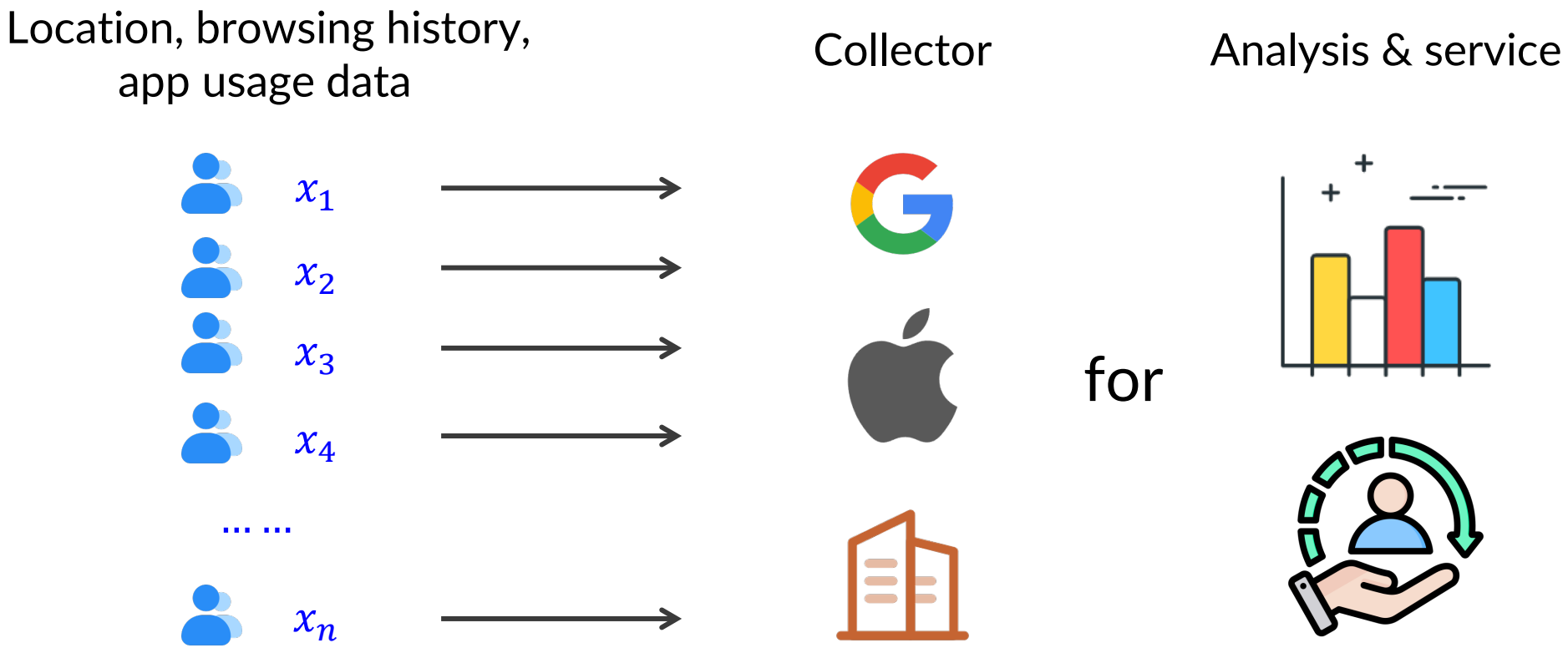
This dissertation

- [PETS'25] Locally Differentially Private Frequency Estimation via **Joint** Randomized Response
  Ye Zheng, Shafizur Rahman Seeam, Yidan Hu, Rui Zhang, Yanchao Zhang
- [PETS'25] **Optimal** Piecewise-based Mechanism for Collecting Bounded Numerical Data under LDP
  Ye Zheng, Sumita Mishra, Yidan Hu
  ★ ***Artifact Award Runner-up***
- [PETS'26] TraCS: Trajectory Collection in **Continuous Space** under Local Differential Privacy
  Ye Zheng, Yidan Hu
- [PETS'26] Quantifying **Classifier Utility** under Local Differential Privacy
  Ye Zheng, Yidan Hu
  ★ ***Andreas Pfitzmann Best Student Paper Award***
- [PETS'26] AudAgent: Automated Auditing of **Privacy Policy Compliance in AI Agents**
  Ye Zheng, Yimin Chen, Yidan Hu

Data Collection

RIT

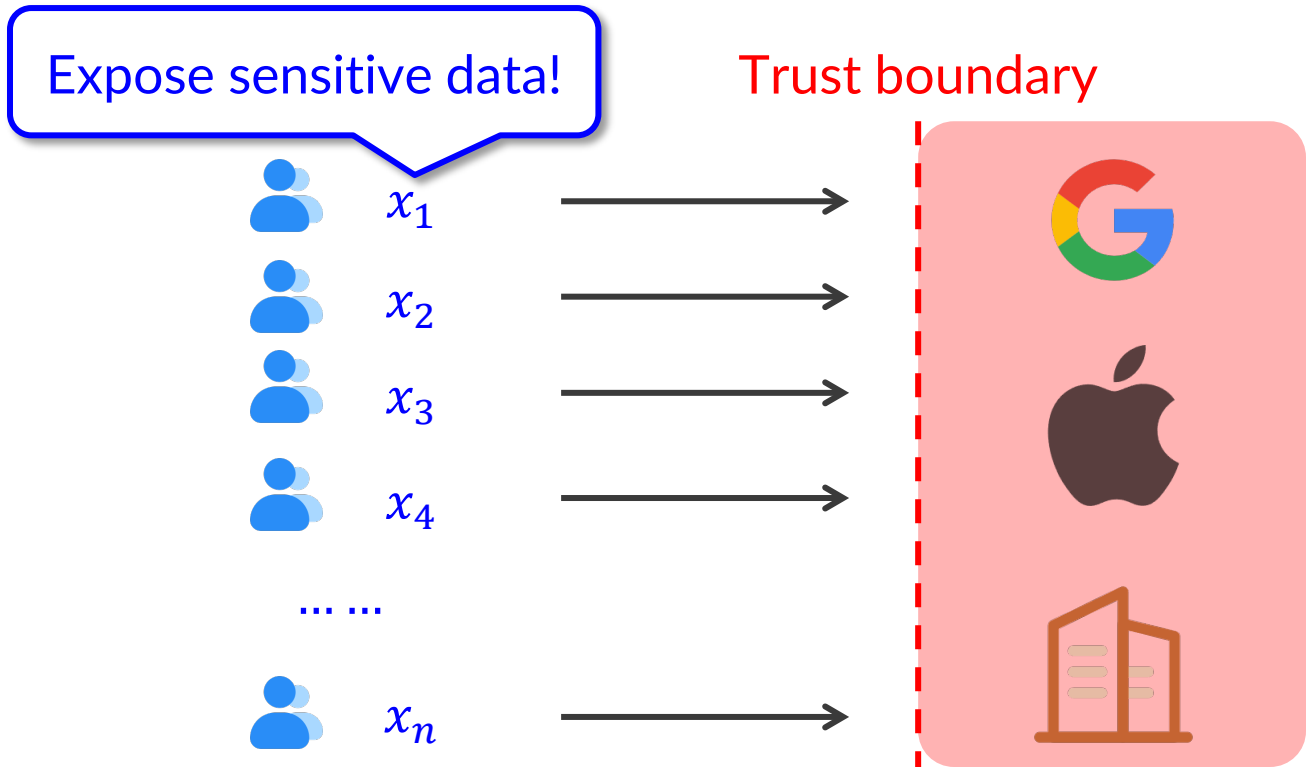
- Users' personal data are collected by companies for analysis or services



Users' Data Privacy

RIT

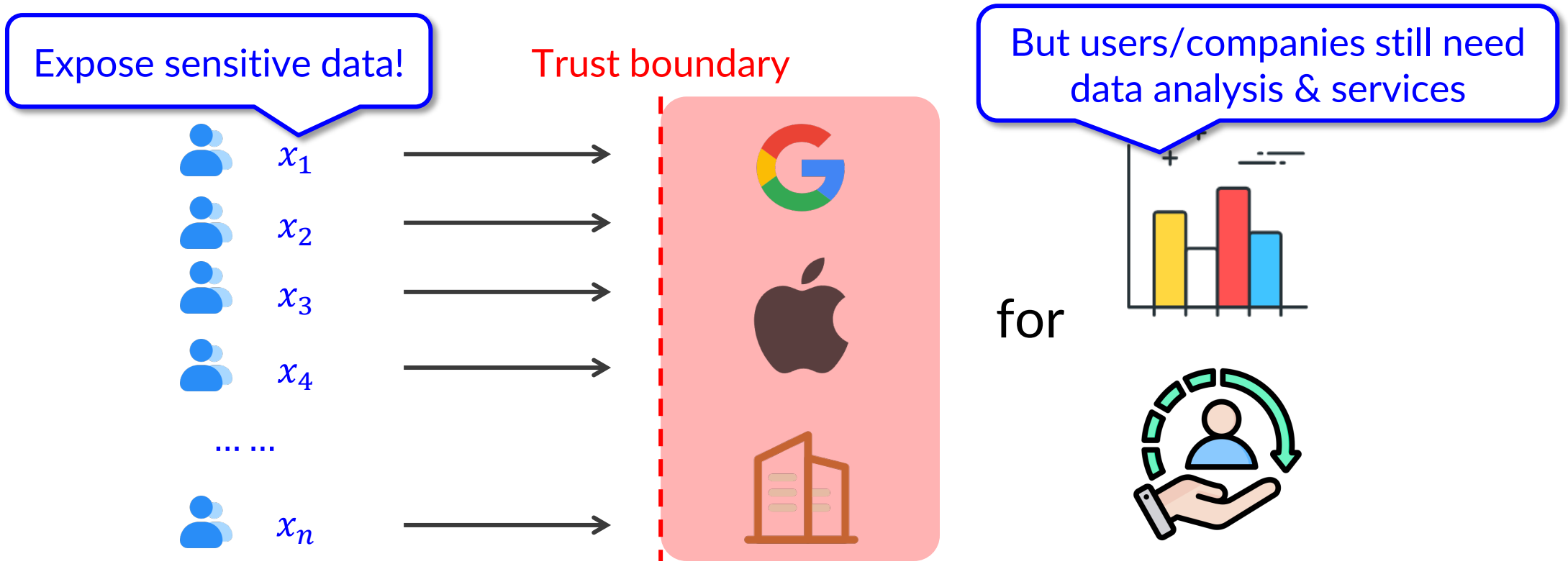
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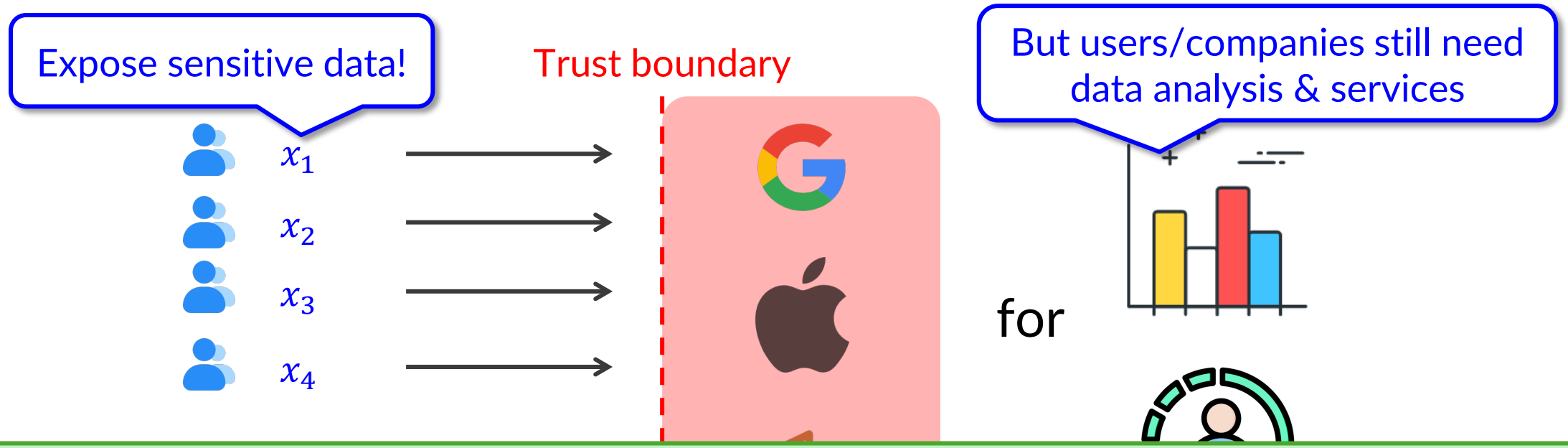
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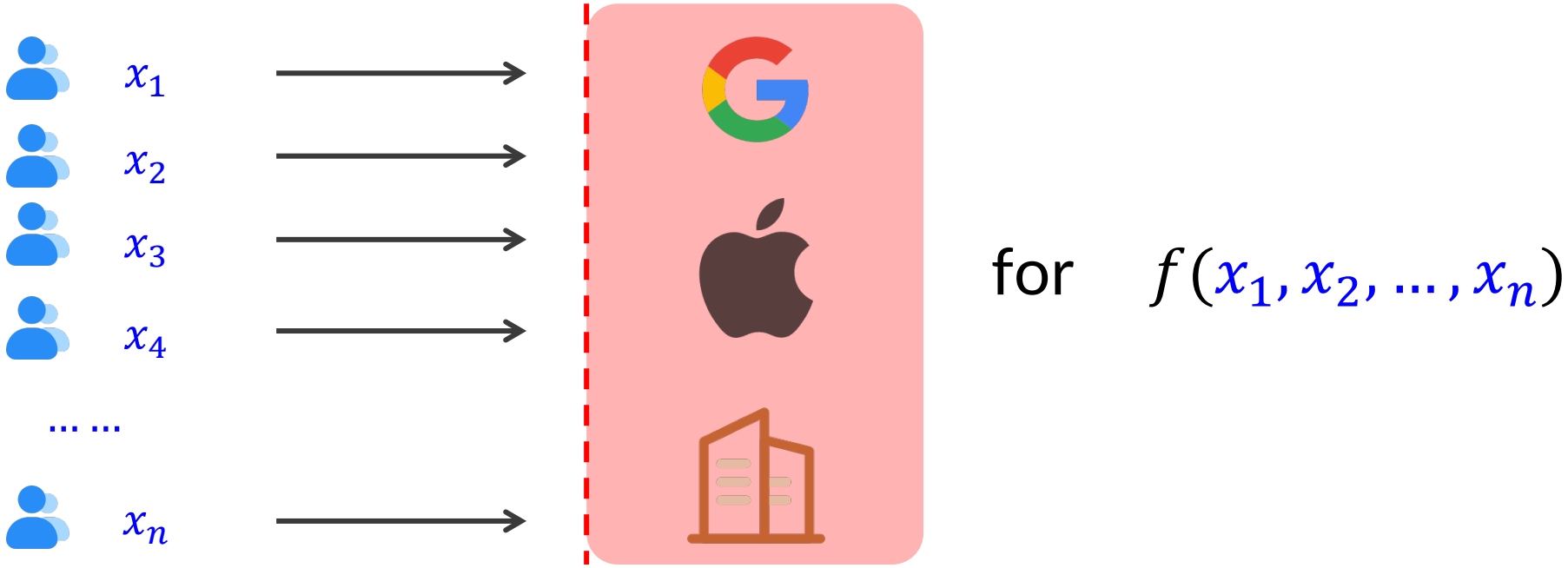


Q: How can we provide data analysis & services **while** protecting users' data privacy?

Privacy-Preserving Computation

RIT

- Users' personal data are collected by companies for analysis or services
  - these companies may be untrusted to collect users' sensitive data
  - how to compute $f(x_1, x_2, \dots, x_n)$ without revealing $x_1, x_2, \dots, x_n$?



for $f(x_1, x_2, \dots, x_n)$

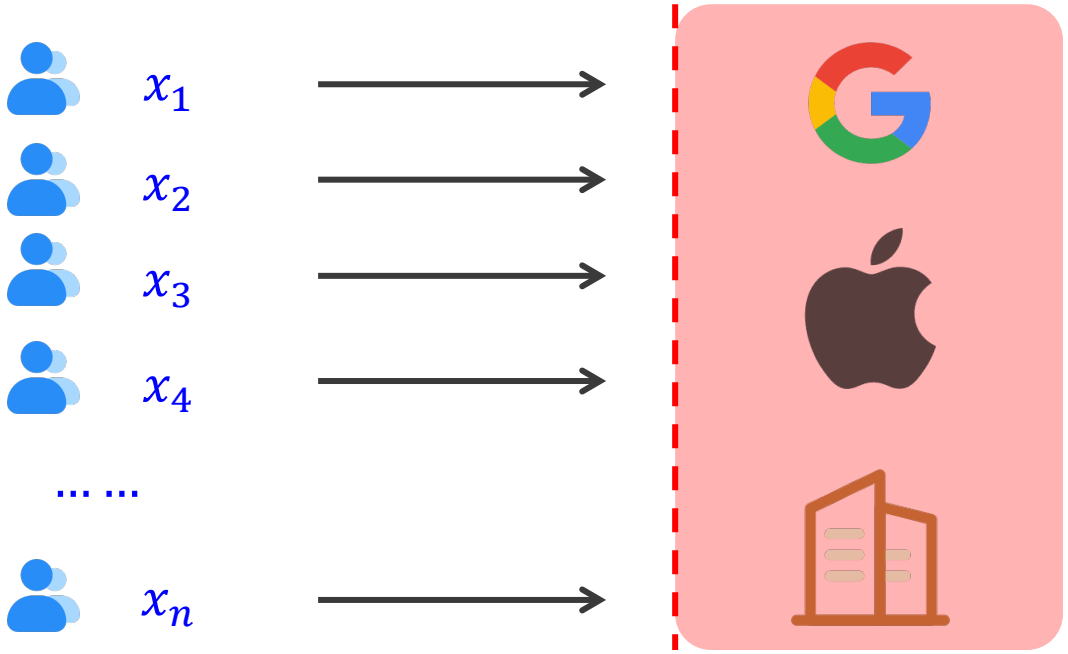
Privacy-Preserving Computation - **Techniques**

- Homomorphic encryption (HE), multi-party computation (MPC), local differential privacy (LDP), etc

Privacy-Preserving Computation - **HE**

RIT

- Homomorphic encryption (HE):
  - "homomorphic": preserving structure
  - design $\{Enc, Dec, Eval\} \rightarrow Dec\left(Eval\left(f, Enc_1(x_1), Enc_2(x_2), \ldots, Enc_n(x_n)\right)\right) = f(x_1, x_2, \ldots, x_n)$

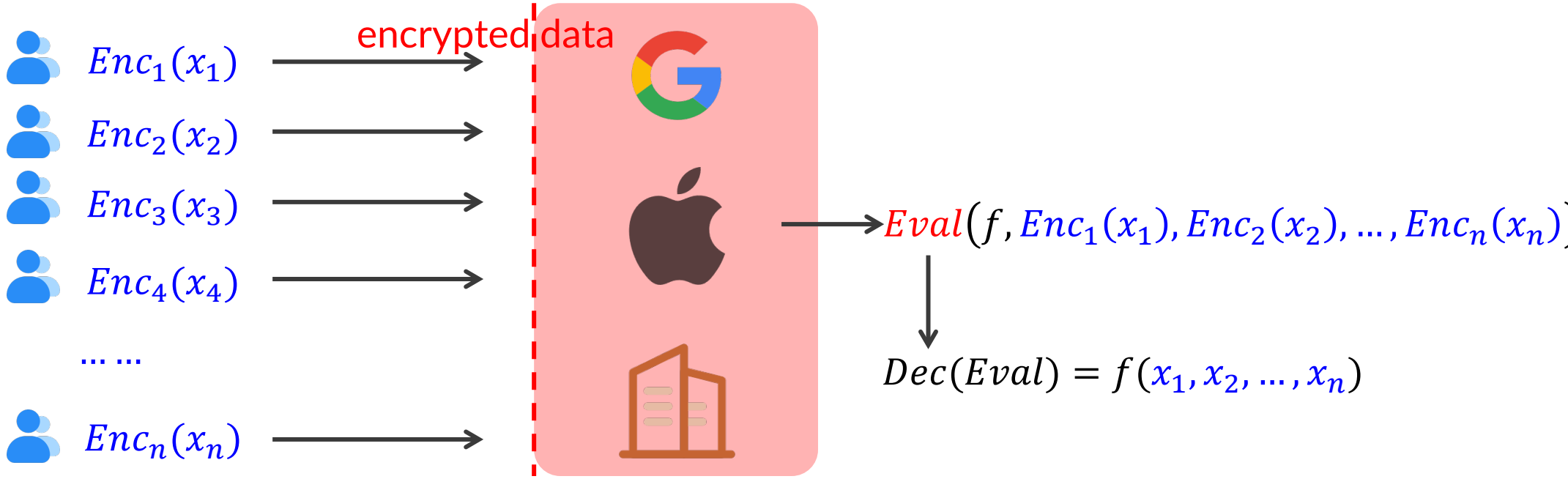


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Privacy-Preserving Computation - **HE**

RIT

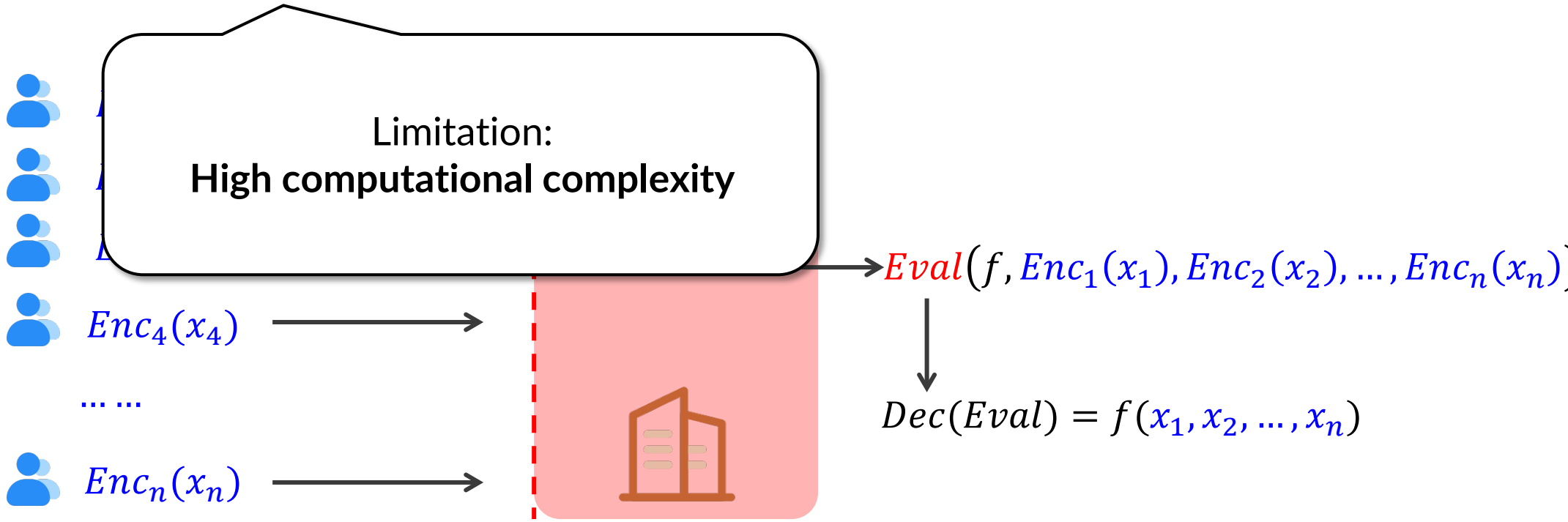
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Privacy-Preserving Computation - **MPC**

RIT

- Multi-party computation (MPC):
  - no central party
  - jointly compute f without revealing x_i
- Example: $f(x_1, x_2) = x_1 + x_2$



x_1



x_2

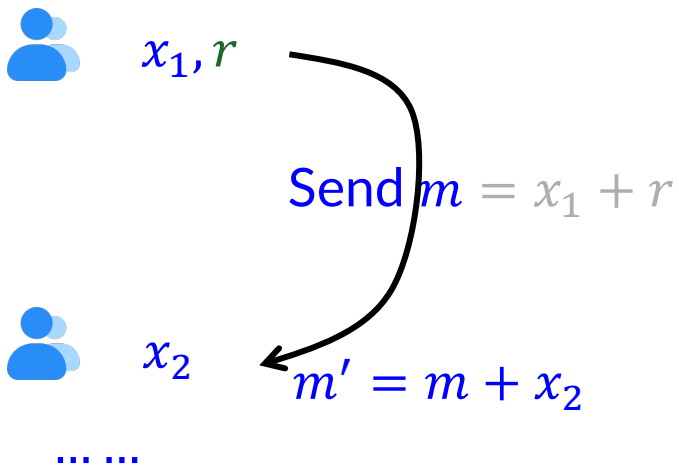
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for $f(x_1, x_2)$

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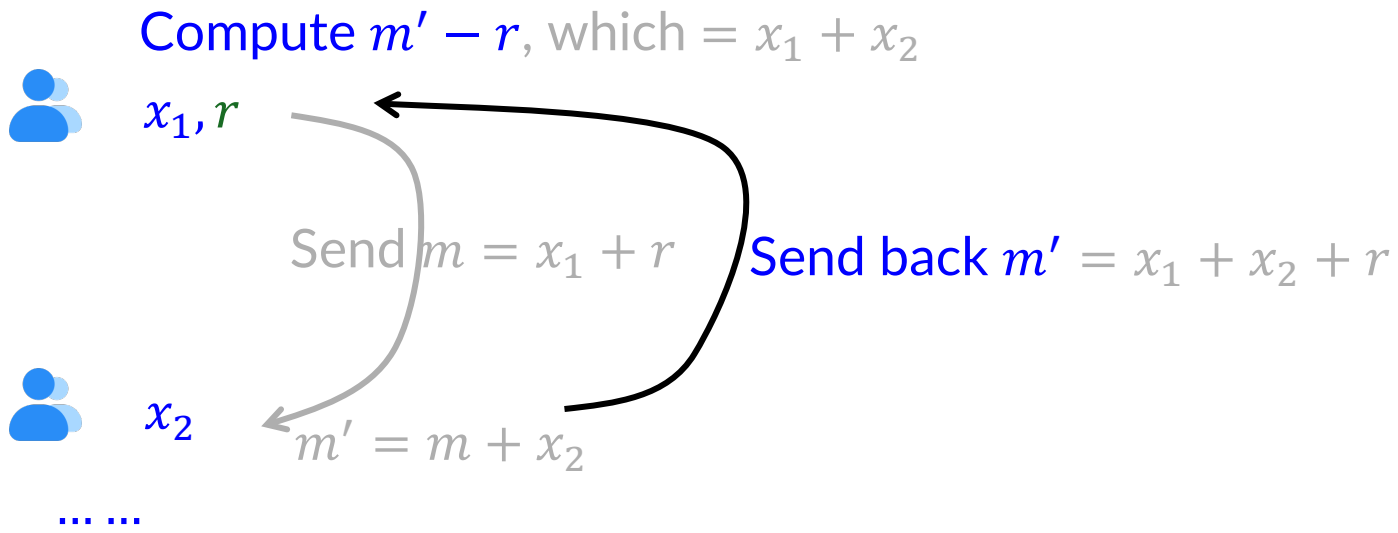


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Privacy-Preserving Computation - MPC

RIT

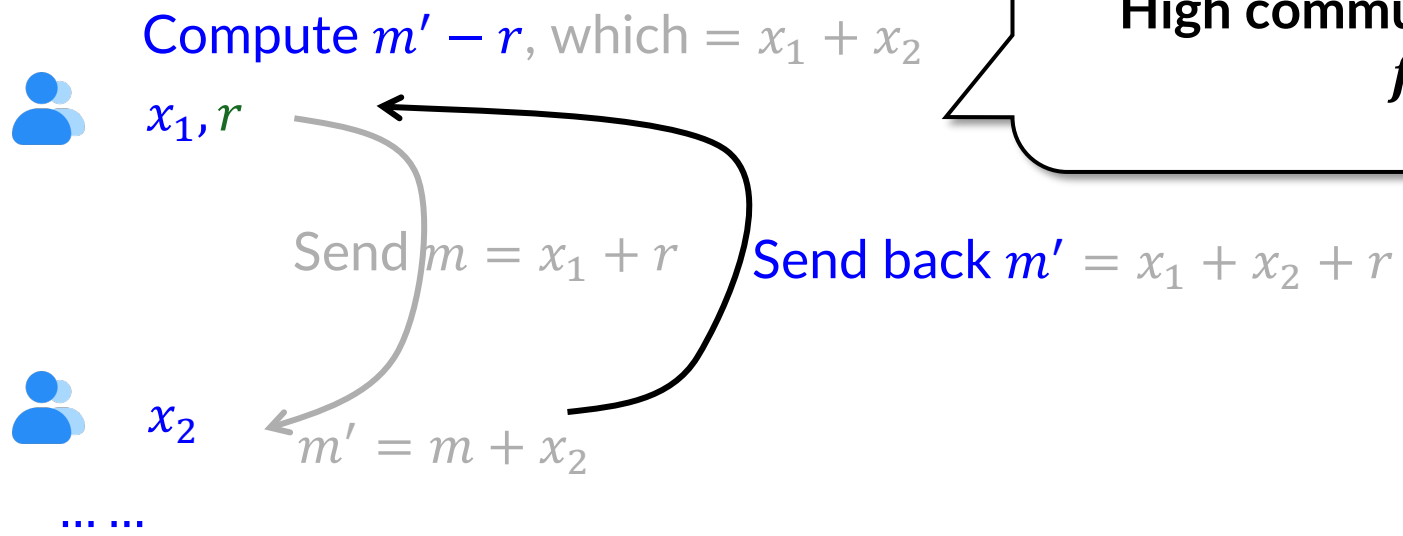
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Limitations:
High communication complexity
f-specific

Privacy-Preserving Computation - **LDP**

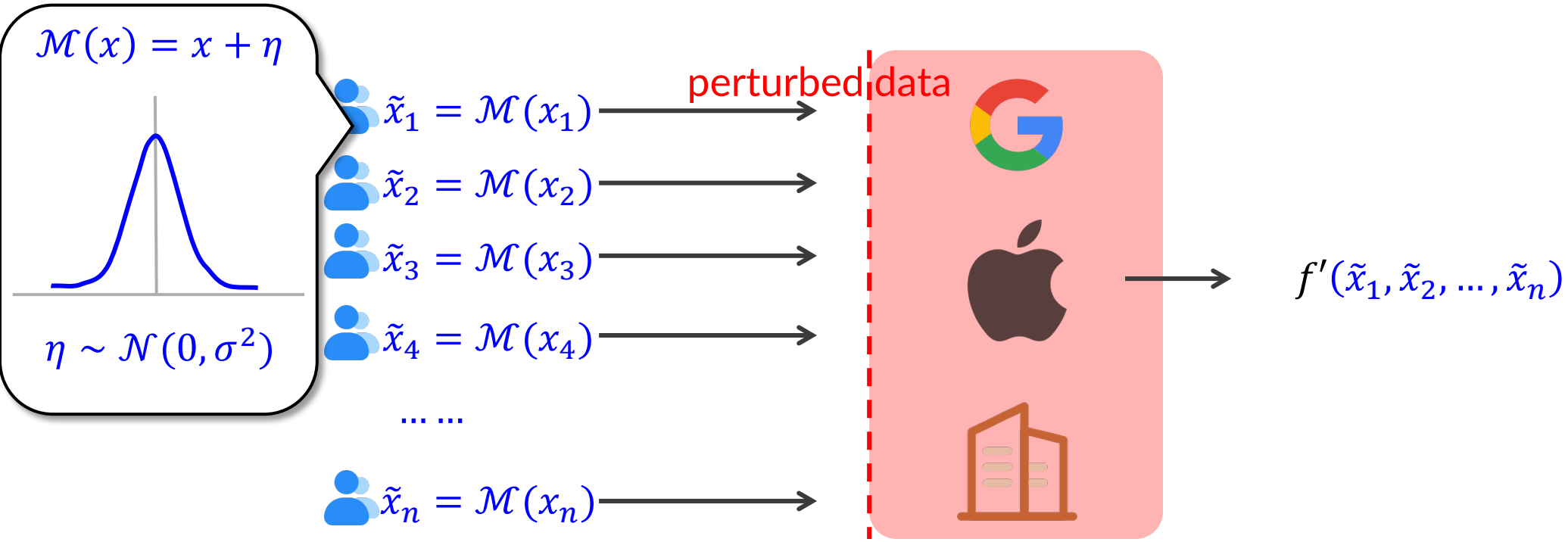
RIT

- Local differential privacy (LDP):
  - hard to differentiate the sensitive data from other data
  - each user locally perturbs x_i to $\tilde{x}_i$ $\rightarrow$ $f'(\tilde{x}_1, \tilde{x}_2, \ldots, \tilde{x}_n) \approx f(x_1, x_2, \ldots, x_n)$

Privacy-Preserving Computation - **LDP**

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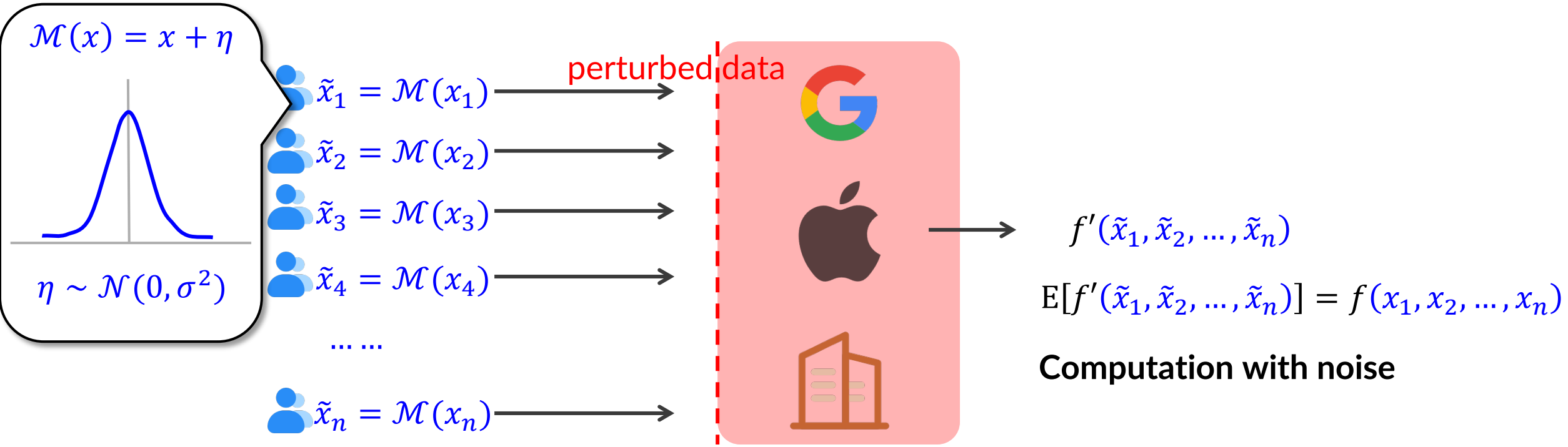
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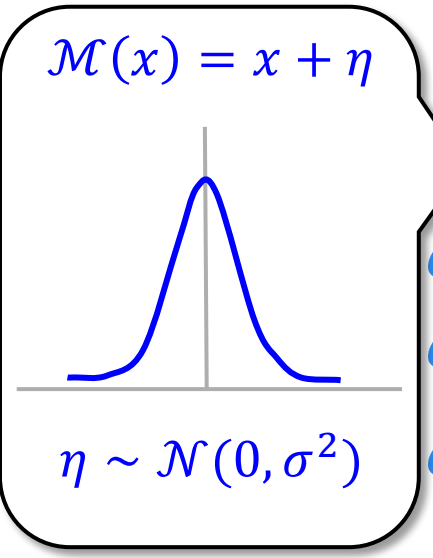
RIT

- Local differential privacy (L
  - hard to differentiate the
  - each user locally perturb

Advantages:
Negligible computational complexity
No communication between users

But approximated f

$\ldots, x_n)$



perturbed data

$\tilde{x}_1 = \mathcal{M}(x_1)$
$\tilde{x}_2 = \mathcal{M}(x_2)$
$\tilde{x}_3 = \mathcal{M}(x_3)$
$\tilde{x}_4 = \mathcal{M}(x_4)$
... ...
$\tilde{x}_n = \mathcal{M}(x_n)$

$f'(\tilde{x}_1, \tilde{x}_2, \ldots, \tilde{x}_n)$

$\mathrm{E}[f'(\tilde{x}_1, \tilde{x}_2, \ldots, \tilde{x}_n)] = f(x_1, x_2, \ldots, x_n)$

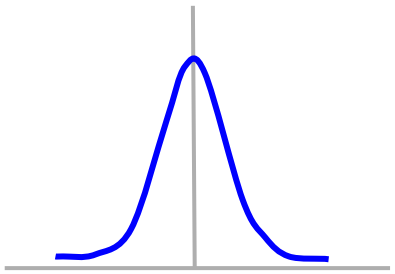
Computation with noise

LDP – Applications

RIT

Chrome uses LDP to collect homepage settings, extension usage, etc

https://developers.googleblog.com/enabling-developers-and-organizations-to-use-differential-privacy/



$\eta \sim \mathcal{N}(0, \sigma^2)$

$\tilde{x}_1 = \mathcal{M}(x_1)$

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perturbed data

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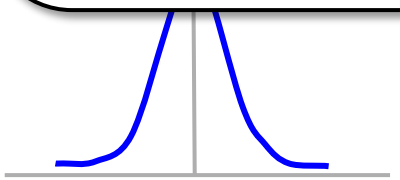
RIT

Chrome uses LDP to collect homepage settings, extension usage, etc

https://developers.googleblog.com/enabling-developers-and-organizations-to-use-differential-privacy/

Emoji usage, new keyboard words, Safari URL statistics, health analytics

https://machinelearning.apple.com/research/learning-with-privacy-at-scale



$\eta \sim \mathcal{N}(0, \sigma^2)$

$\tilde{x}_2 = \mathcal{M}(x_2)$

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Computation with noise

LDP – Formal Privacy

RIT

- After applying $\mathcal{M}$, the confidence of distinguishing sensitive x_1 and x_2 from observation $\tilde{x}$:

$$\forall x_1, x_2 \in \mathcal{D}, \forall y \in \widetilde{\mathcal{D}} \qquad \max \frac{\Pr[\mathcal{M}(x_1) = \tilde{x}]}{\Pr[\mathcal{M}(x_2) = \tilde{x}]} \leq e^{\varepsilon}$$

Distinguishability

LDP – Formal Privacy

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- The collector's / adversary's view: hard to infer the sensitive data

Privacy quantified by ε

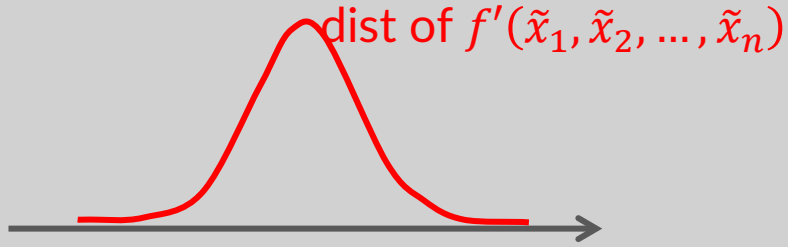
$x_1 \rightarrow \mathcal{M} \rightarrow \tilde{x}$



Provable defense against data inference attacks



Data utility by approximation error



$f'(\tilde{x}_1, \tilde{x}_2, \ldots, \tilde{x}_n) \approx f(x_1, x_2, \ldots, x_n)$

LDP – Formal Privacy

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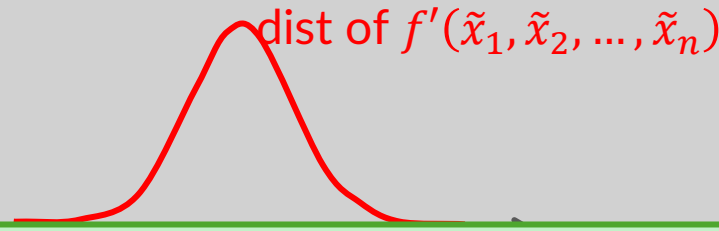
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Data utility by approximation error



Fundamental direction: **Design** of $\mathcal{M}$ to **optimize** the privacy—utility tradeoff

LDP – Formal Privacy

RIT

Utility analysis of $f \circ \mathcal{M}$

$$f(x_1, x_2, \dots, x_n) \coloneqq \sum_{i=1}^{n} x_i \quad \text{or} \quad f(x_1, x_2, \dots, x_n) \coloneqq \{x_1, x_2, \dots, x_n\} \rightarrow \text{Variance, MSE (simple)}$$

$$f(x_1, x_2, \dots, x_n) \coloneqq h: \mathbb{R}^n \rightarrow \{1,2, \dots, K\} \text{ is a classifier} \rightarrow$$



Privacy quantified by ε

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Provable defense against data inference attacks



Data utility by app… …error

dist of $f'(\tilde{x}_1, \tilde{x}_2, \dots, \tilde{x}_n)$



$$f'(\tilde{x}_1, \tilde{x}_2, \dots, \tilde{x}_n) \approx f(x_1, x_2, \dots, x_n)$$

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Data utility by app… …error

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Fundamental direction: **Utility analysis** of complex task f

This Dissertation: LDP Theory

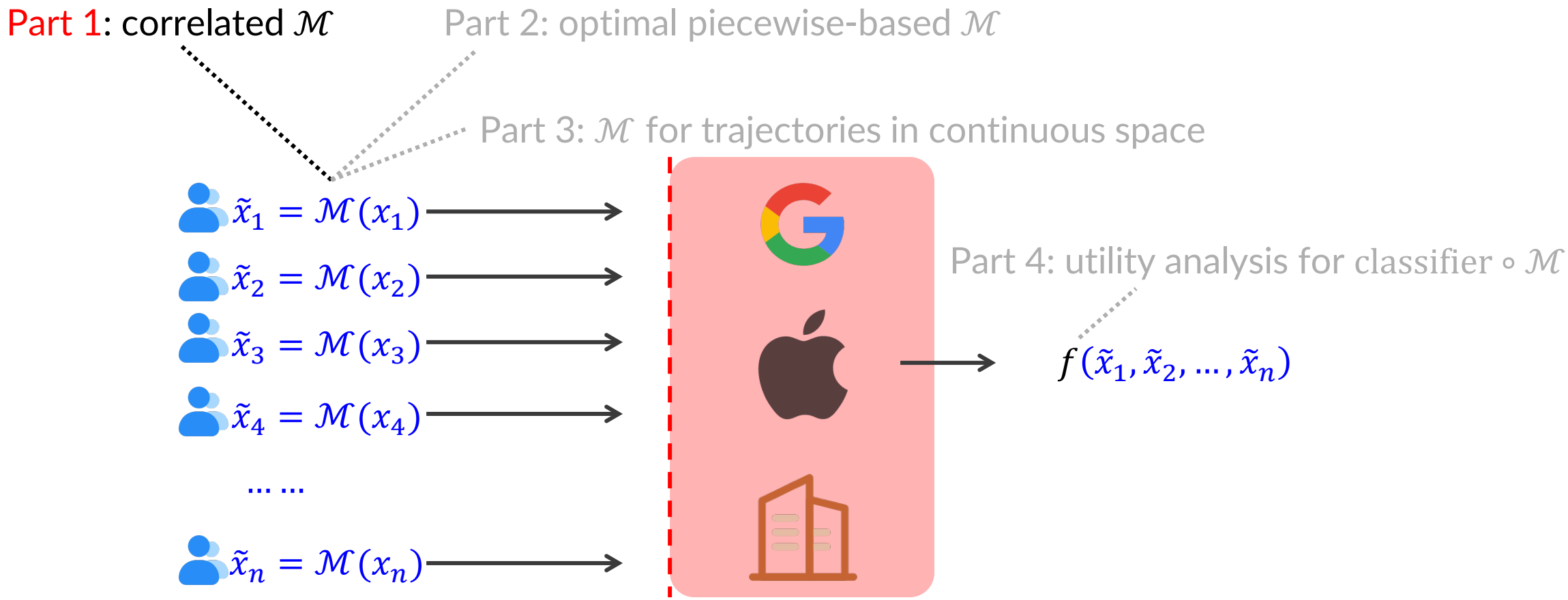
RIT

- Advancing LDP's mechanism design and utility analysis

This Dissertation: LDP Theory

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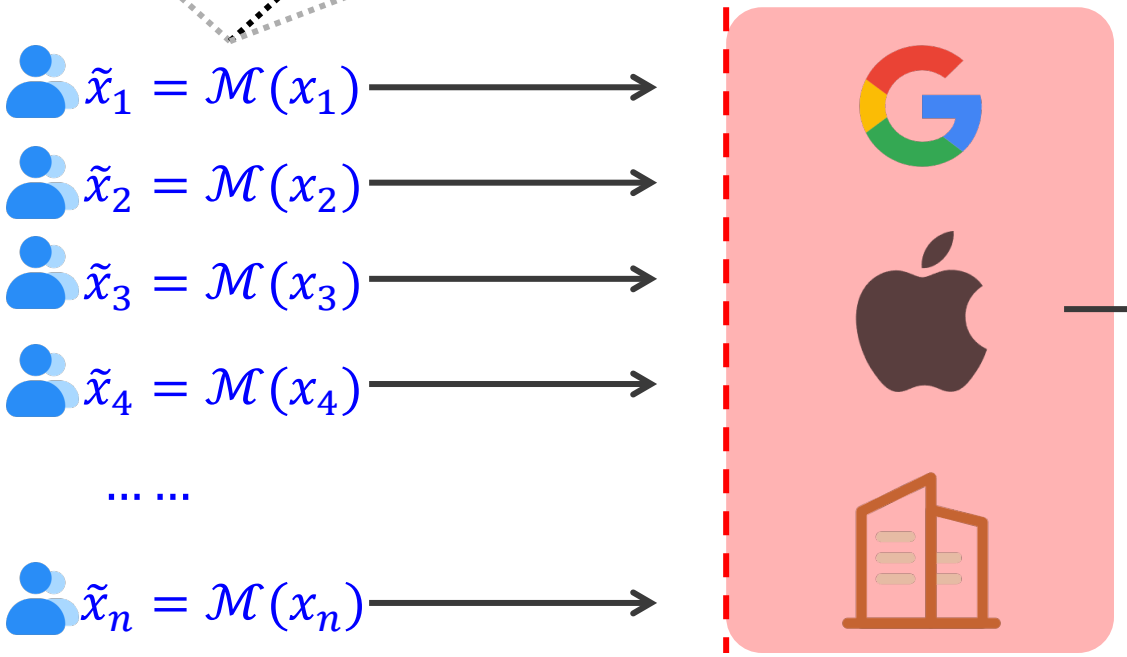
RIT

- Advancing LDP's mechanism design and utility analysis

binary x → numerical x

Part 1: correlated $\mathcal{M}$ → Part 2: optimal piecewise-based $\mathcal{M}$

Part 3: $\mathcal{M}$ for trajectories in continuous space



Part 4: utility analysis for classifier $\circ$ $\mathcal{M}$

f is a classifier

$f(\tilde{x}_1, \tilde{x}_2, \ldots, \tilde{x}_n)$

This Dissertation: LDP Theory

RIT

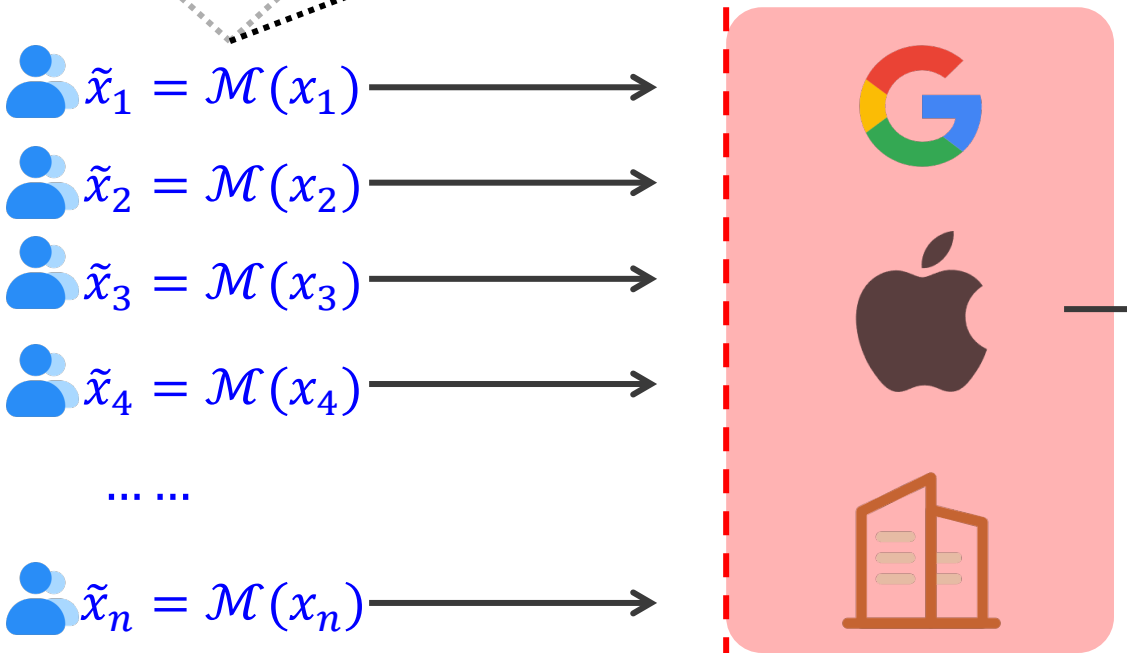
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This Dissertation: LDP Theory

RIT

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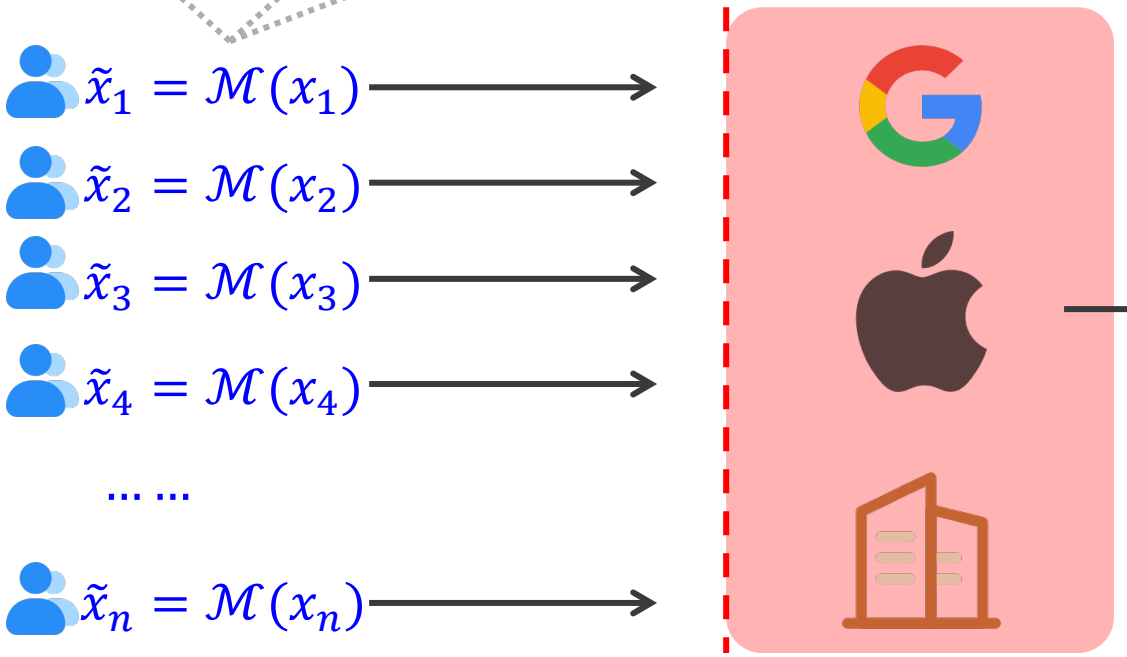
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Mechanism-level ↓ Task-level



Part 4: utility analysis for classifier $\circ$ $\mathcal{M}$

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Societal Impacts

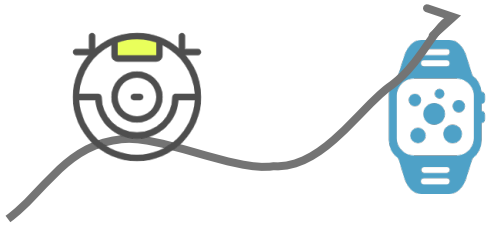
RIT

- **New LDP building blocks**
  - correlated LDP mechanisms
  - optimal piecewise-based mechanisms



Sensor networks & Federated learning, etc

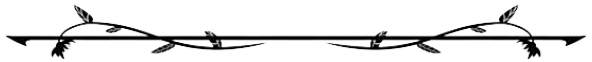
- **Universal trajectory collection mechanisms**
  - applicable to both continuous / discrete space



Smart home & wearable devices' trajectories, etc

- **Analytical view of classifier utility under LDP-perturbed inputs**
  - predicating classifier utility under LDP / choosing best ε

Improving and Quantifying Utility under Local Differential Privacy



Mechanism design

- Part 1: Correlated $\mathcal{M}$
- Part 2: Optimal piecewise-based $\mathcal{M}$
- Part 3: $\mathcal{M}$ for trajectories in continuous space

Utility analysis

- Part 4: Utility analysis for classifier $\circ$ $\mathcal{M}$

More complex data types & tasks

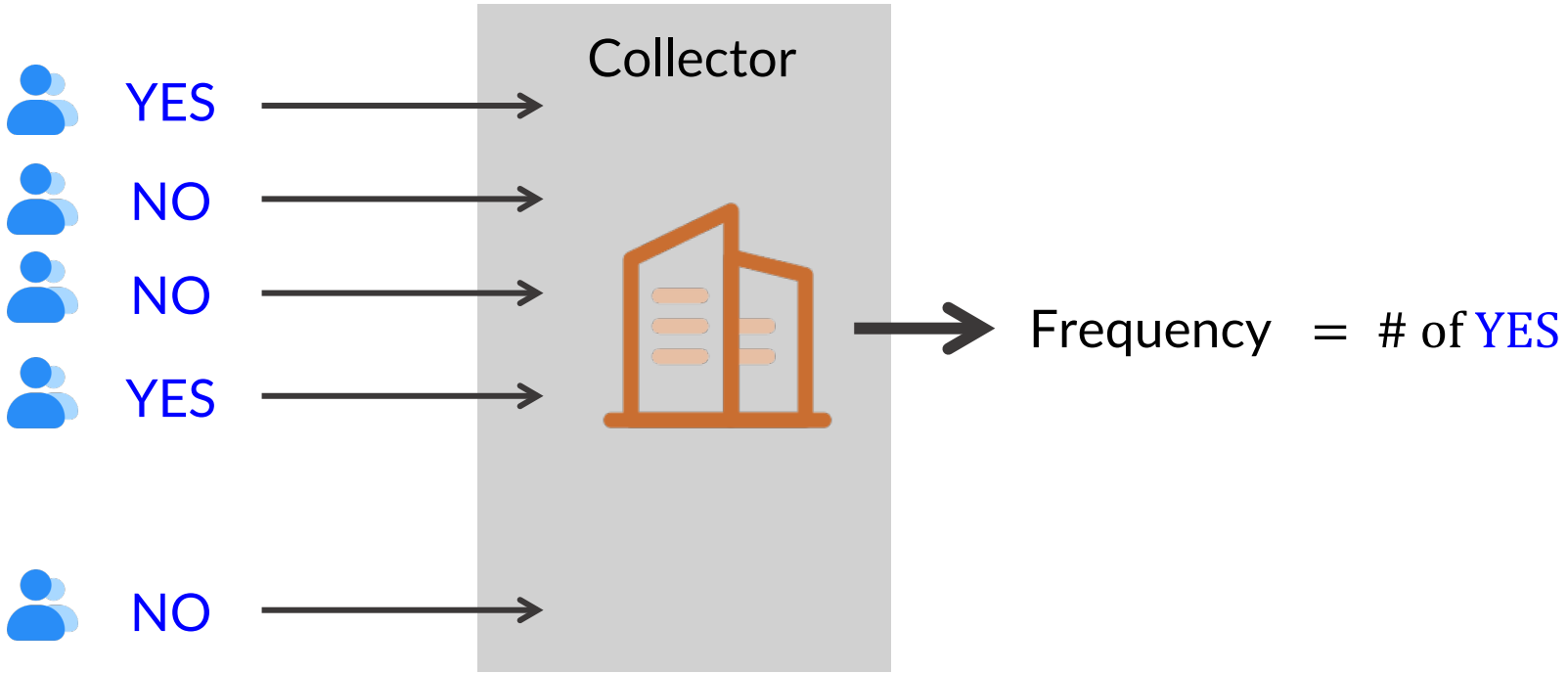
Frequency Estimation of YES/NO

RIT

- Social science: How to know how many people engage in tax evasion?

  - ask one person if they had evaded tax

  - the person answers YES or NO



Formal Privacy

RIT

- People have privacy concerns on sensitive/embarrassing question
  - i.e. don't want to answer → how to collect answers in a privacy-preserving way?

- A privacy mechanism $\mathcal{M}$ satisfies LDP if

For any truth x_1, x_2, and randomized answer y:

$$\max \frac{\Pr[\mathcal{M}(x_1) = y]}{\Pr[\mathcal{M}(x_2) = y]} \leq e^{\varepsilon}$$

Distinguishability of x_1 (YES) and x_2 (NO) from y (randomized answer)

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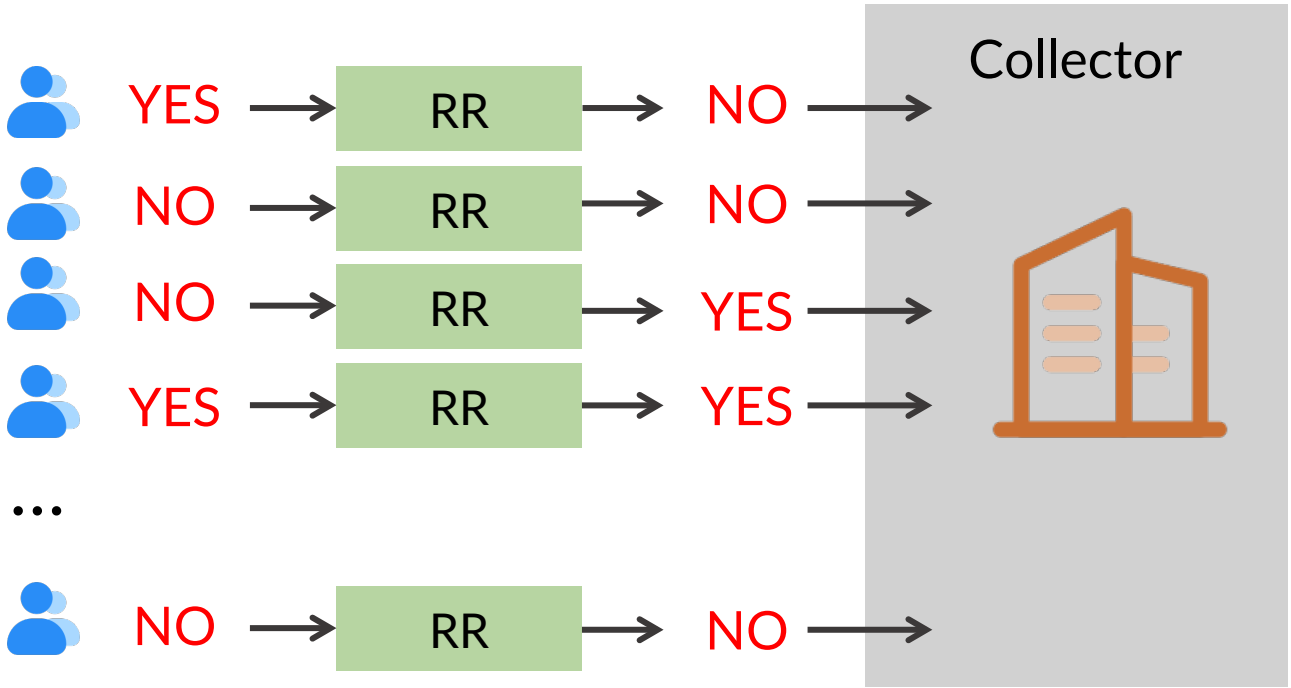
Distinguishability of x_1 (YES) and x_2 (NO) from y (randomized answer)

- **quantifiable hardness** to distinguish x_1 (YES) and x_2 (NO) from the randomized answer y
- defense against inference from data collectors  or adversaries 

Randomized Response for Privacy

RIT

- People have privacy concerns on sensitive/embarrassing question
  - i.e. don't want to answer → how to collect answers in a privacy-preserving way?
- Randomized Response: Randomize the truth before answering the collector



Randomized Response for Privacy

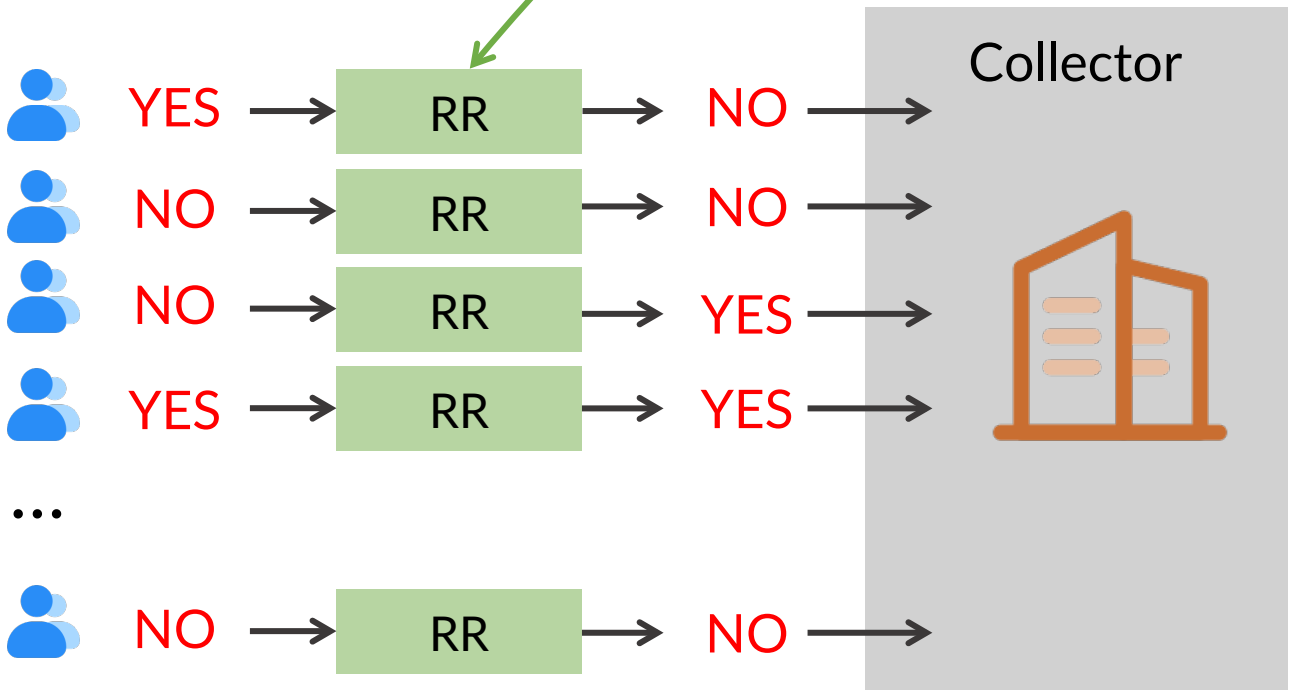
$$\max \frac{\Pr[\mathbf{RR}(x_1) = y]}{\Pr[\mathbf{RR}(x_2) = y]} \leq e^{\ln \frac{p}{1-p}}$$

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Private

RR: [Warner, 1965]
answer truth with probability p

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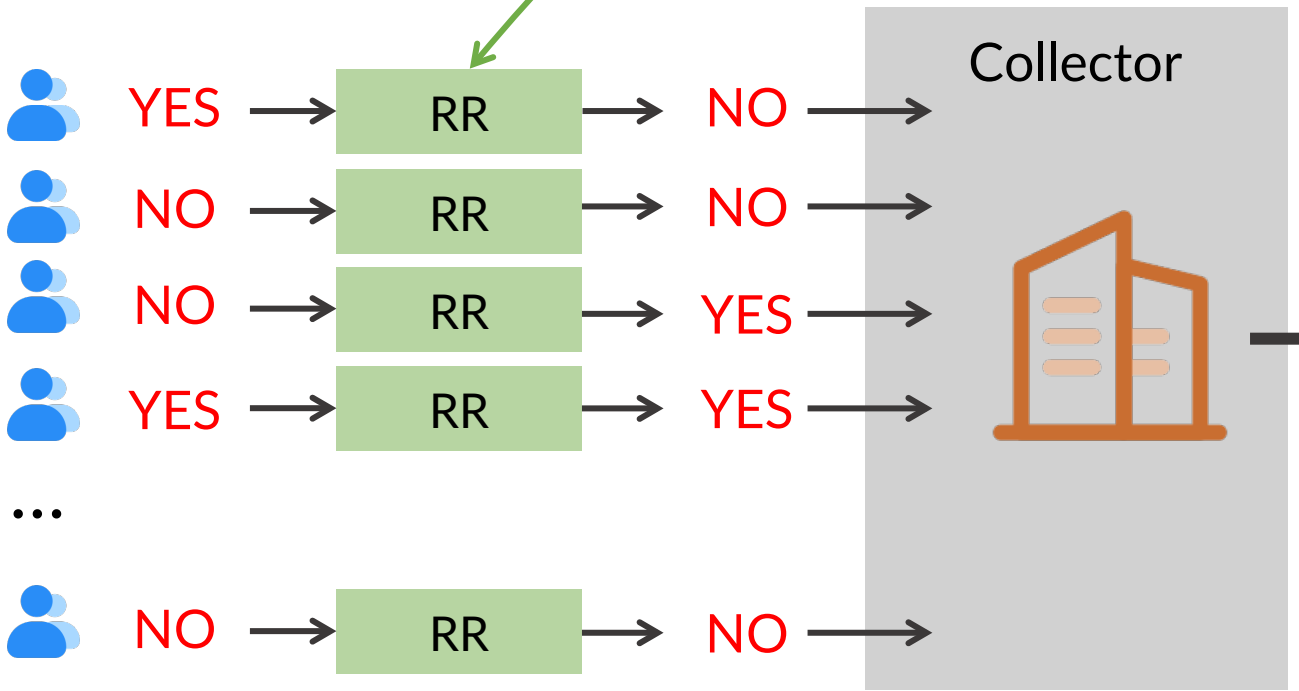
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estimated frequency

$$= \frac{\#\text{ of YES} - \#\,\text{👤} \times q}{p-q}$$

Unbiased:
expectation = truth

Utility: RR's Variance

RIT

- Randomization reduces data utility

$$\mathrm{Var}\left[\frac{\#\text{ of YES} - \#\,\text{👤}\times q}{p-q}\right] = \frac{\mathrm{Var}[\#\text{ of YES}]}{(p-q)^2} = \frac{npq}{(p-q)^2}$$

- classical result: summation of variance from all n **independent** randomization

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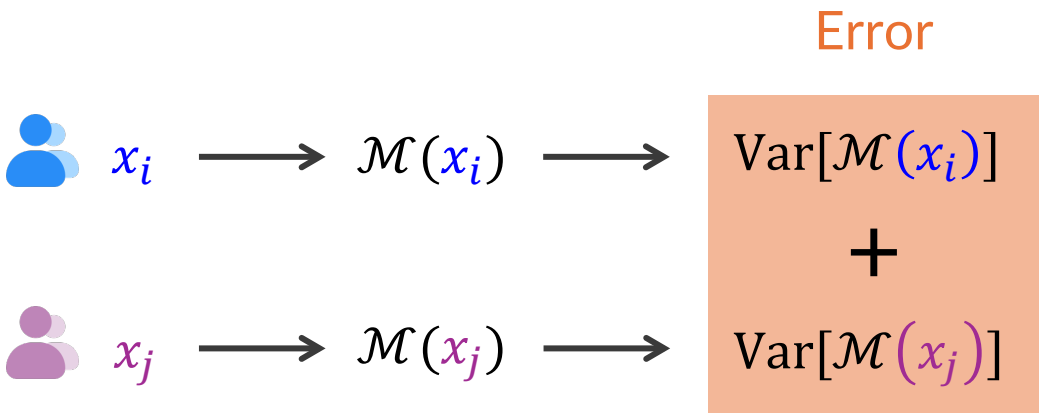
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- Q: Can we do better?
  - **yes, by correlated (joint) randomization**

Joint/Correlated RR (JRR)

RIT

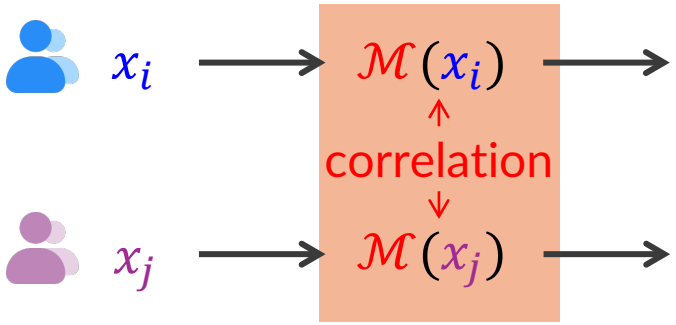
- Existing LDP mechanisms: Each user perturbs their data independently



Joint/Correlated RR (JRR)

RIT

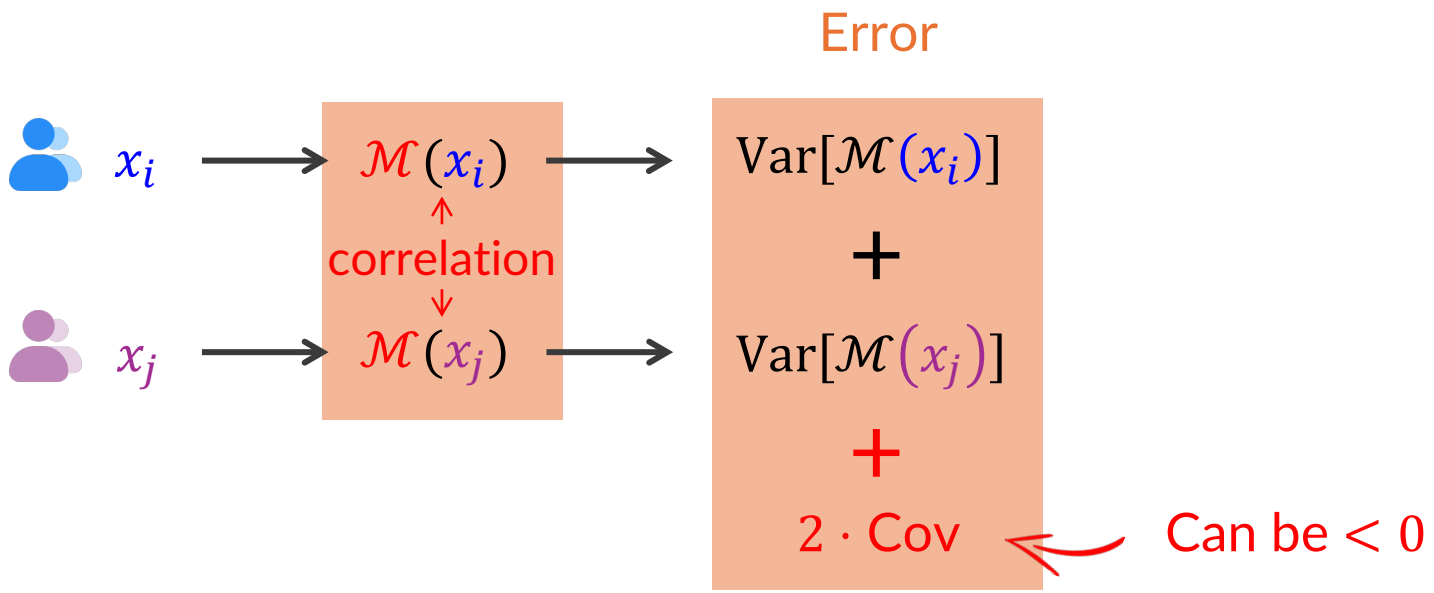
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Joint/Correlated RR (JRR)

RIT

- JRR: Better data utility by joint/correlated randomization
- **Example:** 2-person ($x_1 = \text{YES}$ and $x_2 = \text{YES}$) with $p = 0.8$ ($\text{P}[T = 1] = 0.8$)

RR: Joint distribution

	$T_1 = 1$	$T_1 = 0$
$T_2 = 1$	0.64 ($= p^2$)	0.16 ($= pq$)
$T_2 = 0$	0.16 ($= pq$)	0.04 ($= q^2$)

Truthfulness of x_1

Truthfulness of x_2

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$\mathbf{Cov}[y_1, y_2] = -0.04$

$\text{P}[T_1 = 0 \cap T_2 = \quad] \neq \text{P}[T_1 = 0] \cdot \text{P}[T_2 = 0] = 0.04$

NOT independent T_1 and T_2

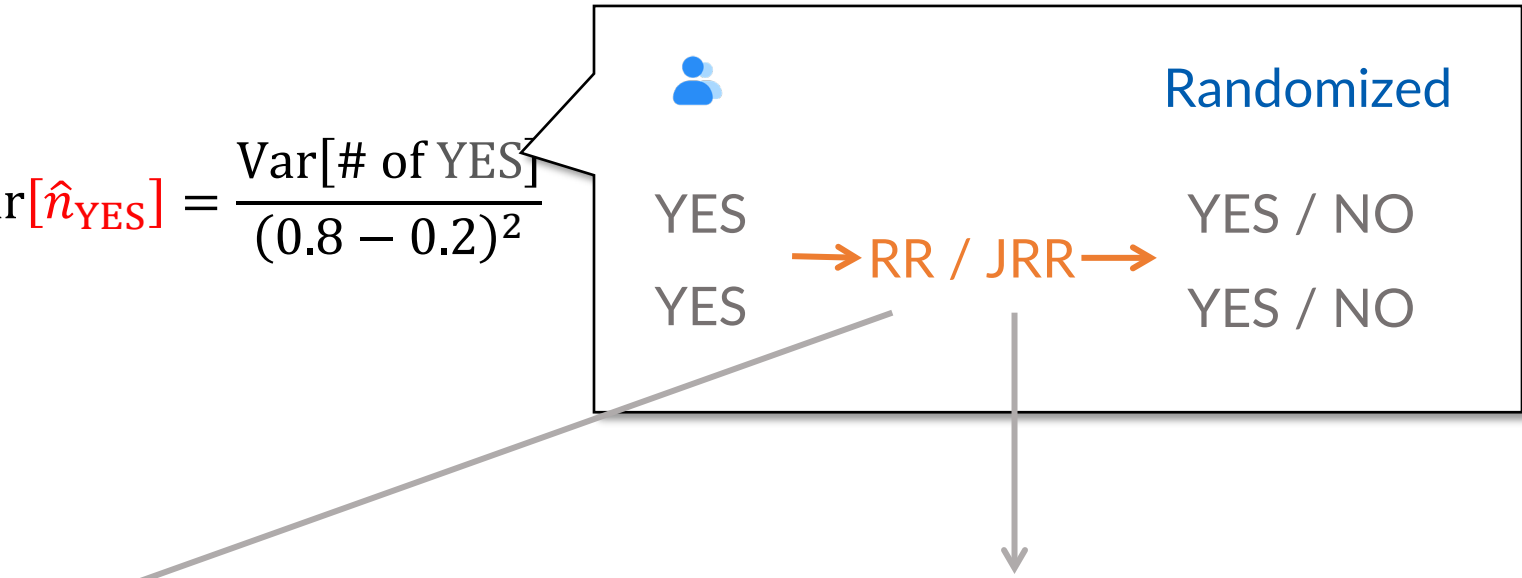
Joint probability ≠ Π of marginal probabilities

Utility: JRR's Variance

RIT

- Variance: (# 👤 = 2, $p = 0.8$)

$$\mathrm{Var}[\hat{n}_{\mathrm{YES}}] = \frac{\mathrm{Var}[\#\ \mathrm{of\ YES}]}{(0.8 - 0.2)^2}$$



RR

$$\mathrm{Var}[\#\ \mathrm{of\ YES}] = 2 \times pq = \mathbf{0.32}$$

JRR

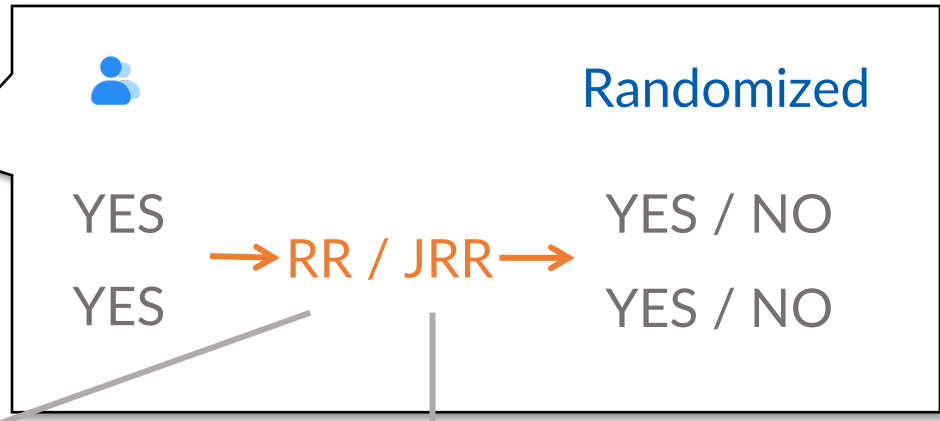
$$\mathrm{Var}[\#\ \mathrm{of\ YES}] = 0.32 + 2 \cdot \mathbf{Cov}[\boldsymbol{y_1}, \boldsymbol{y_2}]$$
$$= 0.32 + 2 \cdot (-0.04)$$
$$= \mathbf{0.24}$$

Utility: JRR's Variance

RIT

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JRR

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$$= 0.32 + 2 \cdot (-0.04)$$
$$= \mathbf{0.24}$$

Better data utility

JRR's General Form

RIT

- Correlated randomization with 2 persons x_{2i-1} and x_{2i}

JRR: Joint distribution

	$T_{2i-1} = 1$	$T_{2i-1} = 0$
$T_{2i} = 1$	$p^2 + \rho pq$	$(1 - \rho)pq$
$T_{2i} = 0$	$(1 - \rho)pq$	$q^2 + \rho pq$

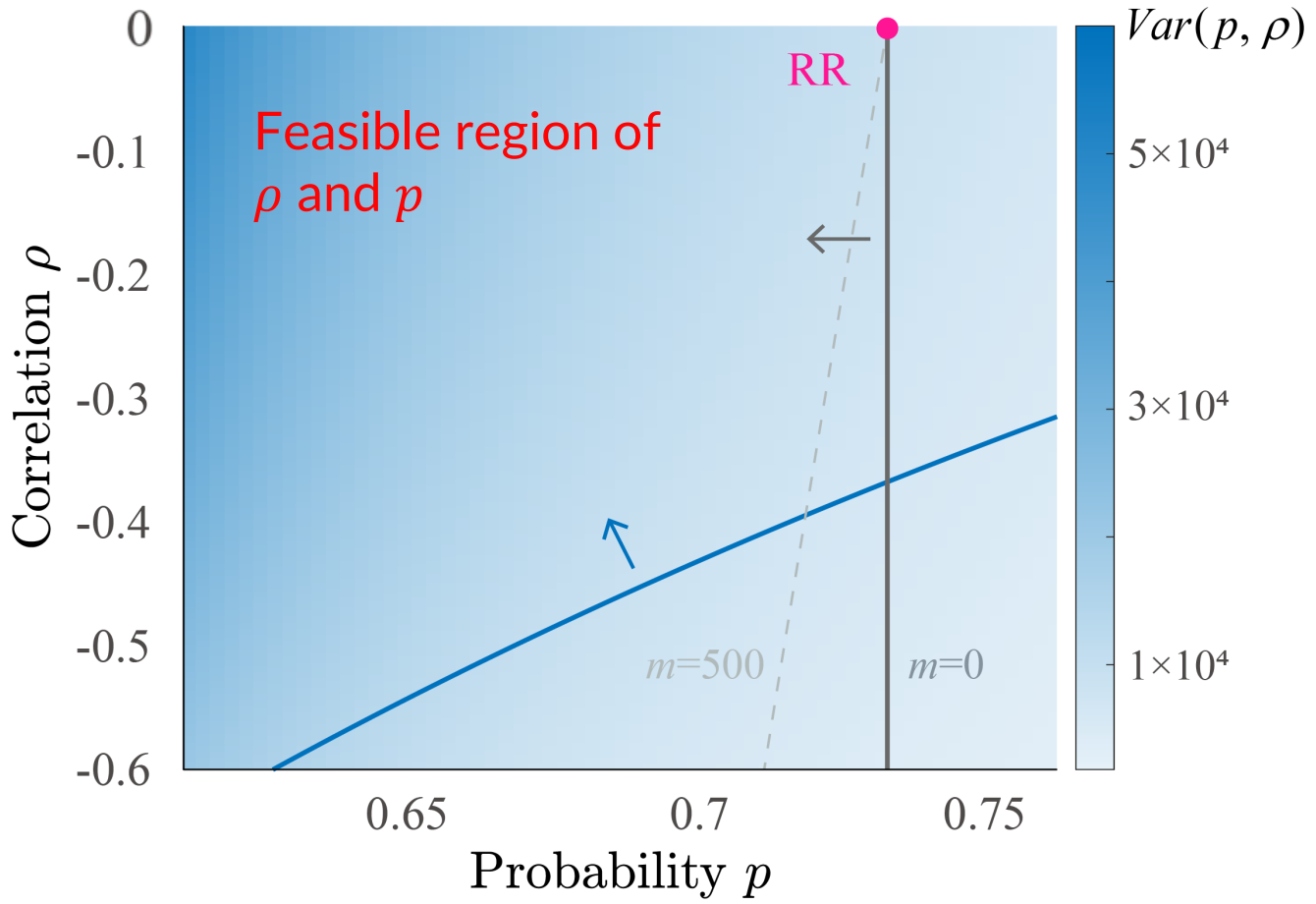
$\rho \in [-1,1]$: correlation coefficient

- RR is a special case of JRR with $\rho = 0$ (no correlation)

JRR – Variance Heatmap

RIT

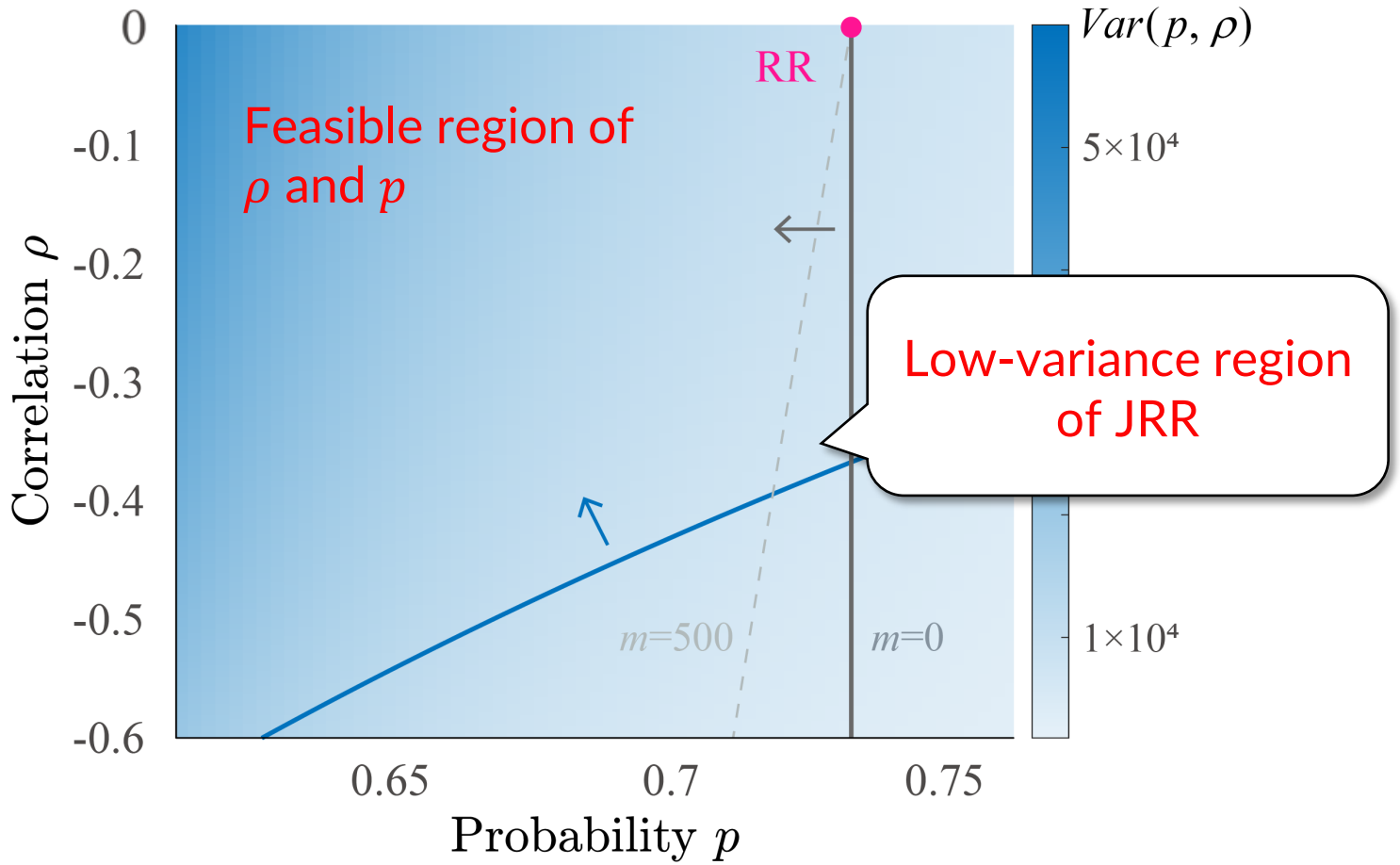
- Effect of ρ and p (when $\varepsilon = 1, n = 10^4, n_{\text{Yes}} = 200$, and malicious users $m = 0$ & 500)



JRR – Variance Heatmap

RIT

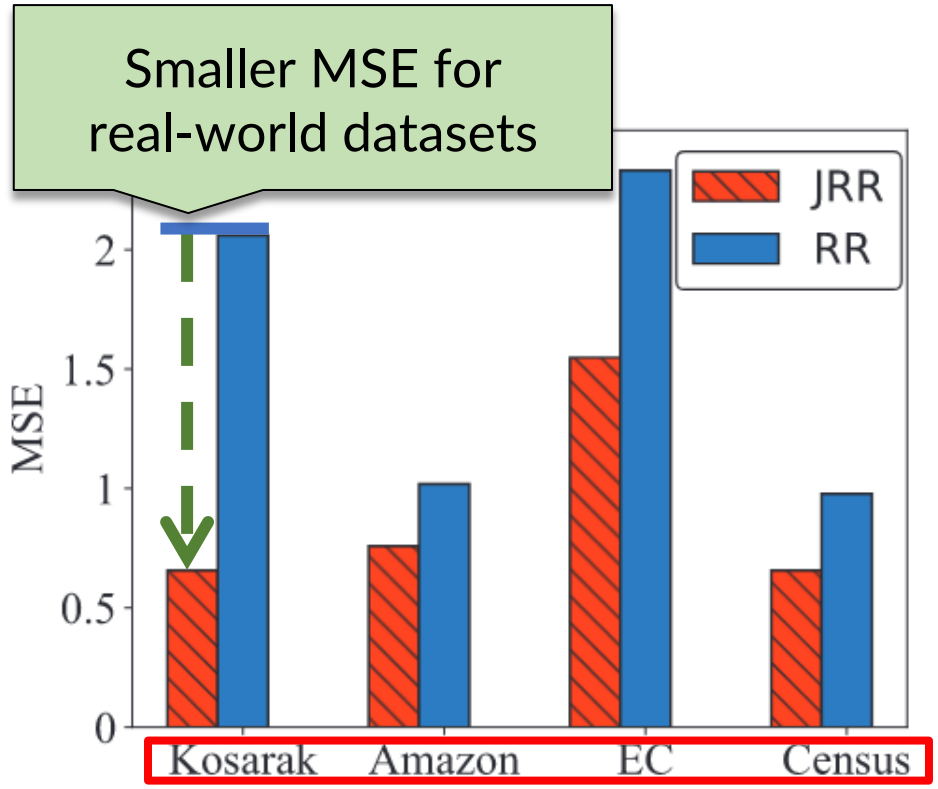
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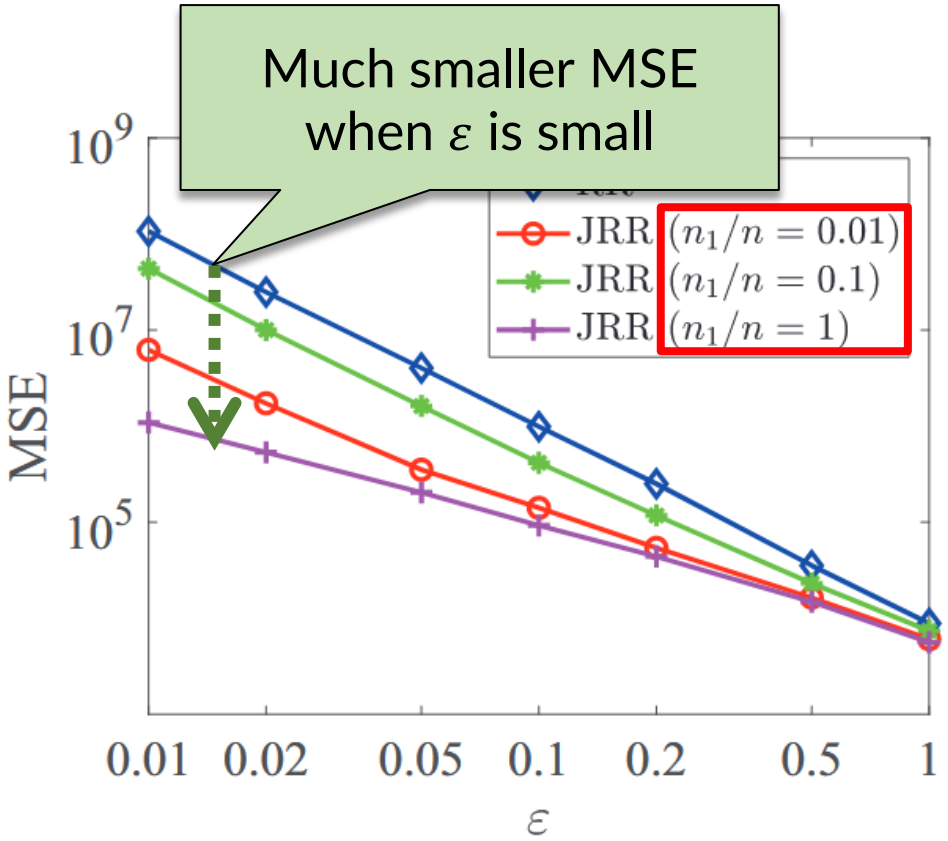
Evaluations

RIT

- Comparison with RR under the same privacy level - JRR: $\varepsilon(n, m, \rho, p)$, RR: $\varepsilon(p)$



Real-world datasets ($\varepsilon = 0.1$)



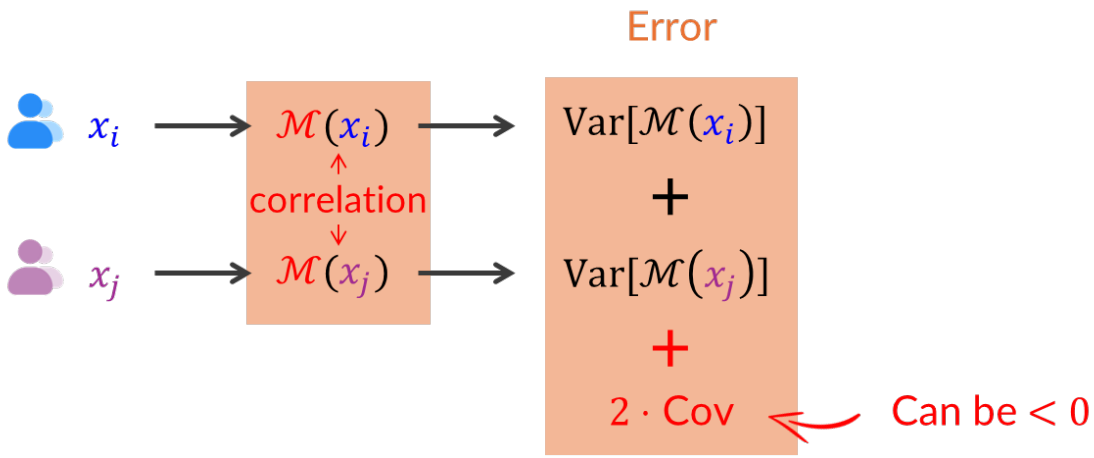
Synthetic datasets ($n = 10^4$)

Summary & Takeaway

RIT

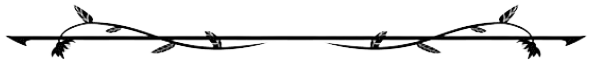
- **RQ: Can the classical RR be better?**

- Contributions:*
  - **introducing correlation into RR**
  - extensible to other LDP mechanisms
  - RR as a special case of $\rho = 0$
  - analytical privacy & data utility analysis



* Locally Differentially Private Frequency Estimation via Joint Randomized Response, PETS'25

Improving and Quantifying Utility under Local Differential Privacy



Mechanism design
- Part 1: Correlated $\mathcal{M}$
- Part 2: Optimal piecewise-based $\mathcal{M}$ — binary x → numerical x
- Part 3: $\mathcal{M}$ for trajectories in continuous space

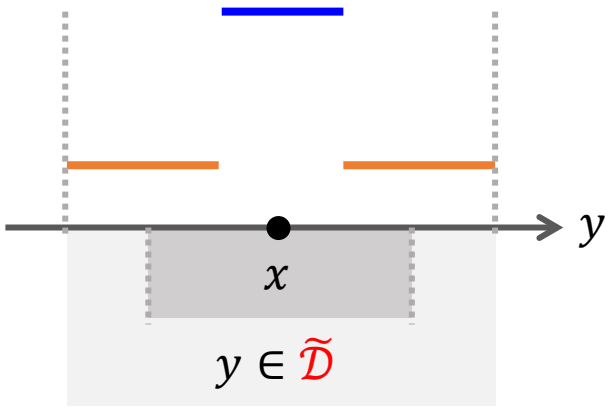
Utility analysis
- Part 4: Utility analysis for classifier $\circ$ $\mathcal{M}$

3-Piecewise Mechanism (Pattern)

RIT

- 3-piecewise distributions on bounded numerical domain $\mathcal{D} \rightarrow \widetilde{\mathcal{D}}$

  - given input x, sample output y from a distribution



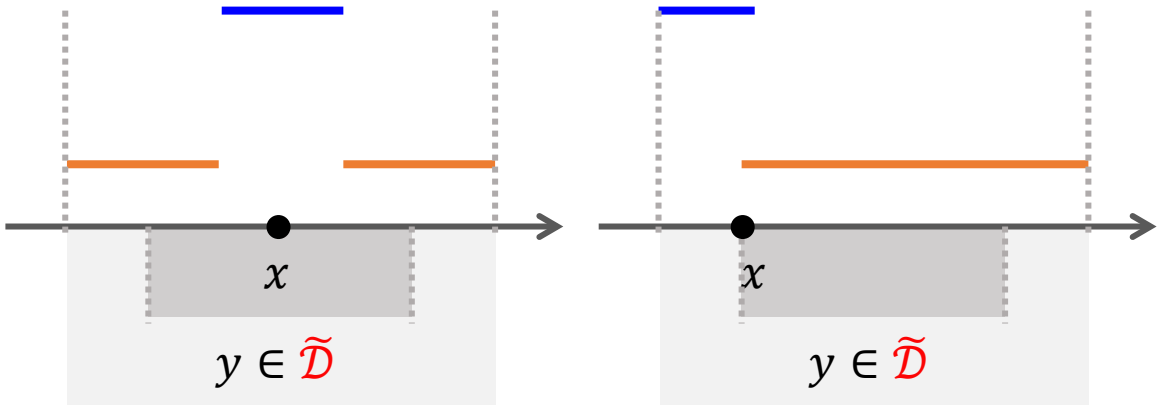
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Sampling probability depends on ε

Sampling interval depends on x and ε

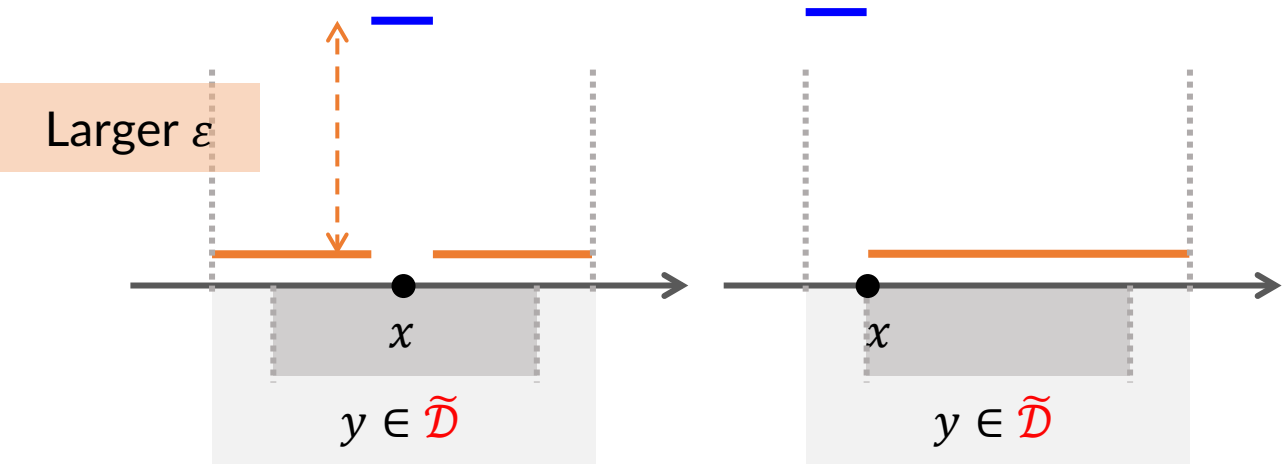
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- Instantiations: PM [ICDE'19], SW [SIGMOD'20], PTT [TMC'24] (design different $p_\varepsilon, l_{x,\varepsilon}, r_{x,\varepsilon}$)

  - different errors, but **without optimality,** which is important for building blocks

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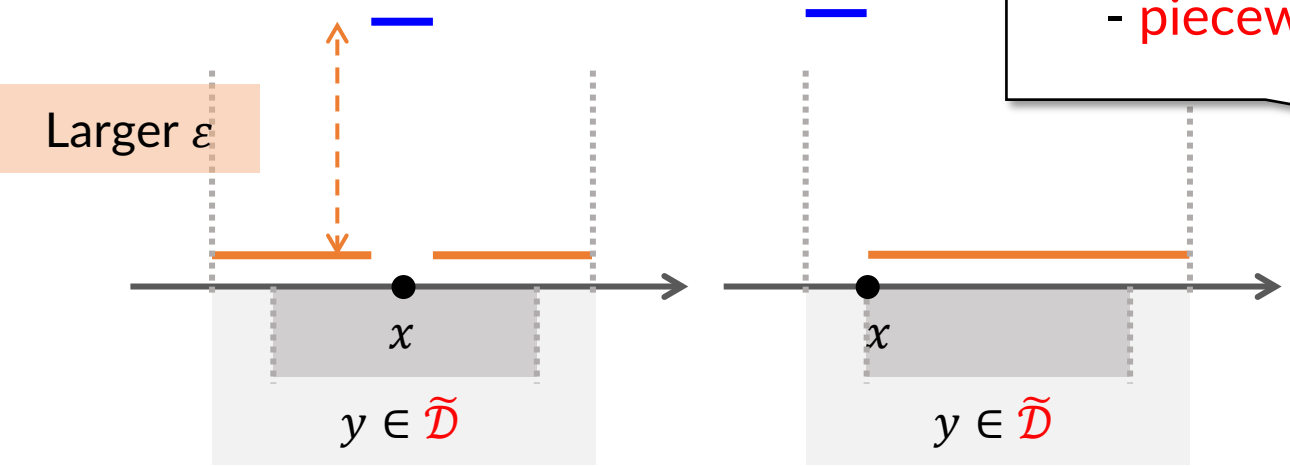
NOT enough to study optimality of piecewise-based mechanism

- only 3 pieces, two probabilities

- piecewise distribution can have more pieces and probabilities

- 3-piecewise distributions on boun

  - given input x, sample output y



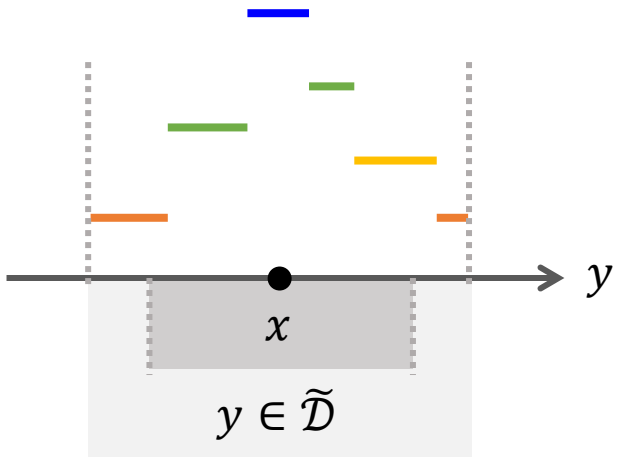
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Generalized Piecewise-based Mechanism

- **Most generalized version:** m-piecewise distribution

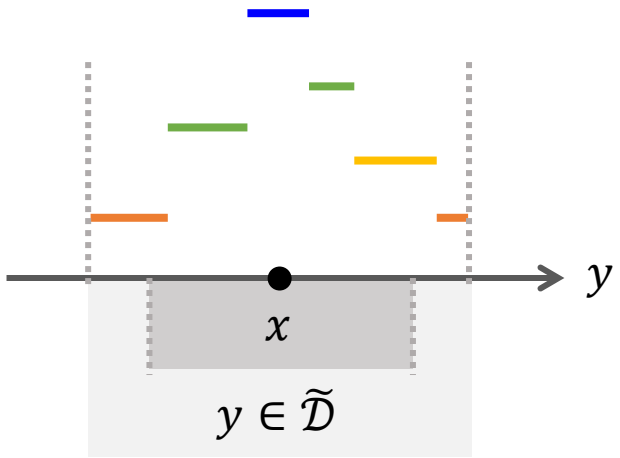


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$$\frac{p_{i,\varepsilon}}{p_{j,\varepsilon}} \leq e^{\varepsilon} \text{ (LDP constraint)}$$

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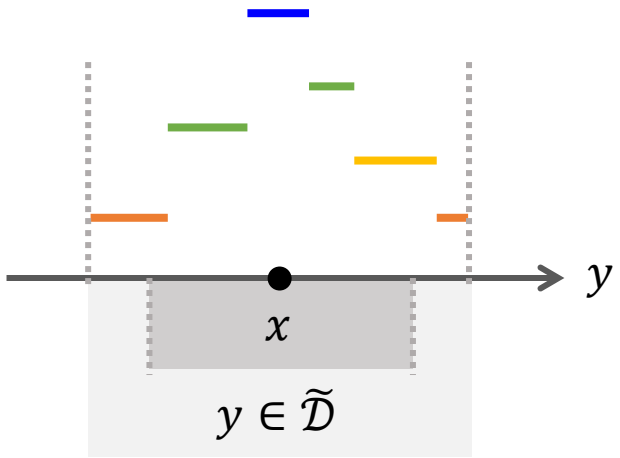
- Error (data utility):

$$\mathcal{L}(y, x)$$

$$\mathcal{L}(y, x) \coloneqq |y - x|^p$$

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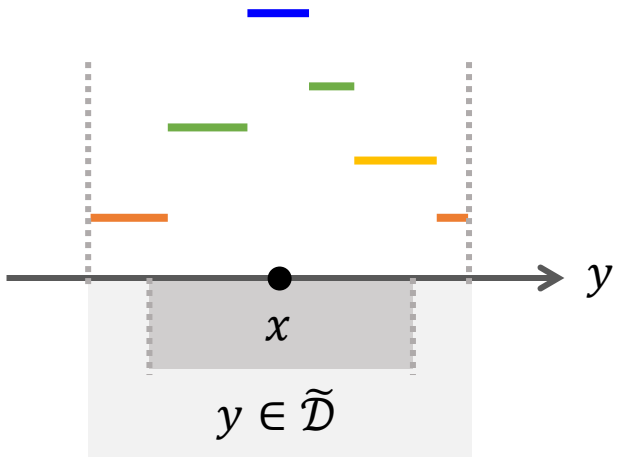
- Expected error:

$$\int_{\widetilde{\mathcal{D}}} \mathcal{L}(y, x) \cdot pdf[\mathcal{M}(x) = y] \mathrm{d}y$$

Optimal Piecewise-based Mechanism

RIT

- **Most generalized version:** m-piecewise distribution



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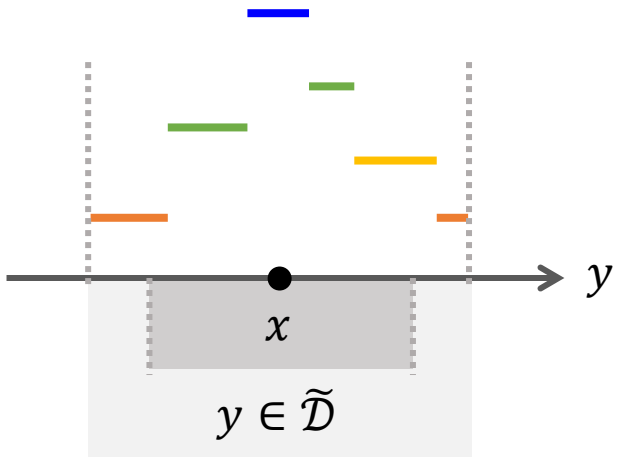
$$\min_{\mathcal{M}:\, p_i, l_i, r_i} \int_{\widetilde{\mathcal{D}}} \mathcal{L}(y, x) \cdot pdf[\mathcal{M}(x) = y] \mathrm{d}y$$

Find $\mathcal{M}$ to minimize the error at x

Optimal Piecewise-based Mechanism

RIT

- **Most generalized version:** m-piecewise distribution



$$pdf[\mathcal{M}(x) = y] = \begin{cases} p_{1,\varepsilon} & \text{if } y \in [l_{1,x,\varepsilon}, r_{1,x,\varepsilon}] \\ p_{2,\varepsilon} & \text{if } y \in [l_{2,x,\varepsilon}, r_{2,x,\varepsilon}] \\ & \dots \\ p_{m,\varepsilon} & \text{if } y \in [l_{m,x,\varepsilon}, r_{m,x,\varepsilon}] \end{cases}$$

$$\frac{p_{i,\varepsilon}}{p_{j,\varepsilon}} \leq e^{\varepsilon} \text{ (LDP constraint)}$$

- Expected error:

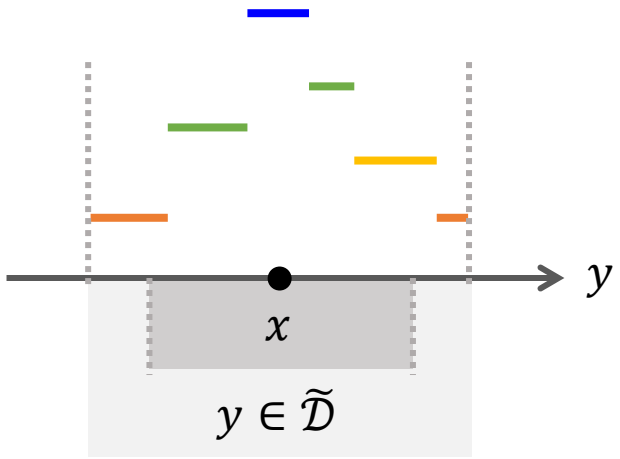
$$\min_{\mathcal{M}:\, p_i, l_i, r_i} \max_{x \in \mathcal{D}} \int_{\widetilde{\mathcal{D}}} \mathcal{L}(y, x) \cdot pdf[\mathcal{M}(x) = y] \mathrm{d}y$$

Find $\mathcal{M}$ to minimize the worst-case error

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$$\frac{p_{i,\varepsilon}}{p_{j,\varepsilon}} \leq e^{\varepsilon} \text{ (LDP constraint)}$$

Solved $\mathcal{M}$ is
the optimal piecewise-based mechanism

Mathematically ≡ to find the **optimal piecewise distribution under the LDP constraint**

$$\min_{\mathcal{M}:\, p_i, l_i, r_i} \max_{x \in \mathcal{D}} \int_{\widetilde{\mathcal{D}}} \mathcal{L}(y, x) \cdot pdf[\mathcal{M}(x) = y] \mathrm{d}y$$

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Challenges & Proofs

RIT

- Challenges
  1. min-max problem & multiple variables
  2. optimal results only for a specific m

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2. $i \in [m]$

1. variables p_i, l_i, r_i

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2. $i \in [m]$ — 1. variables p_i, l_i, r_i

1

Inner max
has a closed form

Worst-case from
x = endpoints of $\mathcal{D}$

Reduced to:
minimization problem

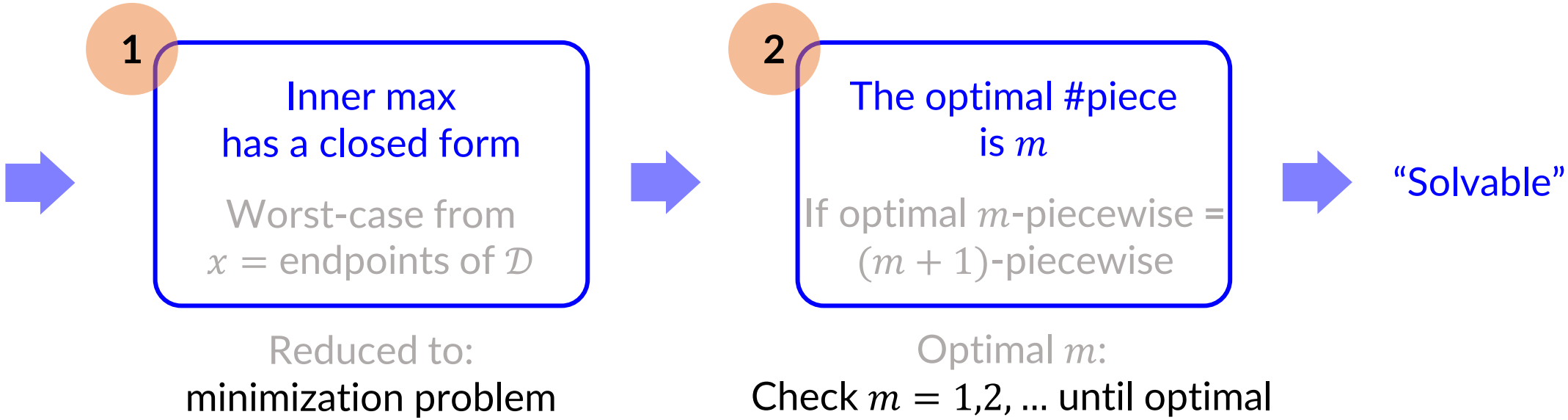
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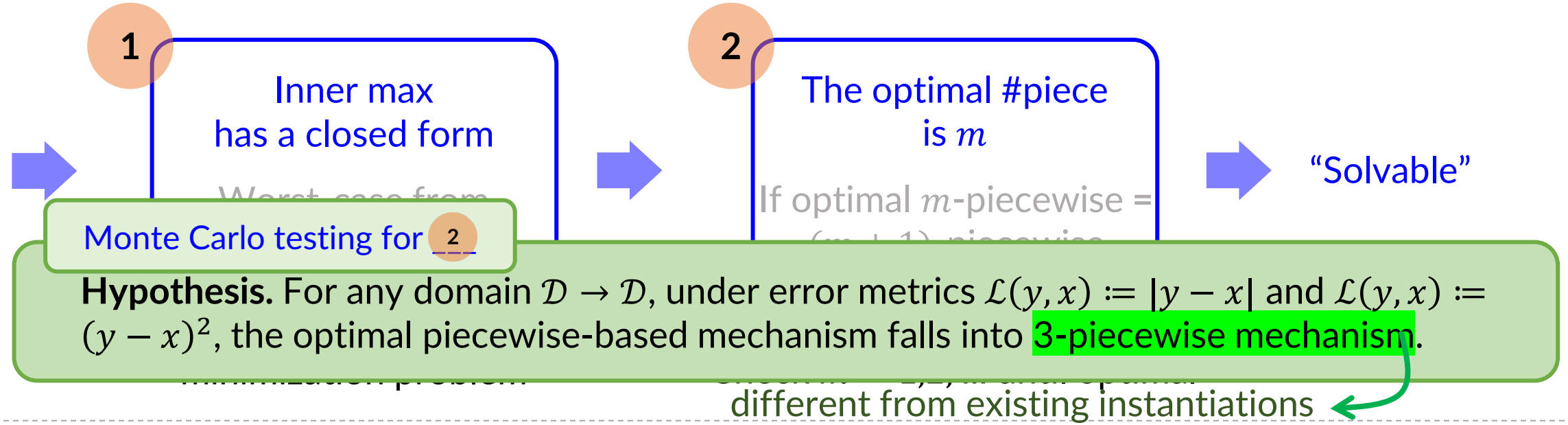
Challenges & Proofs

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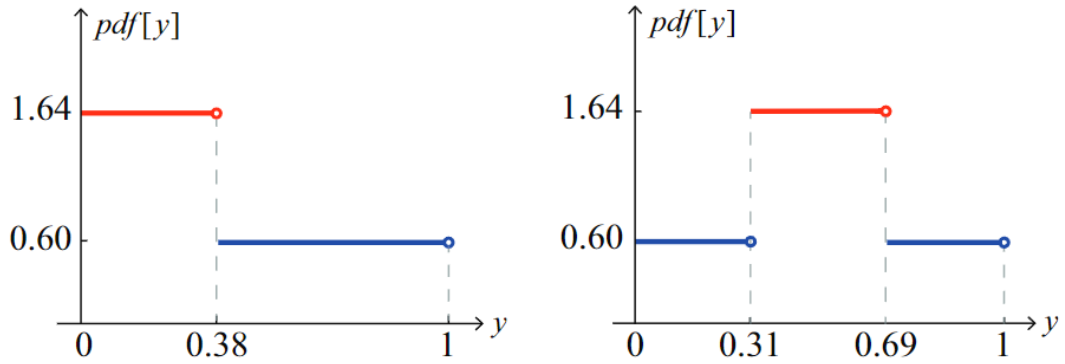
Optimal Closed-Form Mechanism

RIT

- Optimal $\mathcal{M}: [0,1) \to [0,1)$ under $\mathcal{L} := |y - x|$

$$pdf[\mathcal{M}(x) = y] = \begin{cases} \exp\left(\frac{\varepsilon}{2}\right) & \text{if } y \in [l_{x,\varepsilon}, r_{x,\varepsilon}) \\ \exp\left(-\frac{\varepsilon}{2}\right) & \text{if } y \in [0,1) \backslash [l_{x,\varepsilon}, r_{x,\varepsilon}) \end{cases} \leftarrow [l_{x,\varepsilon}, r_{x,\varepsilon}) = \begin{cases} [0, 2C) & \text{if } x \in [0, C) \\ x + [-C, C) & \text{if } x \in [C, 1-C) \\ [1-2C, 1) & \text{otherwise} \end{cases} \leftarrow C = \frac{\exp(\varepsilon/2) - 1}{2(\exp(\varepsilon) - 1)}$$

- When $\varepsilon = 1$:



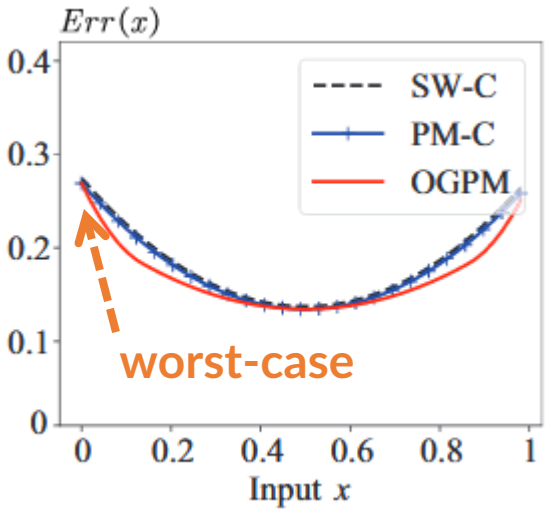
(a) $x = 0$. **(b)** $x = 0.5$.

Comparison of Expected Errors

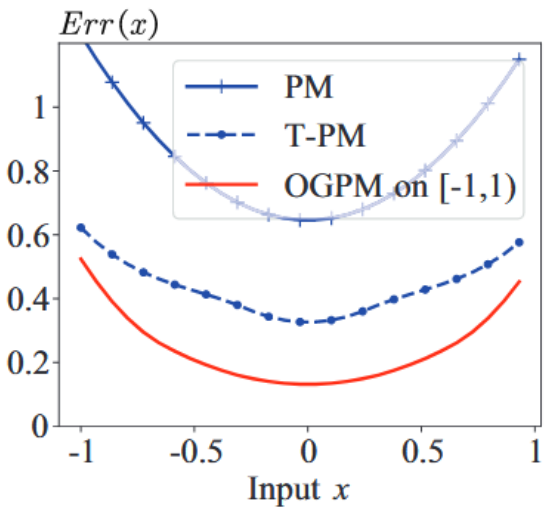
RIT

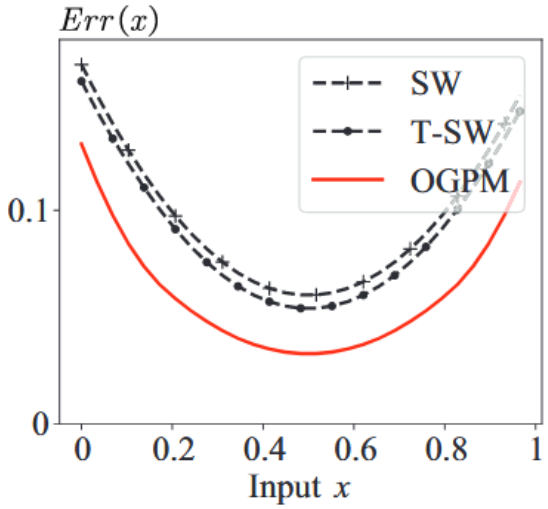
Whole-domain error (i.e. each point in $\mathcal{D}$) **($\varepsilon = 2$)**

Compressed PM, SW



Original and **truncated** PM, SW

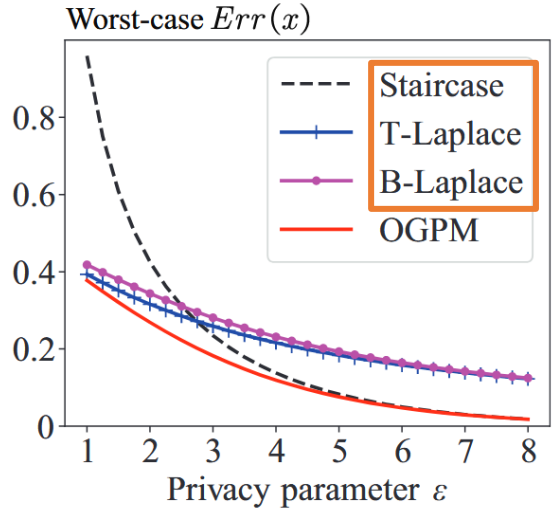




Lowest error

Worst-case error

Non-piecewise-based

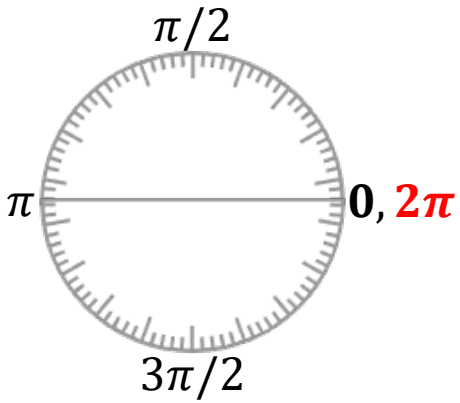


Lowest error

Circular Domain

RIT

- Different meaning of distance, e.g. $\text{distance}(0, 2\pi) = 0 \rightarrow$ requires different mechanisms



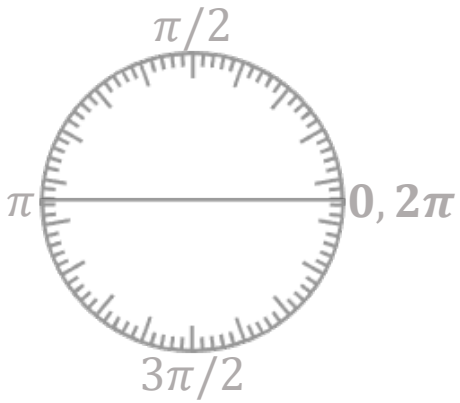
$$\mathcal{L}: |y - x| \rightarrow \mathcal{L}_{\text{mod}}: |y - x|_{\text{mod}}$$

$$\mathcal{L}_{\text{mod}}(y, x) := \min\{|y - x|, 2\pi - |y - x|\}$$

Circular Domain

RIT

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$$\mathcal{L}_{\text{mod}}(y, x) := \min\{|y - x|, 2\pi - |y - x|\}$$

$$\min_{\mathcal{M}: p_i, l_i, r_i} \max_{x \in [0, 2\pi]} \int_{\widetilde{\mathcal{D}}} \mathcal{L}_{\text{mod}}(y, x) \cdot pdf[\mathcal{M}(x) = y] \mathrm{d}y$$

- **Linking** to problems in the classical domain

min-max under $\mathcal{L}_{\text{mod}}$	=	min under $\mathcal{L}$ at π	min under $\mathcal{L}_{\text{mod}}$ at x	= min under $\mathcal{L}$ at π + Transform
p_i			$l_{i,x}^{\text{mod}}, r_{i,x}^{\text{mod}}$	

Optimal Closed-Form Mechanism

RIT

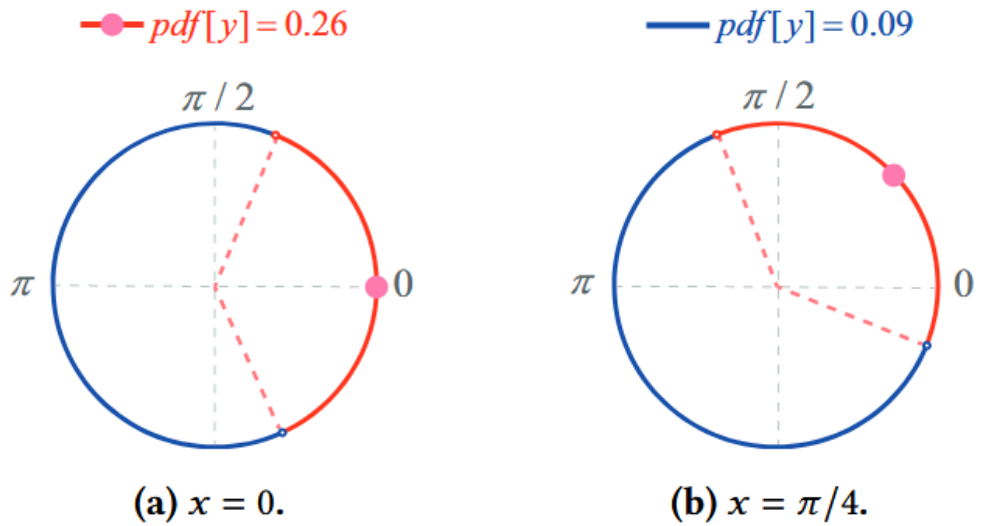
- Optimal $\mathcal{M}: [0,2\pi) \to [0,2\pi)$ under $\mathcal{L}_{\text{mod}}$

$$pdf[\mathcal{M}(x) = y] = \begin{cases} \frac{1}{2\pi}\exp\left(\frac{\varepsilon}{2}\right) & \text{if } y \in [l_{x,\varepsilon}^{\text{mod}}, r_{x,\varepsilon}^{\text{mod}}) \\ \frac{1}{2\pi}\exp\left(-\frac{\varepsilon}{2}\right) & \text{if } y \in [0,2\pi)\backslash[l_{x,\varepsilon}^{\text{mod}}, r_{x,\varepsilon}^{\text{mod}}) \end{cases}$$

$$[l_{x,\varepsilon}^{\text{mod}}, r_{x,\varepsilon}^{\text{mod}}) = [x - C, x + C) \bmod 2\pi$$

$$C = \pi\frac{\exp(\varepsilon/2) - 1}{\exp(\varepsilon) - 1}$$

- When $\varepsilon = 1$:



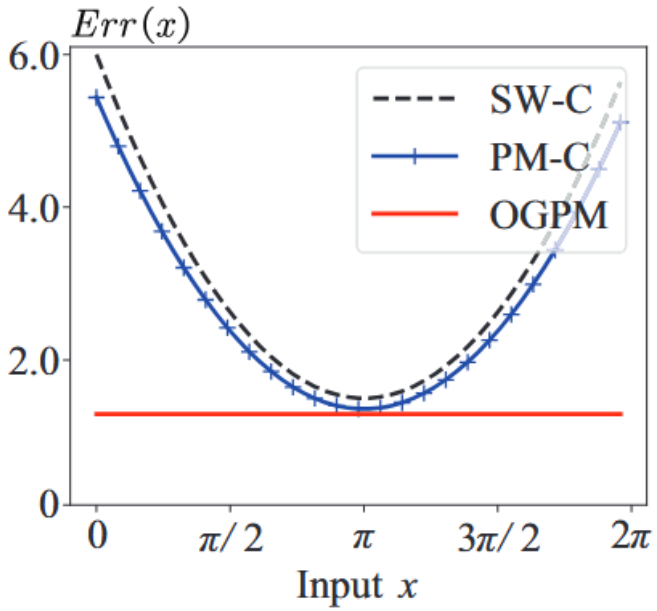
(a) $x = 0$. **(b)** $x = \pi/4$.

Comparison of Expected Errors

RIT

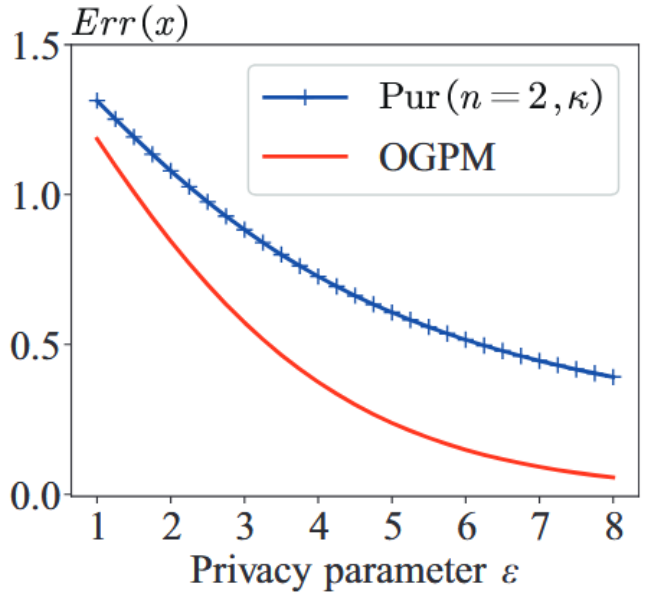
Whole-domain error ($\varepsilon = 2$)

PM, SW on the **flattened** domain



Worst-case error

Purkayastha mechanism [CCS'21]*



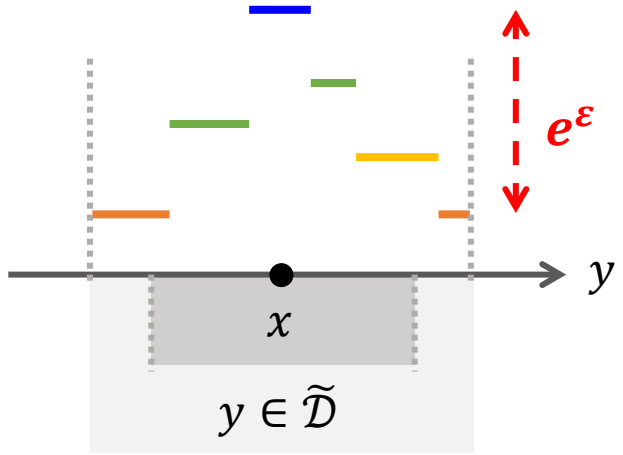
Lowest error

* Differential Privacy for Directional Data, CCS'21

Summary & Takeaway

RIT

- **RQ: What is the optimal piecewise-based mechanism?**



- Contributions:*

  - solving framework for the optimality

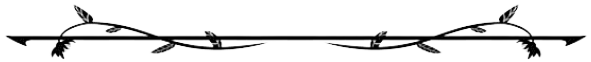
  - closed-form mechanisms for classical domains & circular domains

  - comparison with non-piecewise-based mechanisms

* Optimal Piecewise-based Mechanism for Collecting Bounded Numerical Data under Local Differential Privacy, PETS'25

- Artifact Award Runner-up: Use `uv` package manager for universal and fast reproduction (in 2024, when `uv` was first released by Astra)

Improving and Quantifying Utility under Local Differential Privacy



Mechanism design
- Part 1: Correlated $\mathcal{M}$
- Part 2: Optimal piecewise-based $\mathcal{M}$
- Part 3: $\mathcal{M}$ for trajectories in continuous space

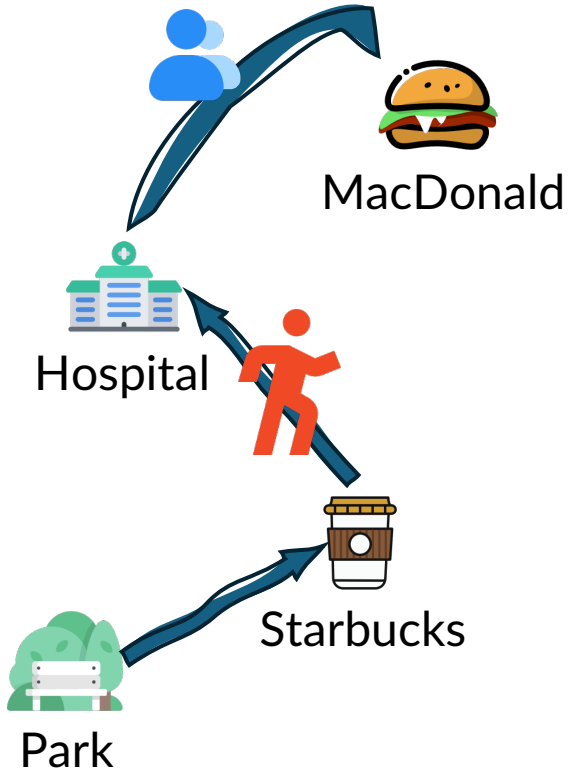
1D x → 2D x

Utility analysis
- Part 4: Utility analysis for classifier $\circ$ $\mathcal{M}$

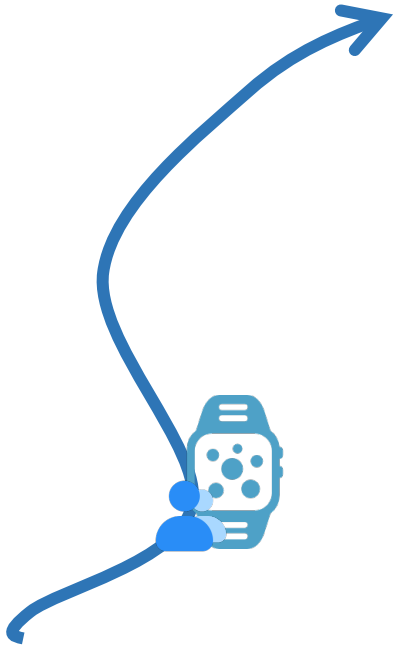
Trajectory Collection

RIT

- Sensitive trajectories: daily routine, wearable-sensor trajectories



Discrete locations w/ labels

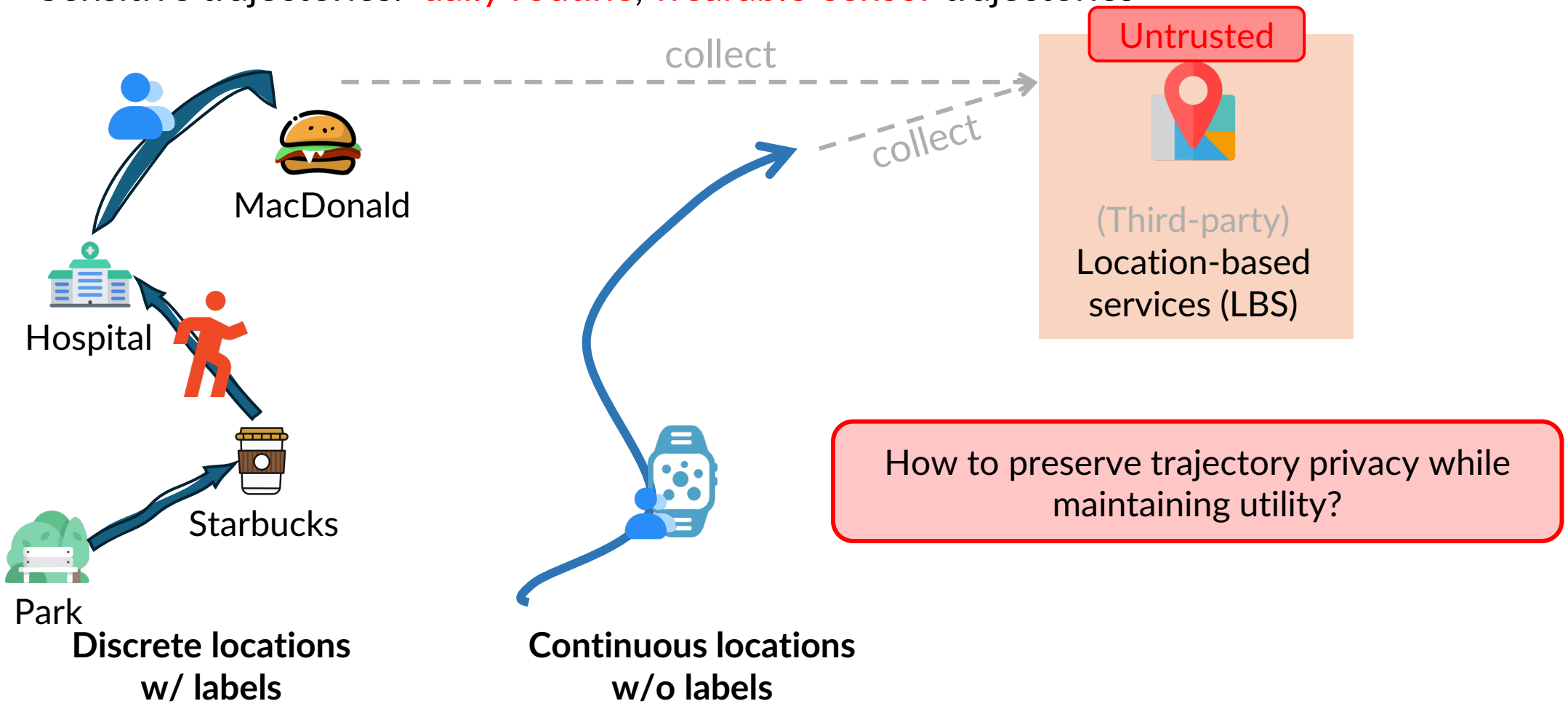


Continuous locations w/o labels

Trajectory Collection

RIT

- Sensitive trajectories: daily routine, wearable-sensor trajectories



LDP-fy a Trajectory

RIT

- LDP-fy: perturb a trajectory with LDP guarantee (provable privacy)
  - cannot distinguish location τ_1 from τ_2 with confidence quantified by e^{ε}

$$\forall \tau_1, \tau_2, \tau_* \in \mathcal{S}: \quad \frac{\Pr[\mathcal{M}(\tau_1) = \tau_*]}{\Pr[\mathcal{M}(\tau_2) = \tau_*]} \leq e^{\varepsilon}$$

location space

distinguishability of τ_1 from τ_2 when observing τ_*

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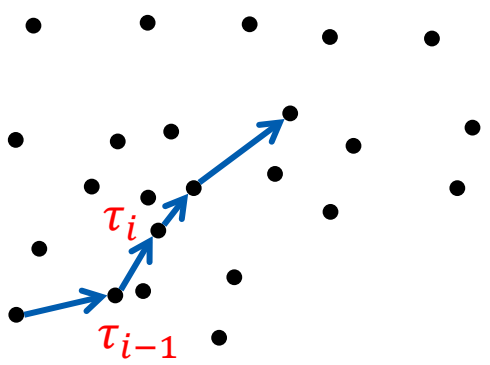
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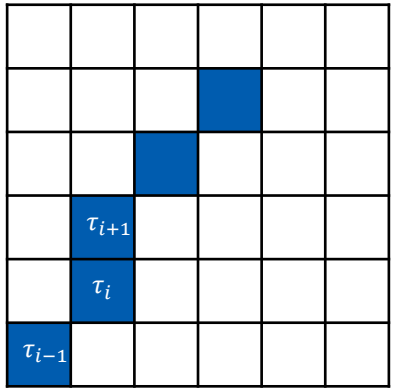
- Target: LDP mechanism $\mathcal{M}$ (for the location space)
  - provable privacy for users' trajectories, in continuous spaces and discrete spaces
  - as high trajectory utility as possible

LDP-fy a Trajectory in A **Discrete Space**

- Discrete location spaces: Point of interests or discretized cells



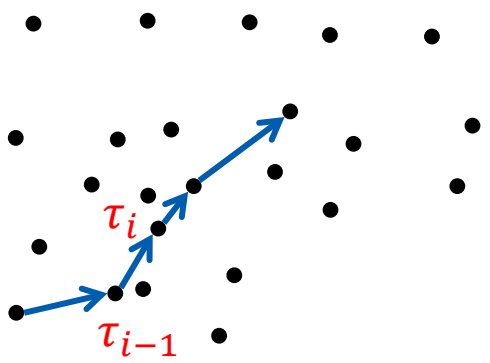
$\mathcal{S} = \{p_1, \dots, p_n\}$



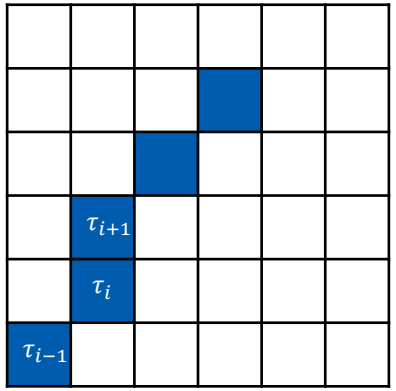
$\mathcal{S} = \{c_1, \dots, c_n\}$

LDP-fy a Trajectory in A **Discrete Space**

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$\mathcal{S} = \{p_1, \dots, p_n\}$



$\mathcal{S} = \{c_1, \dots, c_n\}$

- Exponential mechanism:

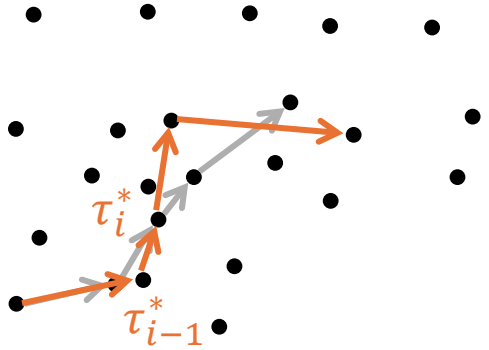
$$\Pr[\mathcal{M}_{\exp}(\tau) = \tau^*] = \frac{\exp\left(\varepsilon d(\tau, \tau^*)\right)}{\sum_{\tau' \in S} \exp\left(\varepsilon d(\tau, \tau')\right)}$$

← Pairwise distance

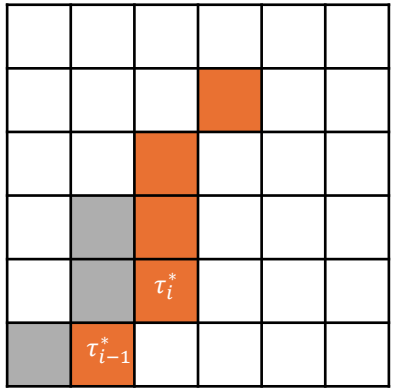
← Sum of distance

LDP-fy a Trajectory in A **Discrete Space**

- Discrete location spaces: Point of interests or discretized cells



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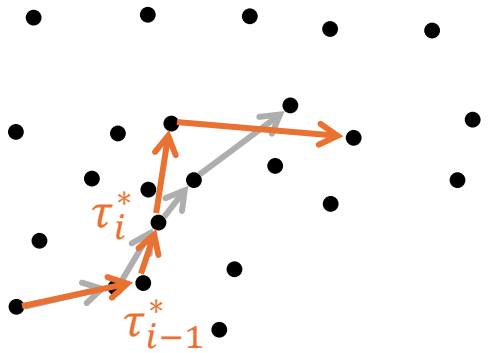
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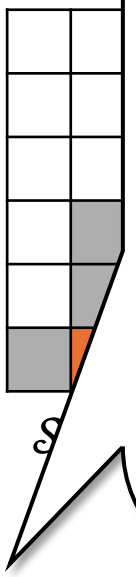
← Sum of distance

LDP-fy a Trajectory in A **Discrete Space**

- Discrete location spaces: Point of interests or discretized c



$\mathcal{S} = \{p_1, \dots, p_n\}$



Limitations

1. Efficiency

- each sample costs $\mathcal{O}(n)$

2. Trajectory Utility

- decreases as n increases

3. Applicability

- cannot apply to continuous $\mathcal{S}$

- Exponential mechanism:

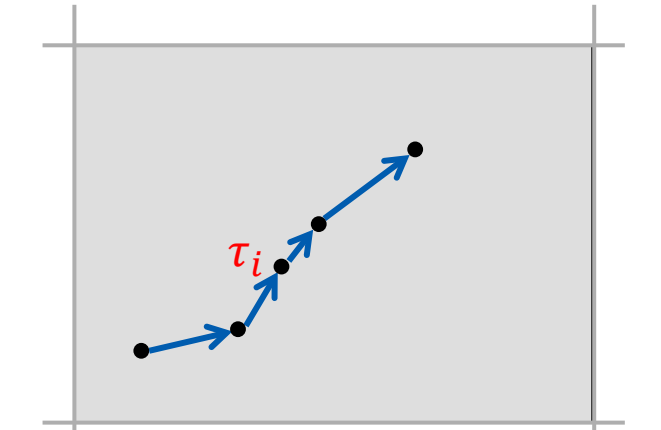
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← Pairwise distance

← Sum of distance

Continuous Spaces: Better & Universal

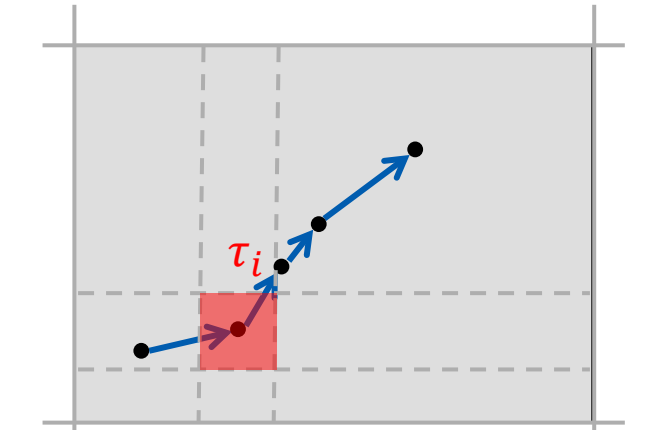
- Operate on a continuous space (covering the discrete space)



$$\mathcal{S} = [a_{\text{sta}}, a_{\text{end}}] \times [b_{\text{sta}}, b_{\text{end}}]$$

Continuous Spaces: Better & Universal

- Operate on a continuous space (covering the discrete space)



$\mathcal{S} = [a_{\text{sta}}, a_{\text{end}}] \times [b_{\text{sta}}, b_{\text{end}}]$

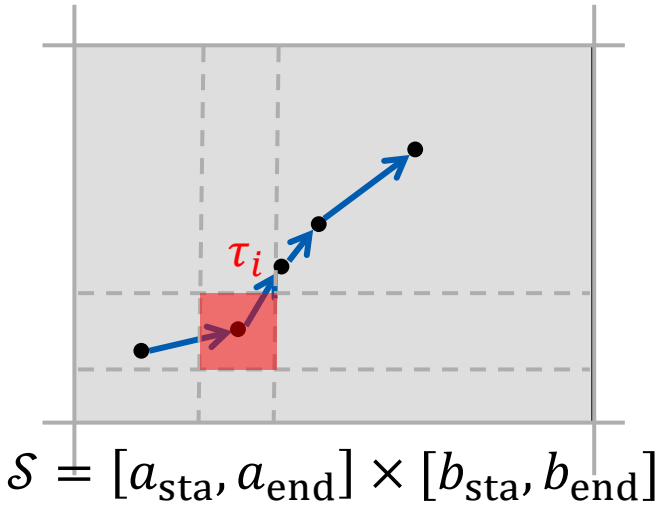
Simple sampling: (satisfying LDP for $\mathcal{S}$)

$$\Pr[\mathcal{M}(x_i) = \tilde{x} \in \blacksquare] = p_{\text{high}}$$

$$\Pr[\mathcal{M}(x_i) = \tilde{x} \in \blacksquare] = p_{\text{low}}$$

Continuous Spaces: Better & Universal

- Operate on a continuous space (covering the discrete space)



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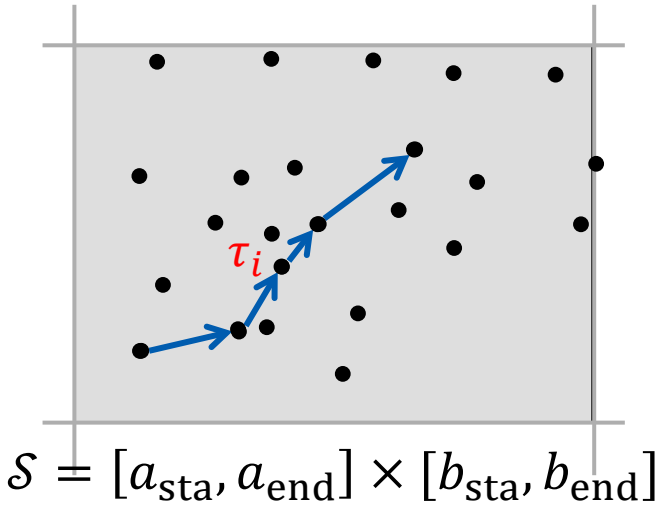
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$$\Pr[\mathcal{M}(x_i) = \tilde{x} \in \blacksquare] = p_{\text{low}}$$

- Benefits:
  1. **Efficiency:** $\mathcal{O}(1)$ sampling complexity
  2. **Trajectory utility:** "n-independent"
  3. **Applicability:** Both continuous spaces & discrete spaces

Continuous Spaces: Better & Universal

- Operate on a continuous space (covering the discrete space)



Simple sampling: (satisfying LDP for $\mathcal{S}$)

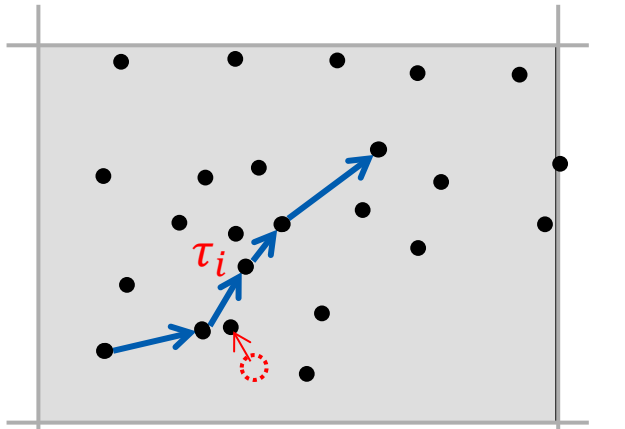
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  1. **Efficiency:** $\mathcal{O}(1)$ sampling complexity
  2. **Trajectory utility:** "n-independent"
  3. **Applicability:** Both continuous spaces & discrete spaces

Continuous Spaces: Better & Universal

- Operate on a continuous space (covering the discrete space)



$\mathcal{S} = [a_{\text{sta}}, a_{\text{end}}] \times [b_{\text{sta}}, b_{\text{end}}]$

Simple sampling: (satisfying LDP for $\mathcal{S}$)

$\Pr[\mathcal{M}(x_i) = \tilde{x} \in \blacksquare] = p_{\text{high}}$

$\Pr[\mathcal{M}(x_i) = \tilde{x} \in \blacksquare] = p_{\text{low}}$

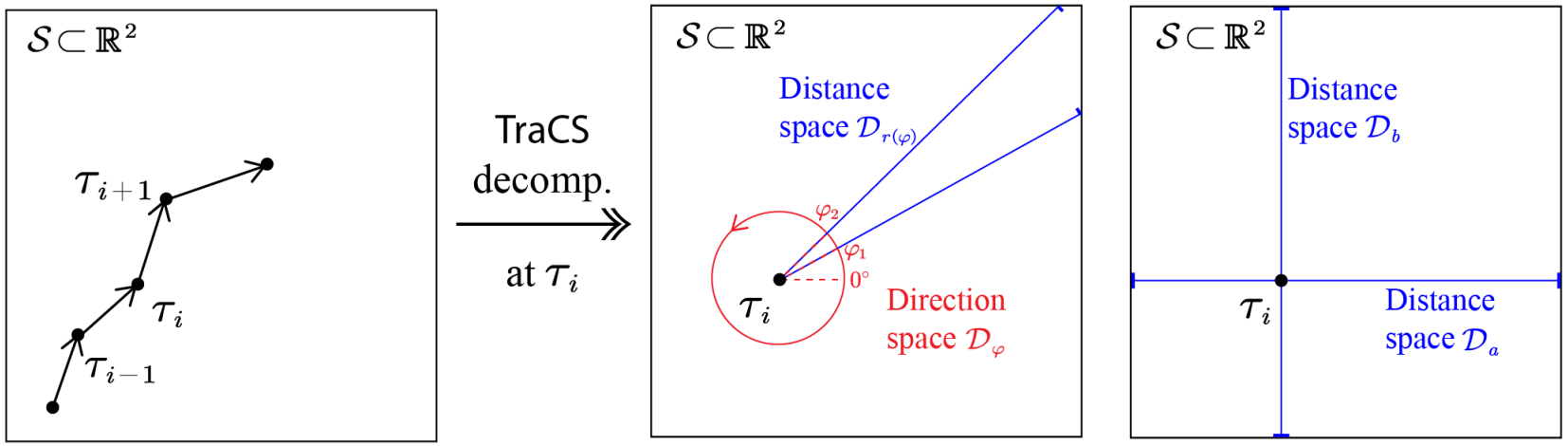
- Benefits:
  1. **Efficiency:** $\mathcal{O}(1)$ sampling complexity
  2. **Trajectory utility:** "n-independent"
  3. **Applicability:** Both continuous spaces & discrete spaces

Rounding-to-the-nearest (post-processing)

TraCS: Trajectory Collection in Continuous Spaces

- TraCS-D: direction-distance perturbation
- TraCS-C: coordinates (Cartesian) perturbation

- **Key idea:** decomposes $\mathcal{S}$ into two subspaces

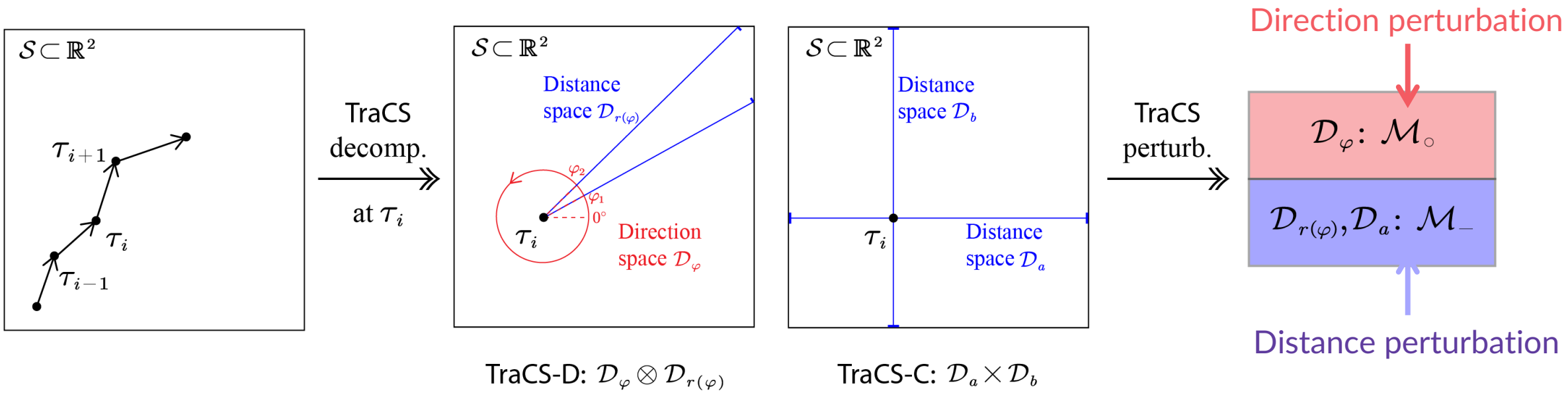


TraCS-D: $\mathcal{D}_\varphi \otimes \mathcal{D}_{r(\varphi)}$ TraCS-C: $\mathcal{D}_a \times \mathcal{D}_b$

TraCS: Trajectory Collection in Continuous Spaces

- TraCS-D: direction-distance perturbation
- TraCS-C: coordinates (Cartesian) perturbation

- **Key idea:** decomposes $\mathcal{S}$ into two subspaces → design $\mathcal{M}$ for each subspace



TraCS-D: $\mathcal{D}_\varphi \otimes \mathcal{D}_{r(\varphi)}$

TraCS-C: $\mathcal{D}_a \times \mathcal{D}_b$

LDP Mechanisms $\mathcal{M}_\circ$ and $\mathcal{M}__$

RIT

- Leverage piecewise-based mechanisms $\mathcal{M}_\circ$ and $\mathcal{M}__$

$[0,2\pi) \to [0,2\pi)$ $\quad$ $[0,1) \to [0,1)$

  - other LDP mechanisms for bounded numerical domain also applicable

LDP Mechanisms $\mathcal{M}_\circ$ and $\mathcal{M}_$

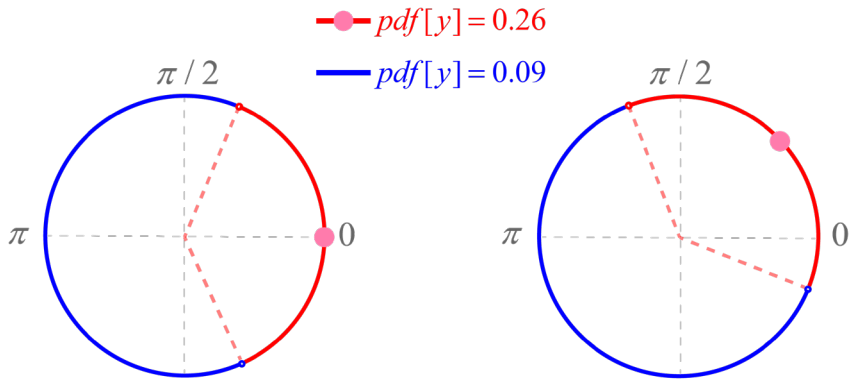
RIT

- Leverage piecewise-based mechanisms $\mathcal{M}_\circ$ and $\mathcal{M}_$

$[0,2\pi) \rightarrow [0,2\pi)$ $\quad$ $[0,1) \rightarrow [0,1)$

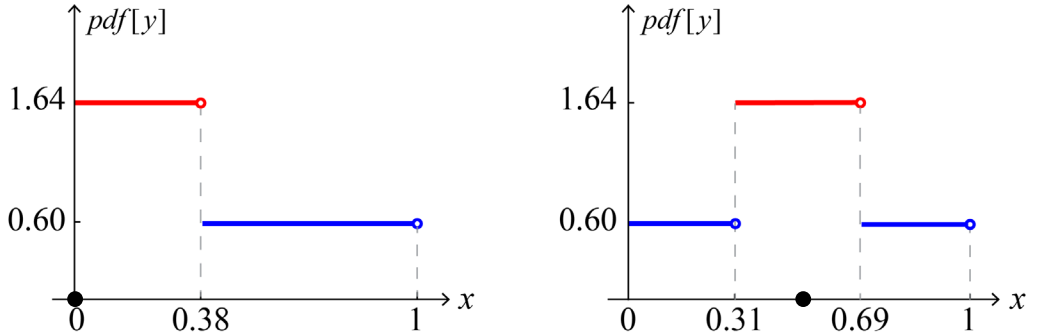
  - other LDP mechanisms for bounded numerical domain also applicable

- Examples of $\mathcal{M}_\circ$ and $\mathcal{M}_$*



(a) $x = 0$ (b) $x = \pi / 4$

$\mathcal{M}_\circ(x; \varepsilon = 1)$



(a) $x = 0$ (b) $x = 0.5$

$\mathcal{M}_(x; \varepsilon = 1)$

* Optimal Piecewise-based Mechanism for Collecting Bounded Numerical Data under Local Differential Privacy, PETS'25

Evaluations

RIT

- Claims
  - continuous spaces: better utility than naïve baselines, e.g. planar Laplace + truncation
  - discrete spaces: better efficiency and utility than existing methods, ATP*, NGram**, L-SRR***

* Trajectory Data Collection with Local Differential Privacy, VLDB'23

** Real-World Trajectory Sharing with Local Differential Privacy, VLDB'21

*** L-SRR: Local Differential Privacy for Location-Based Services with Staircase Randomized Response, CCS'22

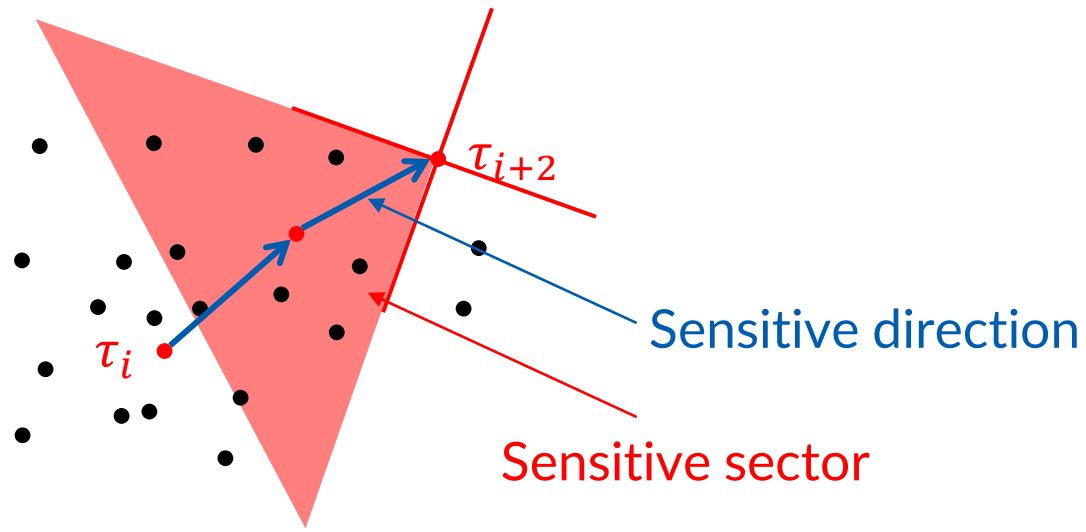
Evaluations

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- **ATP:** direction perturbation

  1. divide direction sectors, e.g. $k = 4$



* Trajectory Data Collection with Local Differential Privacy, VLDB'23

** Real-World Trajectory Sharing with Local Differential Privacy, VLDB'21

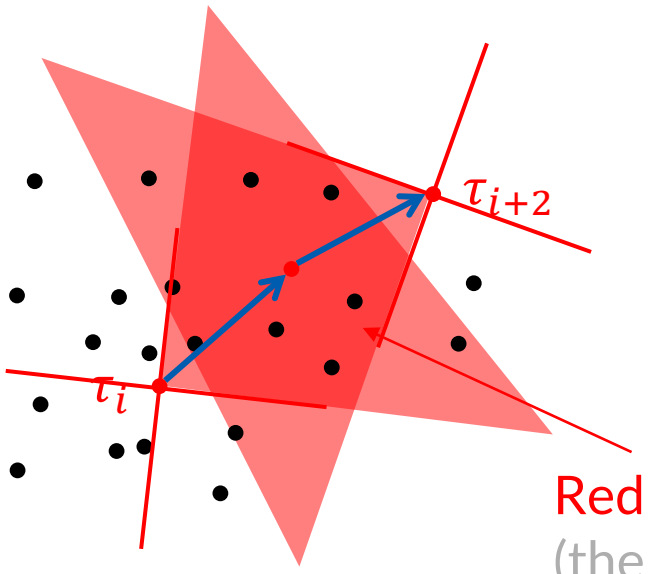
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Evaluations

RIT

- Claims
  - continuous spaces: better utility than naïve baselines, e.g. planar Laplace + truncation
  - discrete spaces: better efficiency and utility than existing methods, ATP*, NGram**, L-SRR***

- **ATP:** direction perturbation
  1. divide direction sectors, e.g. $k = 4$
  2. perturb sector
  3. $\mathcal{M}_{\text{exp}}(\tau)$ on reduced $\mathcal{S}$



Reduced location space $\mathcal{S}$
(then use EM)

* Trajectory Data Collection with Local Differential Privacy, VLDB'23

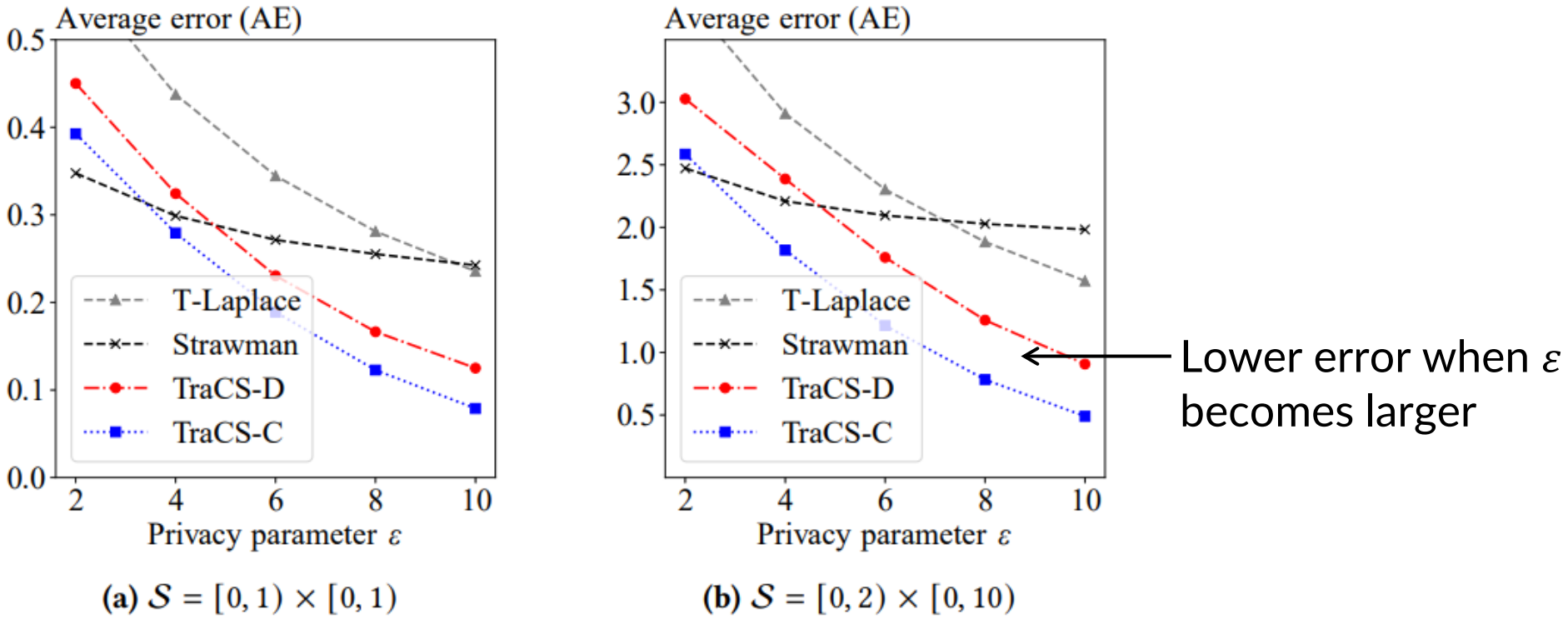
** Real-World Trajectory Sharing with Local Differential Privacy, VLDB'21

*** L-SRR: Local Differential Privacy for Location-Based Services with Staircase Randomized Response, CCS'22

Evaluations – Continuous Spaces

RIT

- Trajectory utility metric: Average error (AE)
  - T-Laplace: Truncated Laplace; Strawman: k-RR + uniform sampling for direction perturbation (similar to SRR)

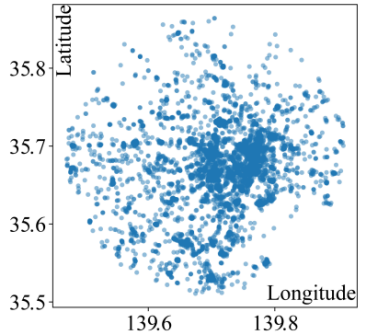


(a) $\mathcal{S} = [0, 1) \times [0, 1)$ **(b)** $\mathcal{S} = [0, 2) \times [0, 10)$

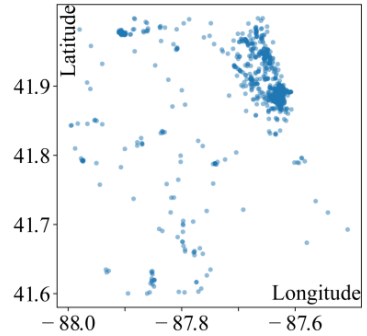
Evaluations – Discrete Spaces

RIT

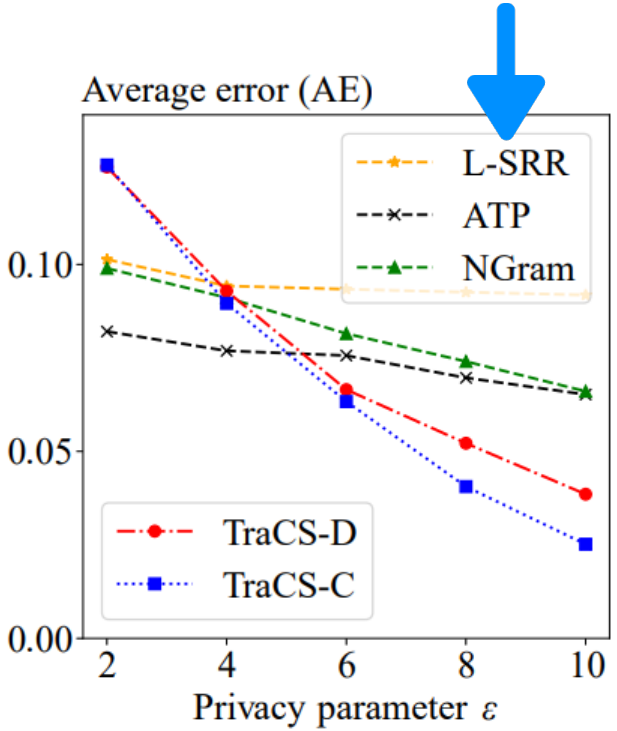
- TKY and CHI datasets
  - TraCS for the rectangular area + rounding-to-the-nearest



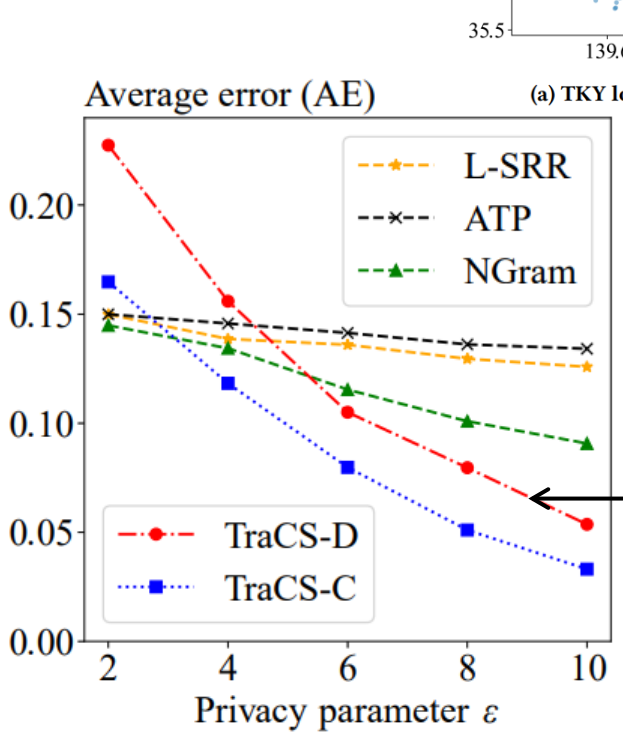
(a) TKY location space



(b) CHI location space



(a) TKY dataset



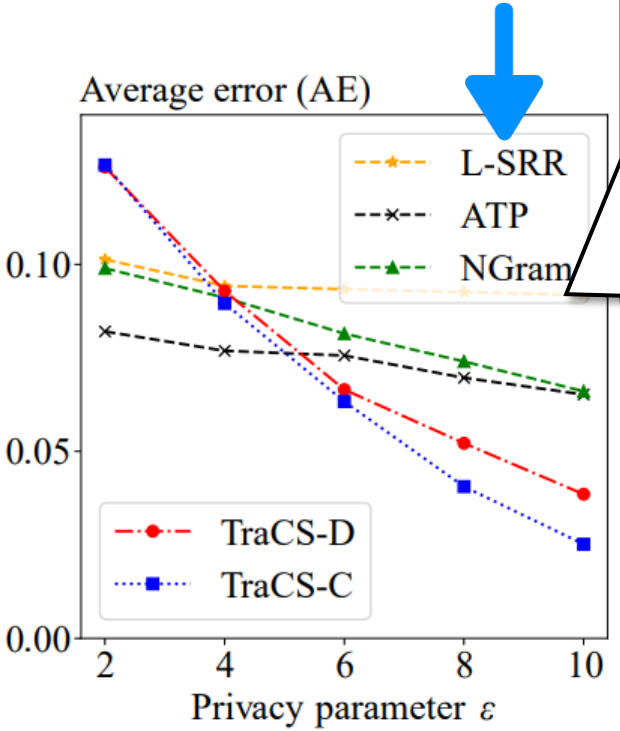
(b) CHI dataset

Lower error when ε becomes larger

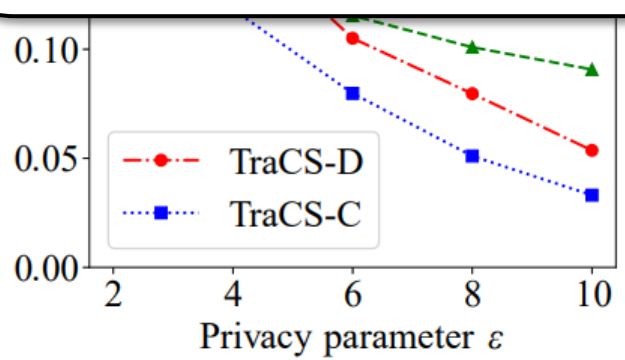
Evaluations – Discrete Spaces

RIT

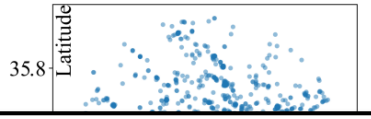
- TKY and CHI datasets
  - TraCS for the rectangular area + roundin



(a) TKY dataset



(b) CHI dataset



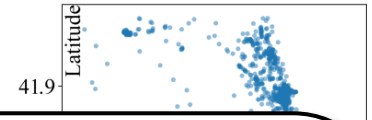


Table 3: Time cost comparison (in milliseconds).
averaged

	ATP	NGram	L-SRR	TraCS-D	TraCS-C
Total	145.7	100.9	6.2	0.06	0.05
Perturb	125.8	92.8	0.003	0.018	0.003

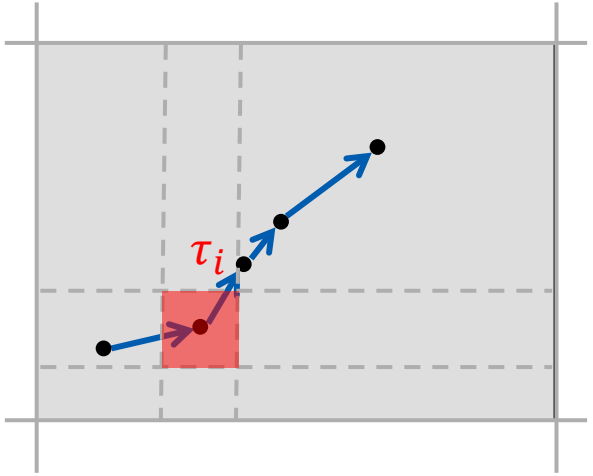
Negligible time cost

Summary & Takeaway

RIT

- **RQ: Trajectory collection under LDP**

- Our results:*
  - **operating in continuous spaces can do better**
  - better efficiency: $\mathcal{O}(1)$ sampling complexity
  - better trajectory utility, especially for larger ε
  - better applicability, for both continuous and discrete $\mathcal{S}$



$$\Pr[\mathcal{M}(x_i) = \tilde{x} \in \blacksquare] = p_{\text{high}}$$

$$\Pr[\mathcal{M}(x_i) = \tilde{x} \in \blacksquare] = p_{\text{low}}$$

* TraCS: Trajectory Collection in Continuous Space under Local Differential Privacy, PETS'26

Improving and Quantifying Utility under Local Differential Privacy

Mechanism design
- Part 1: Correlated $\mathcal{M}$
- Part 2: Optimal piecewise-based $\mathcal{M}$
- Part 3: $\mathcal{M}$ for trajectories in continuous space

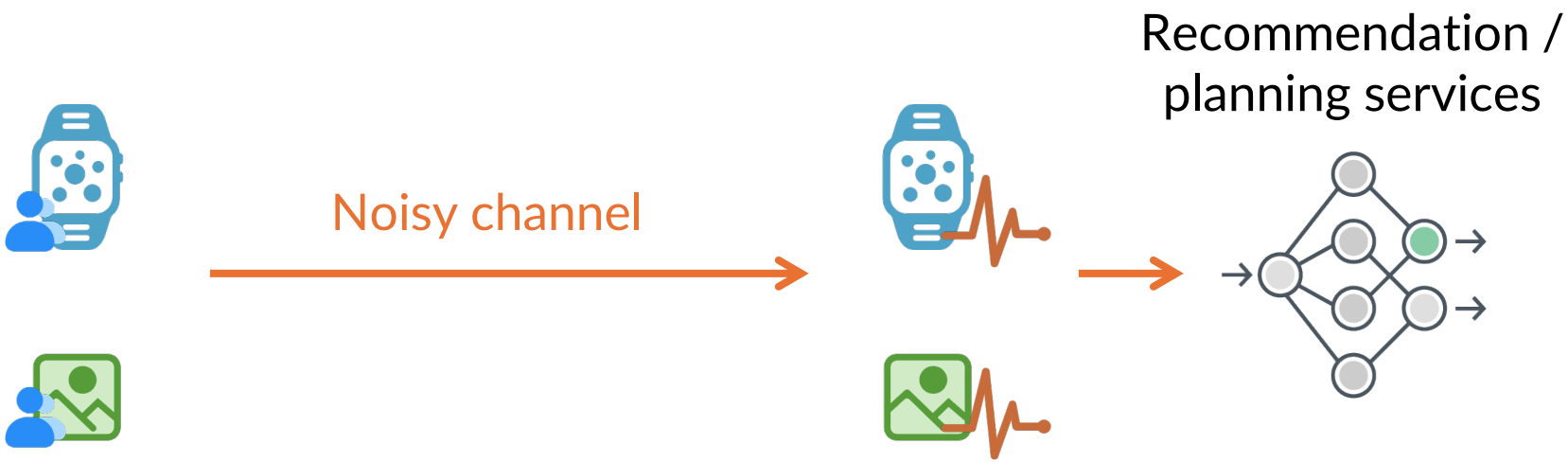
Utility analysis
- Part 4: Utility analysis for classifier $\circ$ $\mathcal{M}$

Mechanism-level utility
↓
Task-level utility

Classifier Utility under Noisy Inputs

RIT

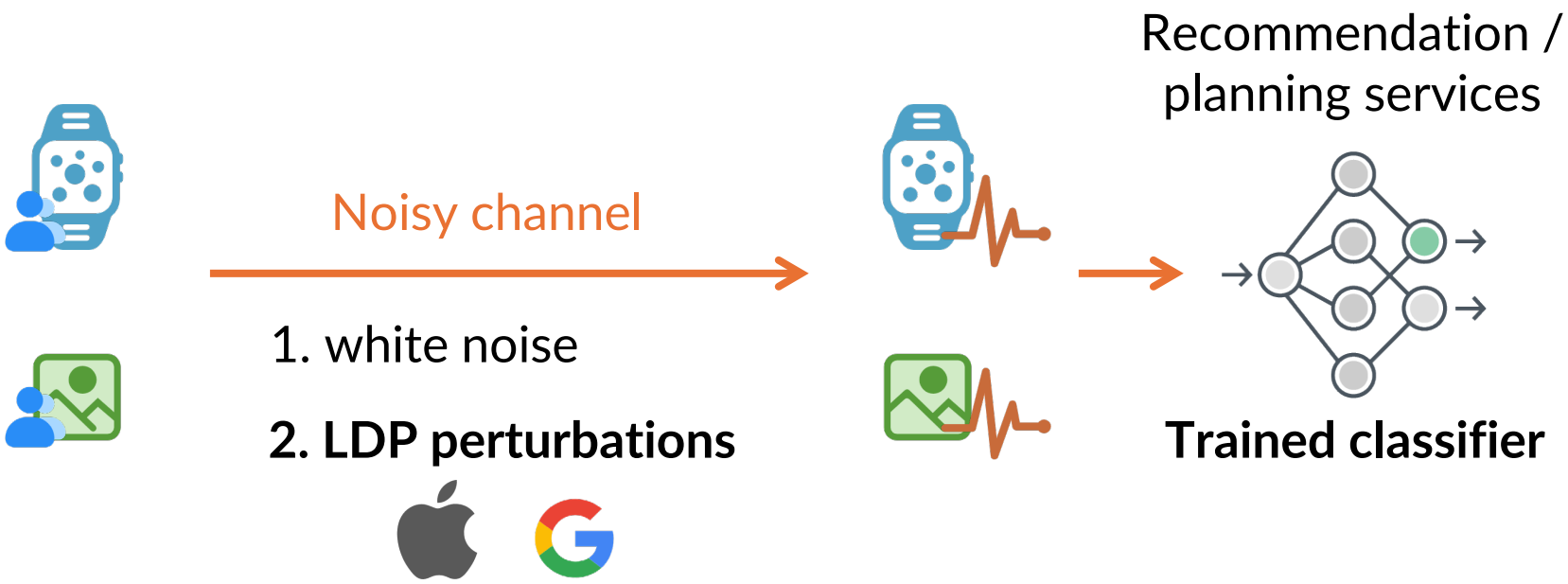
- Motivation: Input data for classifiers may be noisy



Classifier Utility under Noisy Inputs

RIT

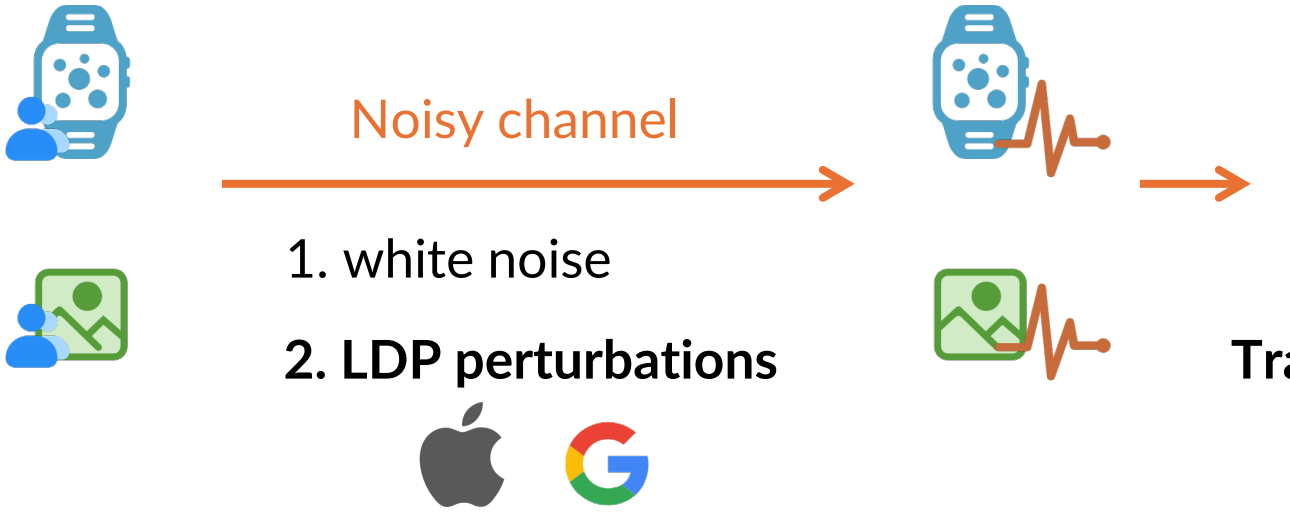
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Classifier Utility under Noisy Inputs

RIT

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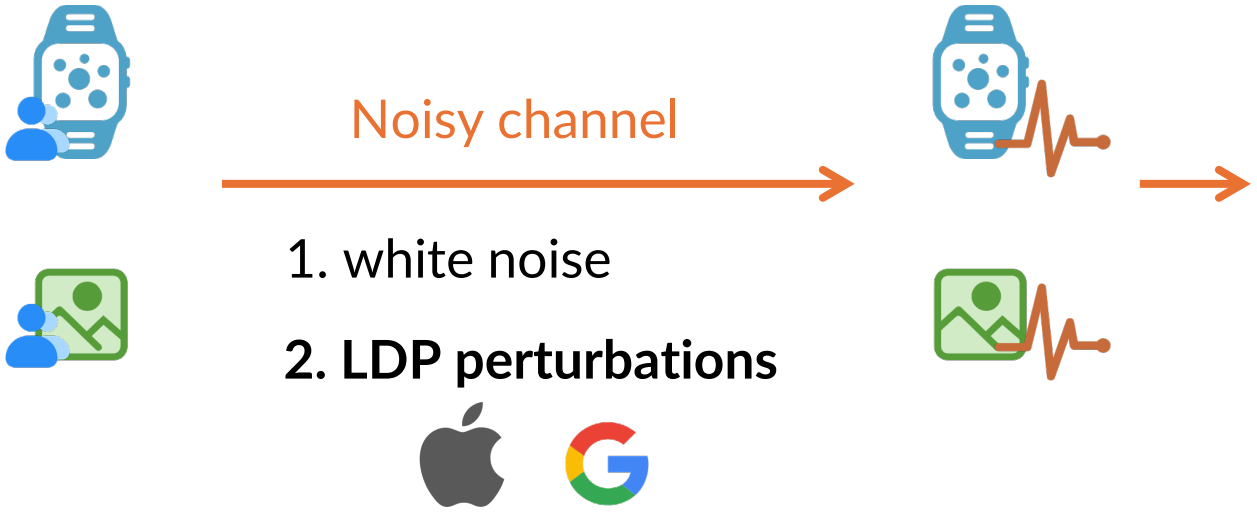


What's the utility of the trained classifier under a noisy channel, e.g. LDP perturbation?

Classifier Utility under Noisy Inputs

RIT

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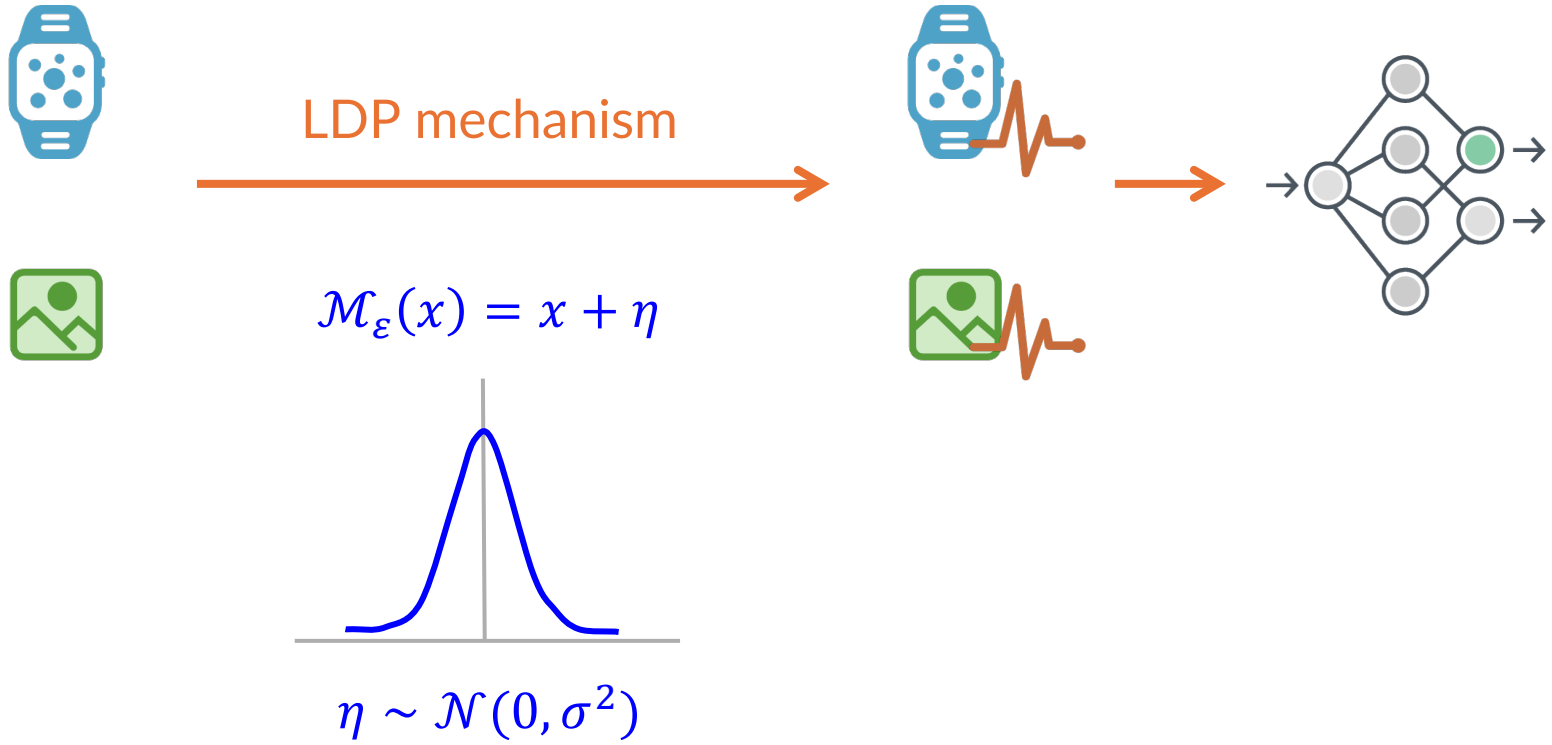


What's the utility of the trained classifier under a noisy channel, e.g. LDP perturbation?

- Q: How can classifier designers/users know the classifier's accuracy under LDP-perturbed data?
  - For an LDP-friendly classifier
  - For a better privacy-utility balance

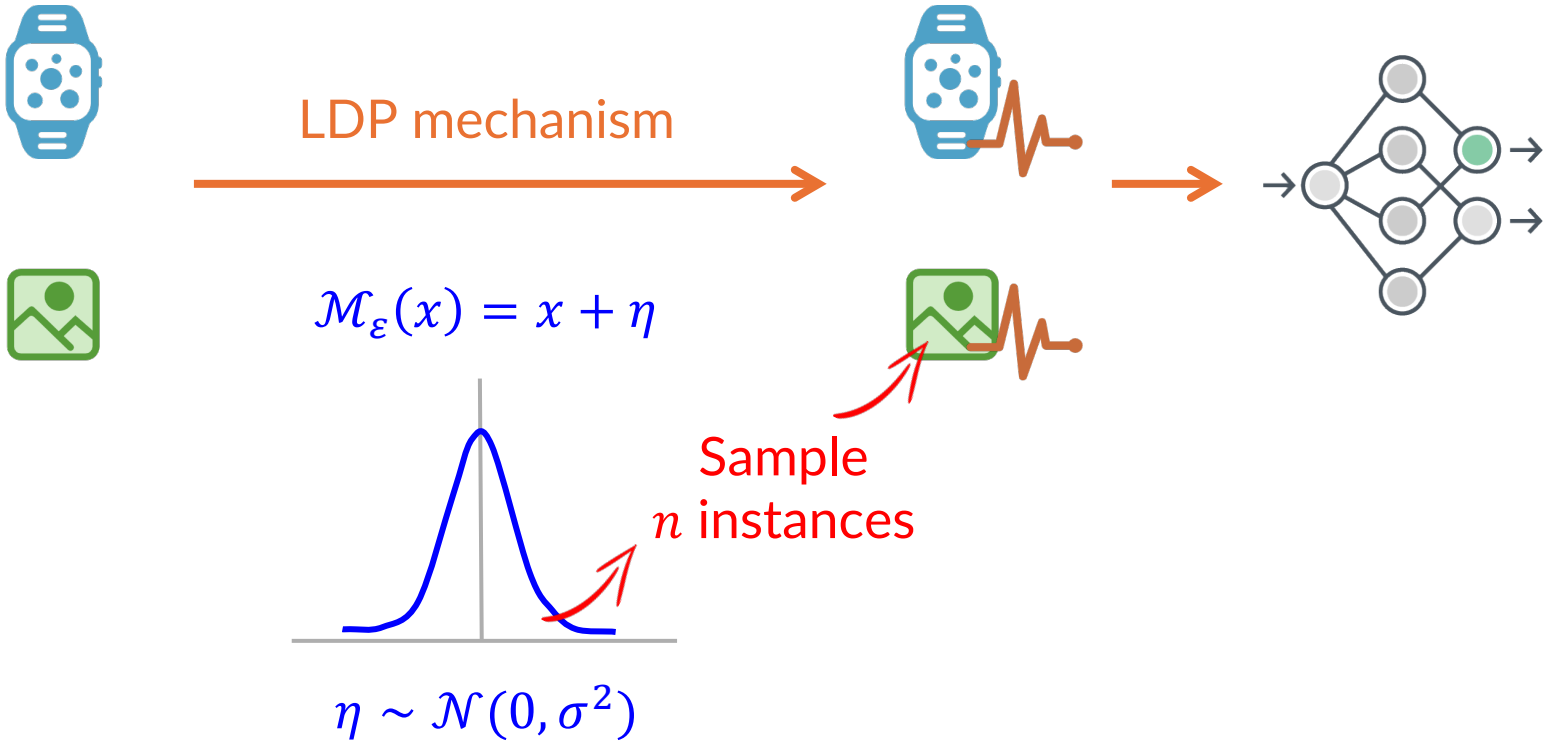
Empirical Classifier Utility under LDP-Data

- Empirical approach: Sample and then test
  - $\mathcal{M}$'s variance or MSE **doesn't help** – cannot provide a classifier accuracy



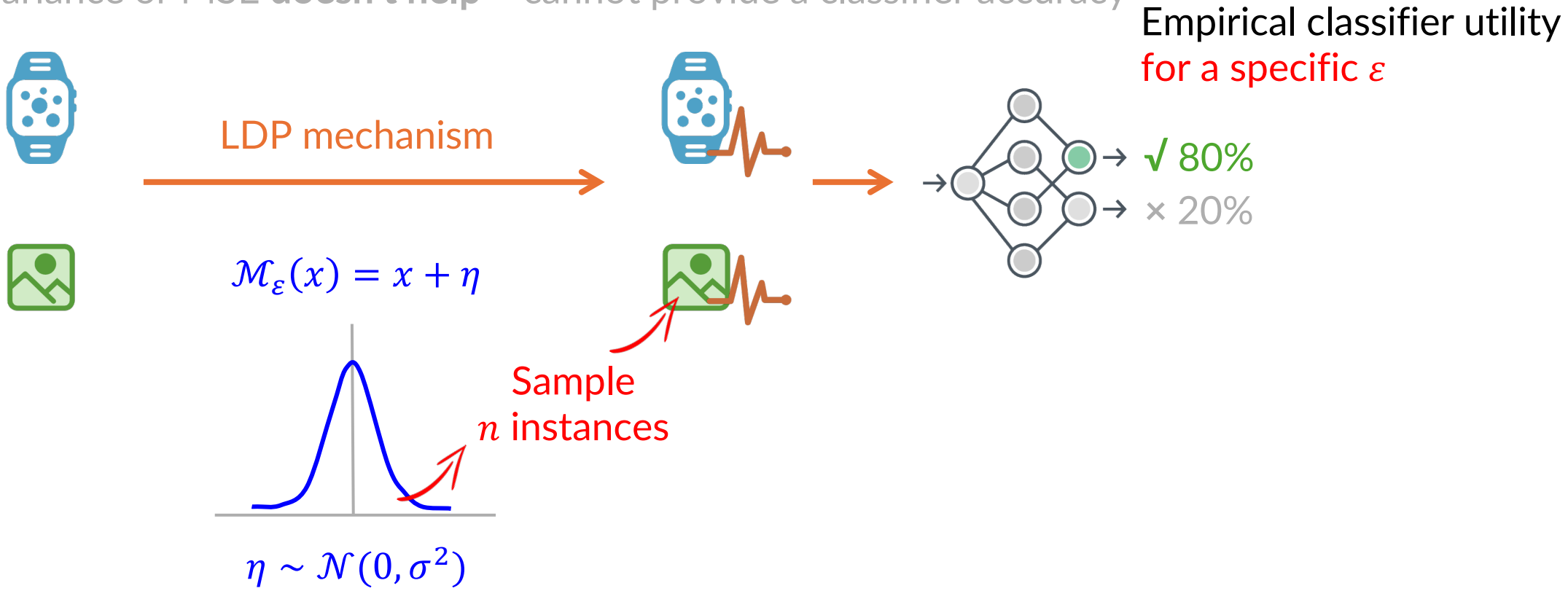
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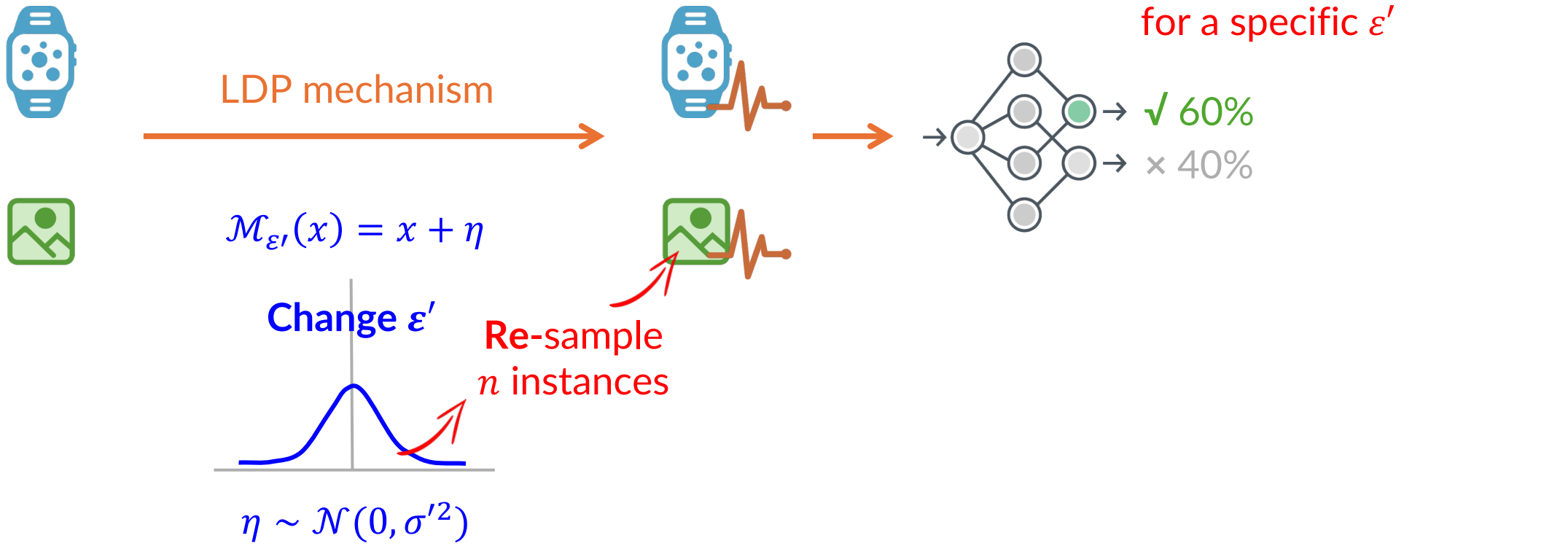
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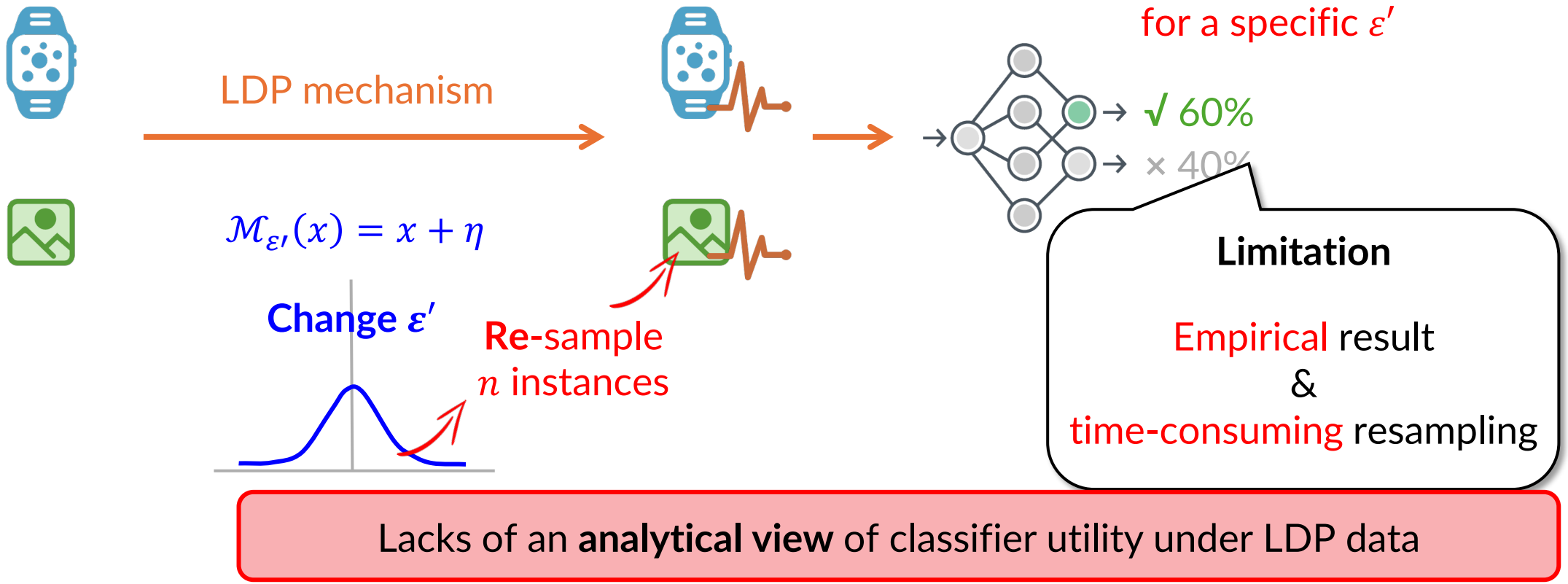
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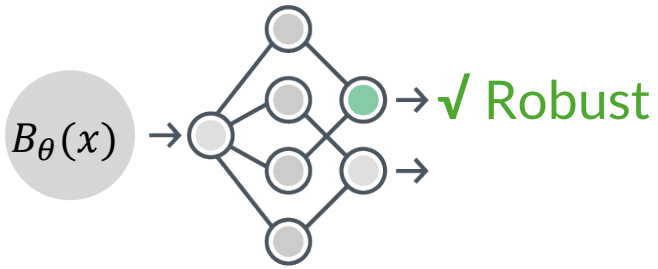
Lacks of an **analytical view** of classifier utility under LDP data

Empirical Utility → Analytical Utility

RIT

- Analytical approach: connecting LDP with robustness

$B_\theta(x)$ is robust



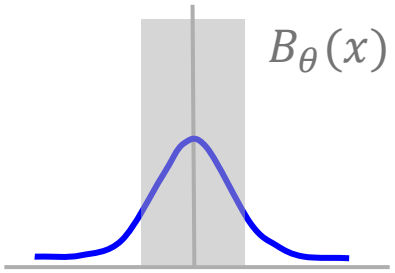
Robustness:
A θ-ball around x that doesn't change classification

Empirical Utility → Analytical Utility

RIT

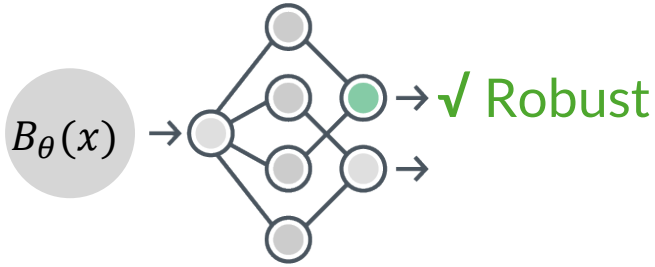
- Analytical approach: connecting LDP with robustness

$\mathcal{M}_\varepsilon(x)$ concentrates in $B_\theta(x)$



Concentration:
Probability of falling into a region
$\Pr[\mathcal{M}_\varepsilon(x) \in B_\theta(x)] \coloneqq p(\varepsilon, \theta)$

$B_\theta(x)$ is robust



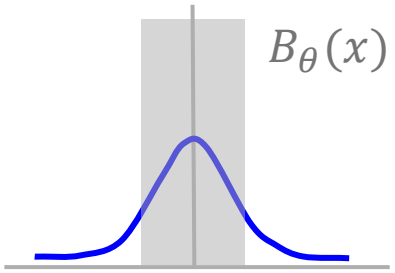
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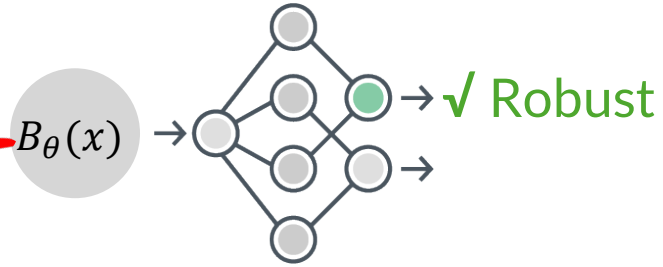
RIT

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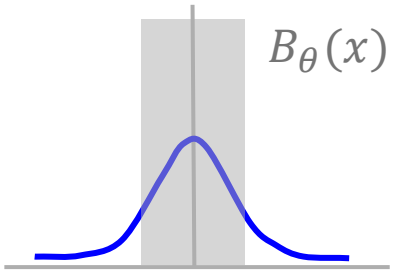
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Empirical Utility → Analytical Utility

RIT

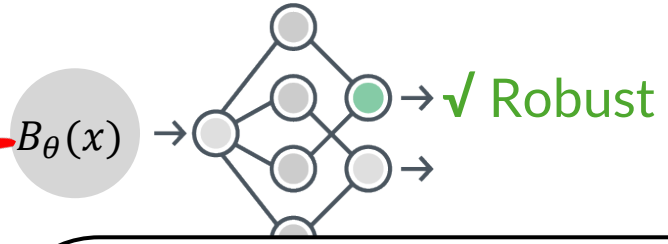
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$B_\theta(x)$ is robust



Analytical classifier utility

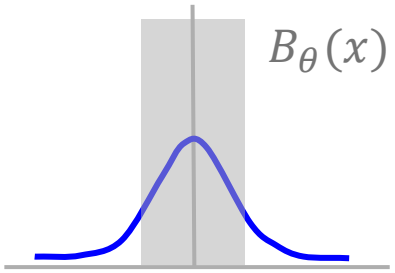
With probability at least $p(\varepsilon, \theta)$, the classifier preserves its correct classification under input $\mathcal{M}_\varepsilon(x)$.

Empirical Utility → Analytical Utility

RIT

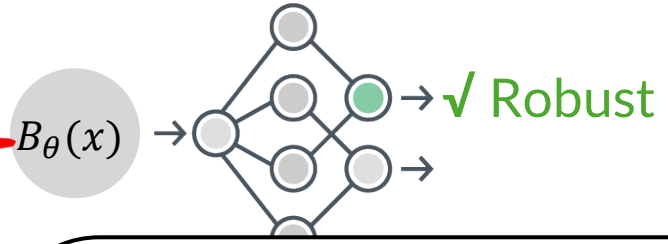
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Concentration:
Probability of falling into a region
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Analytical classifier utility

With probability at least $p(\varepsilon, \theta)$, the classifier preserves its correct classification under input $\mathcal{M}_\varepsilon(x)$.

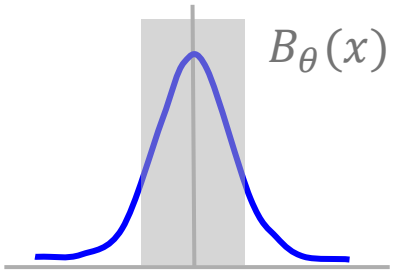
- Counterpart to the empirical "80%" accuracy

Empirical Utility → Analytical Utility

RIT

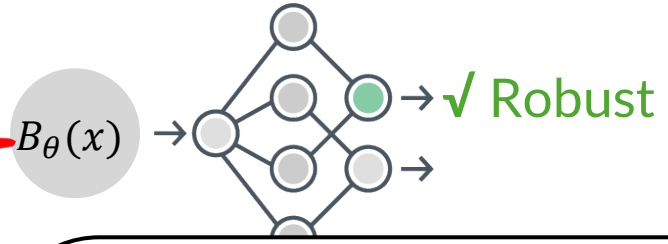
- Analytical approach: connecting LDP with robustness

$\mathcal{M}_{\varepsilon\prime}(x)$ concentrates in $B_\theta(x)$



Concentration:
Probability of falling into a region
$\Pr[\mathcal{M}_{\varepsilon\prime}(x) \in B_\theta(x)] \coloneqq p(\varepsilon', \theta)$

$B_\theta(x)$ is robust



Analytical classifier utility

With probability at least $p(\varepsilon', \theta)$, the classifier preserves its correct classification under input $\mathcal{M}_{\varepsilon\prime}(x)$.

Analytical & systematic view for any ε without resampling

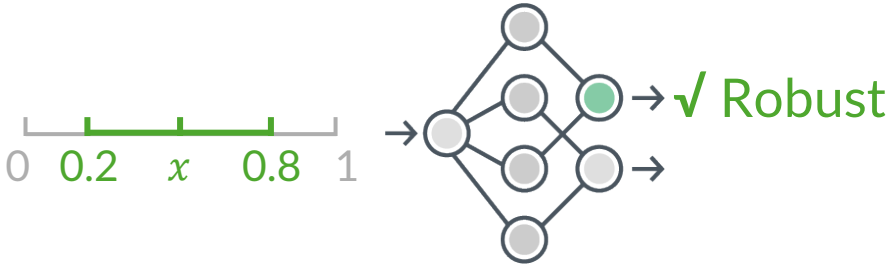
One-Dimension Example

RIT

- Classifier h: $[0,1] \to \{1, 2\}$ under Laplace mechanism $\mathcal{M}_{\text{Lap}}(x)$

$B_{0.3}(x)$ is robust

Classifier:
(robustness analysis)



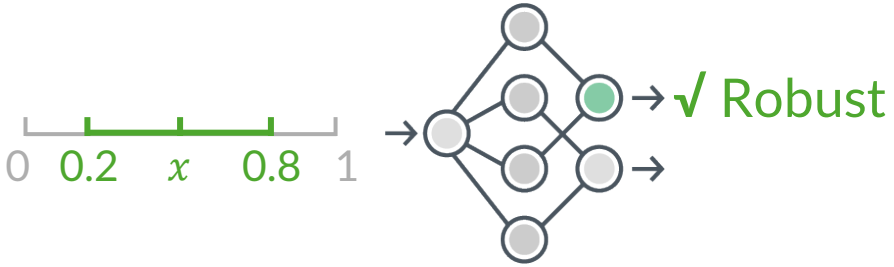
One-Dimension Example

RIT

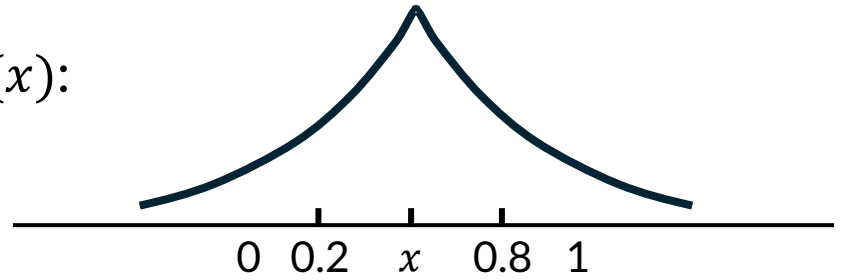
- Classifier h: $[0,1] \to \{1, 2\}$ under Laplace mechanism $\mathcal{M}_{\text{Lap}}(x)$

$B_{0.3}(x)$ is robust

Classifier:
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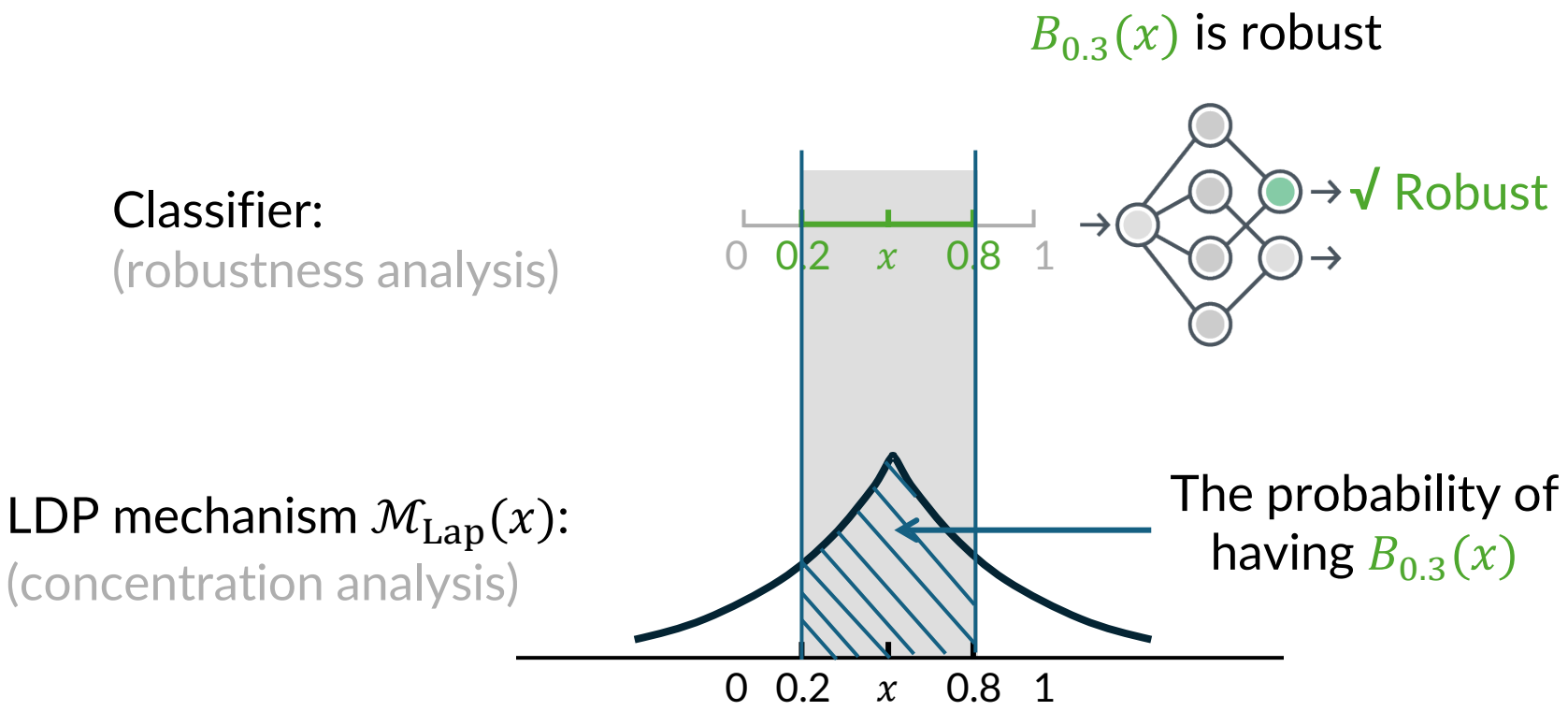
LDP mechanism $\mathcal{M}_{\text{Lap}}(x)$:



One-Dimension Example

RIT

- Classifier h: $[0,1] \to \{1, 2\}$ under Laplace mechanism $\mathcal{M}_{\text{Lap}}(x)$



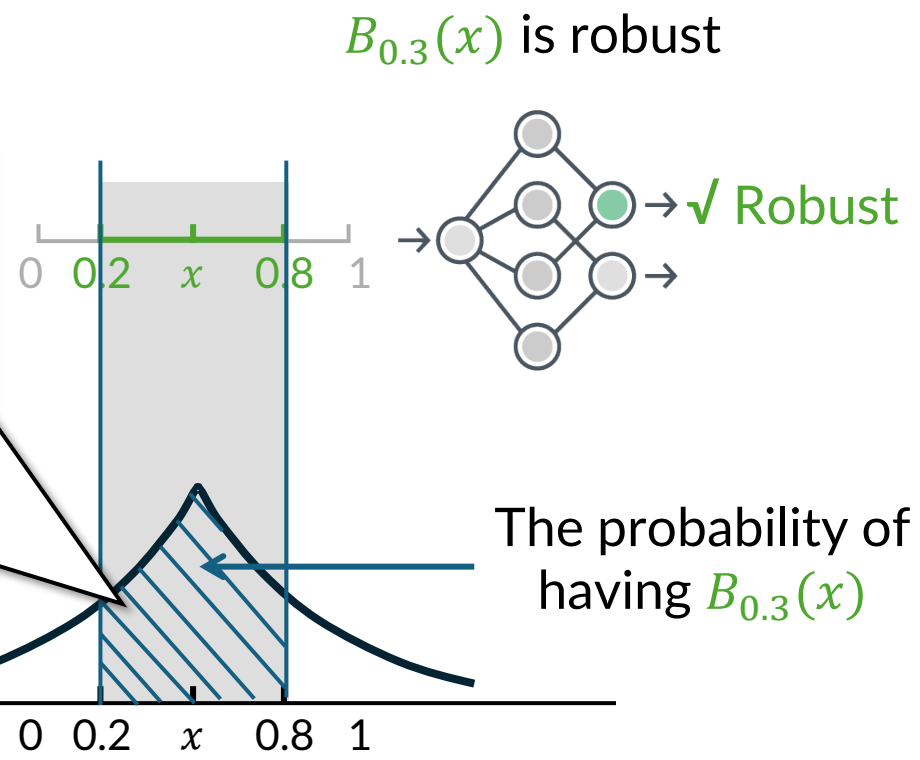
One-Dimension Example

RIT

- Classifier h: $[0,1] \to \{1,2\}$ under Laplace mechanism $\mathcal{M}_{\text{Lap}}(x)$

Concentration analysis

$$p(\varepsilon, \theta) = \text{cdf}_{\text{Lap}}[\theta] - \text{cdf}_{\text{Lap}}[-\theta]$$
$$= 1 - e^{-\theta\varepsilon}$$



One-Dimension Example

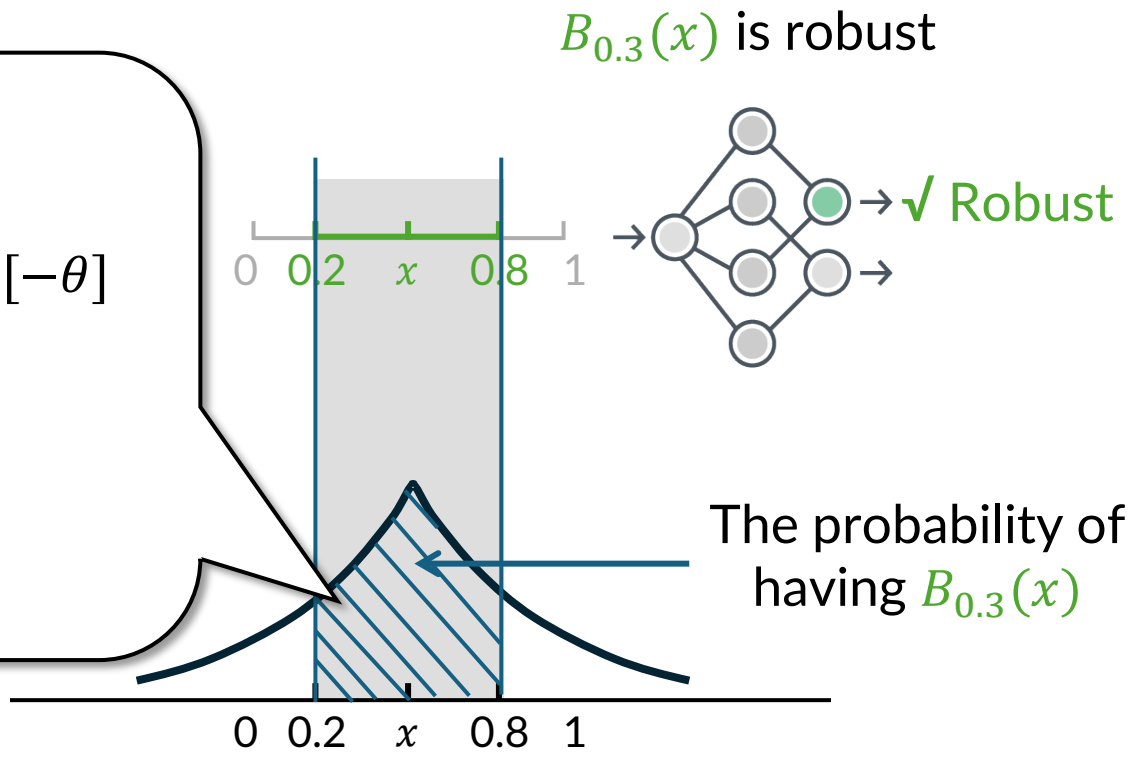
RIT

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$$p(\varepsilon = 2, \theta = 0.3) = 0.46$$



One-Dimension Example

RIT

- Classifier h: $[0,1] \to \{1,2\}$ under Laplace mechanism $\mathcal{M}_{\text{Lap}}(x)$

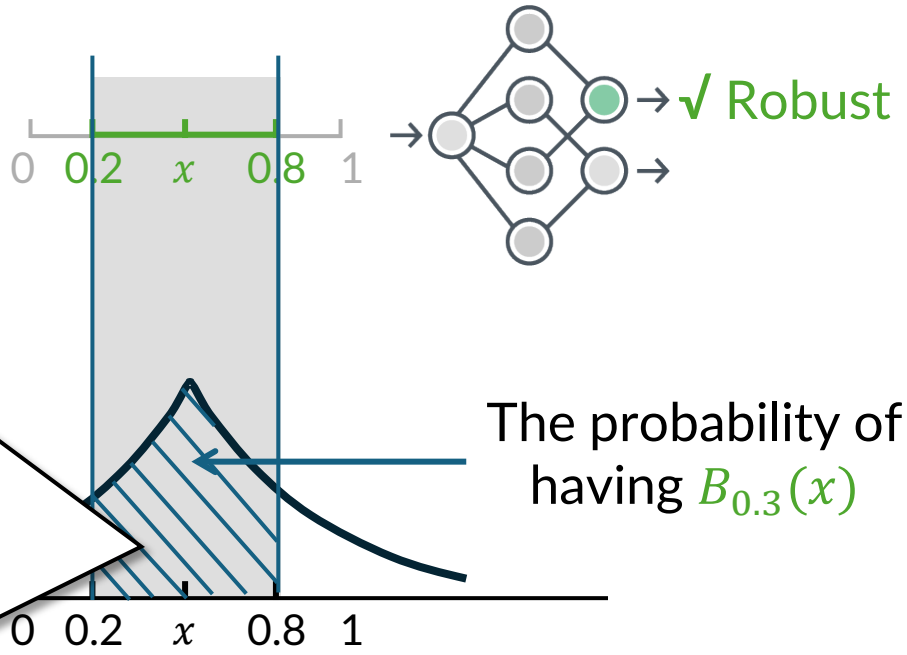
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$$p(\varepsilon = 2, \theta = 0.3) = 0.46$$

With probability at least $p(2,0.3) = 0.46$, the classifier preserves its correct classification under input $\mathcal{M}_{\varepsilon=2}(0.5)$.

$B_{0.3}(x)$ is robust



One-Dimension Example

RIT

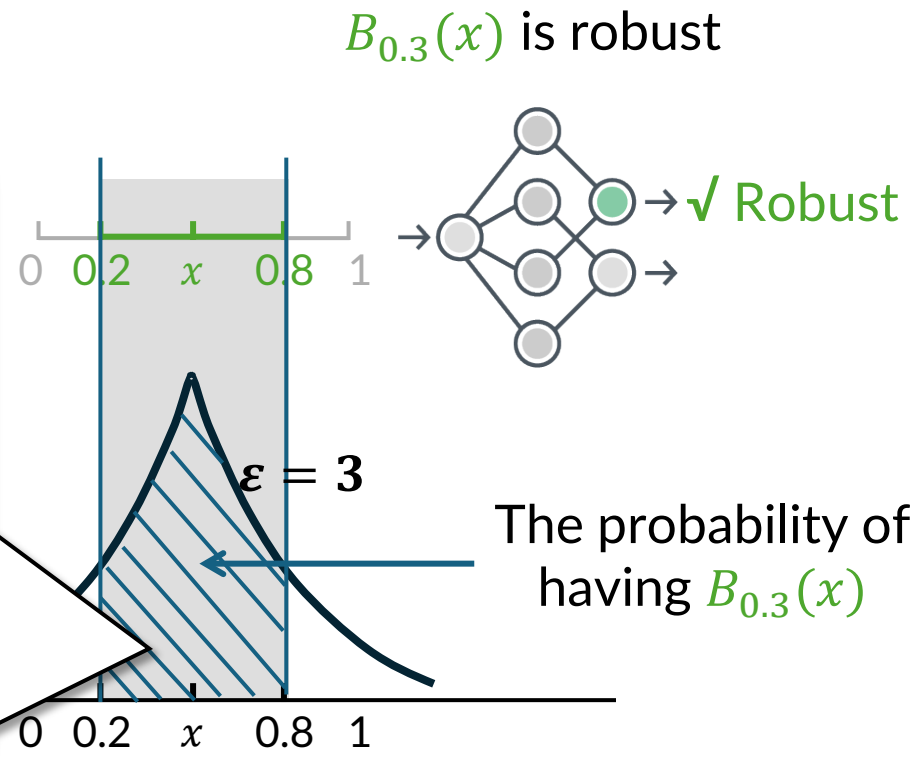
- Classifier h: $[0,1] \to \{1, 2\}$ under Laplace mechanism $\mathcal{M}_{\text{Lap}}(x)$

Concentration analysis

$$p(\varepsilon, \theta) = \text{cdf}_{\text{Lap}}[\theta] - \text{cdf}_{\text{Lap}}[-\theta]$$
$$= 1 - e^{-\theta\varepsilon}$$

For any ε:

With probability at least $p(\varepsilon, \theta)$, the classifier preserves its correct classification under input $\mathcal{M}_{\varepsilon}(x)$.

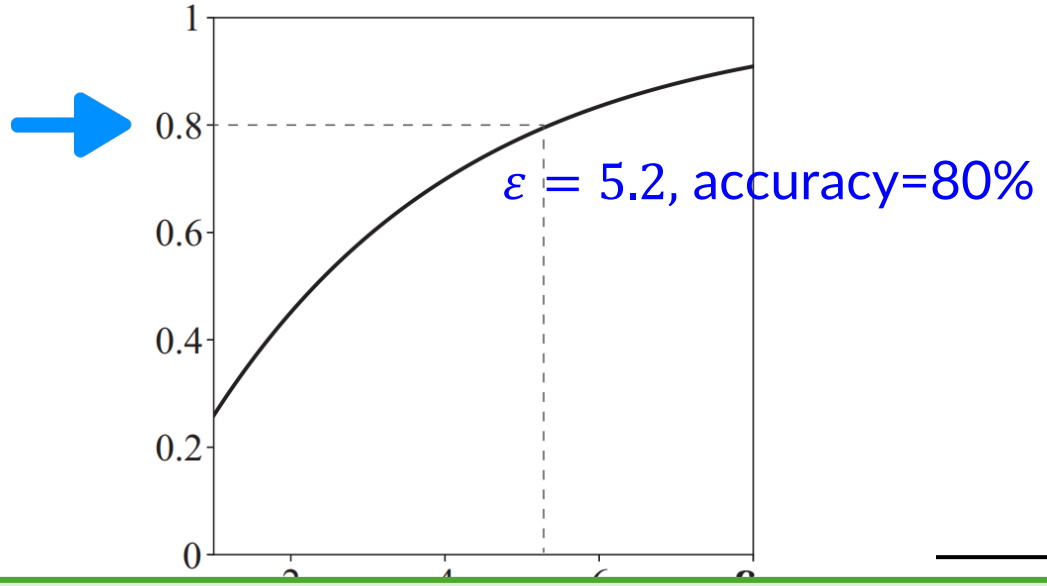


Fixed Classifier (θ)

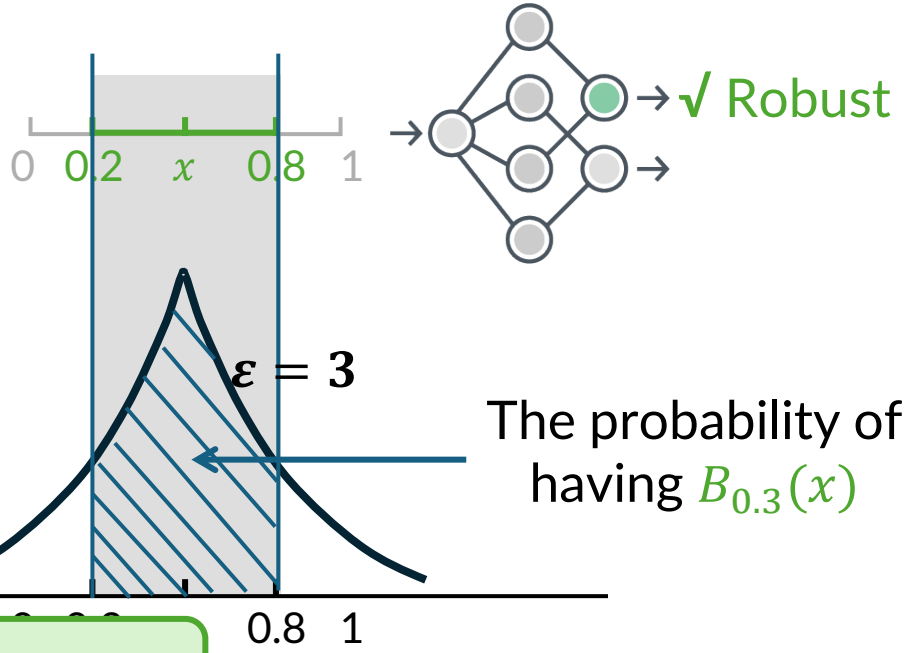
RIT

- Classifier h: $[0,1] \rightarrow \{1, 2\}$ under Laplace mechanism $\mathcal{M}_{\text{Lap}}(x)$

This classifier's utility $p(\varepsilon, 0.3)$



$B_{0.3}(x)$ is robust

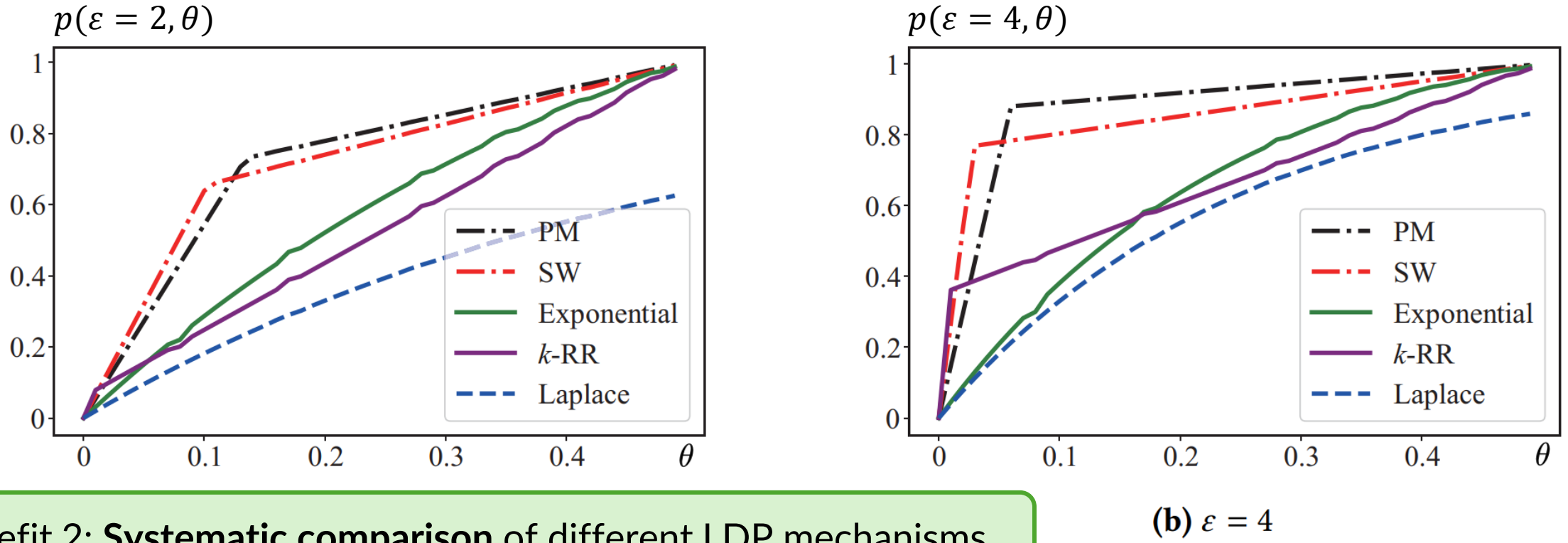


Benefit 1: Choose the **best ε** for a desired classifier utility

Different $\mathcal{M}$ & Different Classifiers

RIT

- No universally optimal LDP mechanism for all ε and θ



(b) $\varepsilon = 4$

Benefit 2: **Systematic comparison** of different LDP mechanisms

High-Dimensional Cases

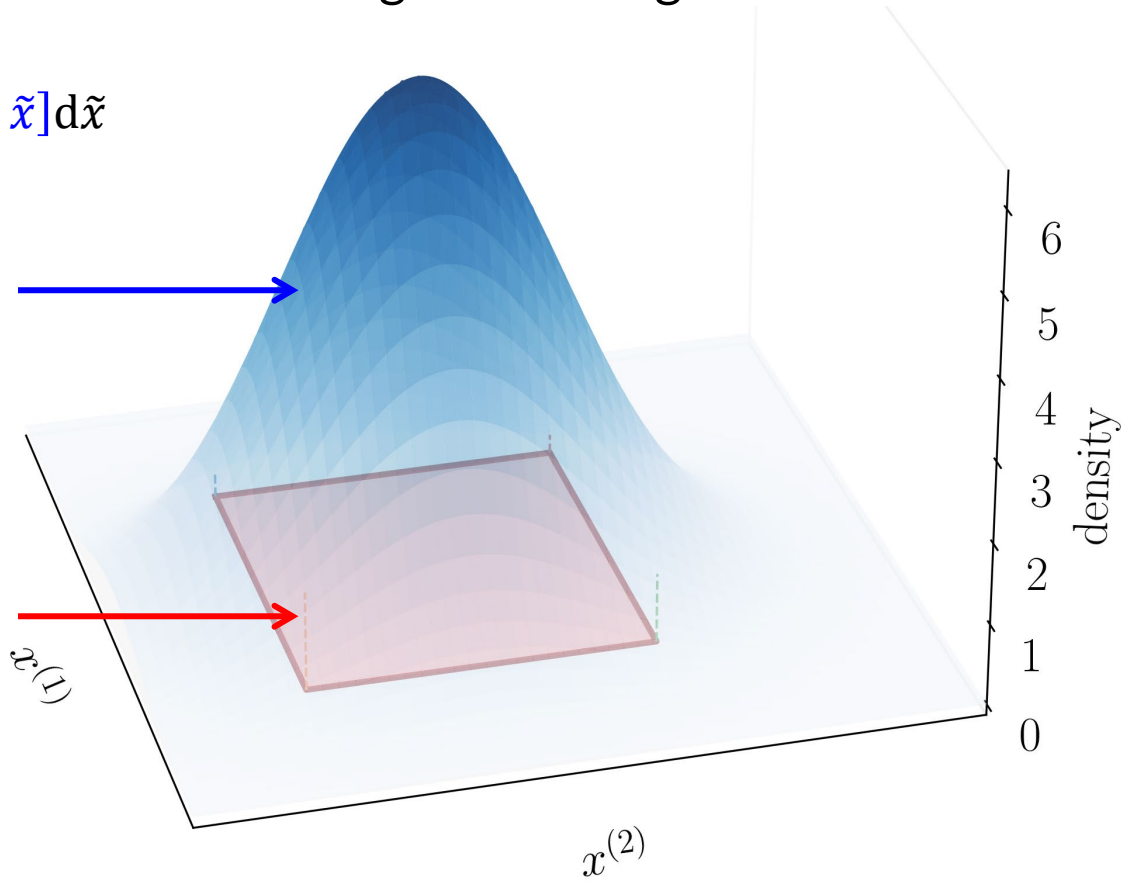
RIT

- Concentration analysis on the robustness region S: Integration over S

$$p(\varepsilon, S) = \int_S pdf[\mathcal{M}_\varepsilon(x) = \tilde{x}]\mathrm{d}\tilde{x}$$

LDP mechanism $\mathcal{M}_\varepsilon(x)$

Robustness region S



- **How robust a system is against LDP perturbation?**

Case Studies – Quality of The Analytical Utility

- Claim: Accurate (compared with the empirical utility) & efficient
- Classifiers:
  - low-dimensional: Logistic Regression, Random Forest
  - high-dimensional: Neural Network
- Datasets: Stroke Prediction (medical data), Bank Customer Attrition (financial data), MNIST-7×7 (image data)

2 sensitive dims

49 sensitive dims

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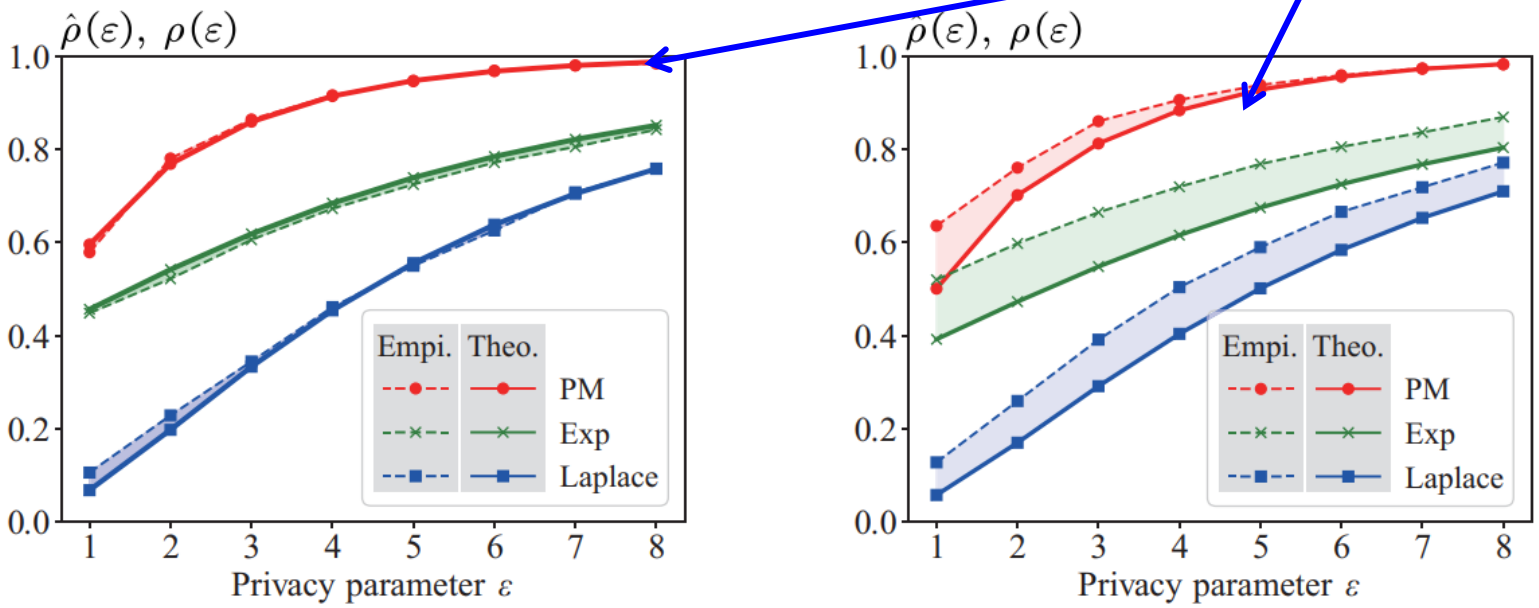
- Classifier utility:
  - analytical utility $p(\varepsilon, S)$: approximated S for black-box classifiers
  - empirical utility $\hat{p}(\varepsilon)$: 2000 samples from $\mathcal{M}_\varepsilon$ and then used for testing

Case Studies – Stoke Predication

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- $x =$ the first record, with noisy "Age" and "BMI"

Accurate analytical utility



(a) Logistic Regression. **(b) Random Forest.**

Case Studies – Stoke Predication

RIT

- $x =$ the first record, with noisy "Age" and "BMI"

Time cost comparison (ms)

	PM	Exponential	Laplace
Empirical[a]	$6.56 + 1.38$	$859.83 + 1.53$	$11.29 + 1.53$
Theoretical[b]	0.24	0.94	0.30

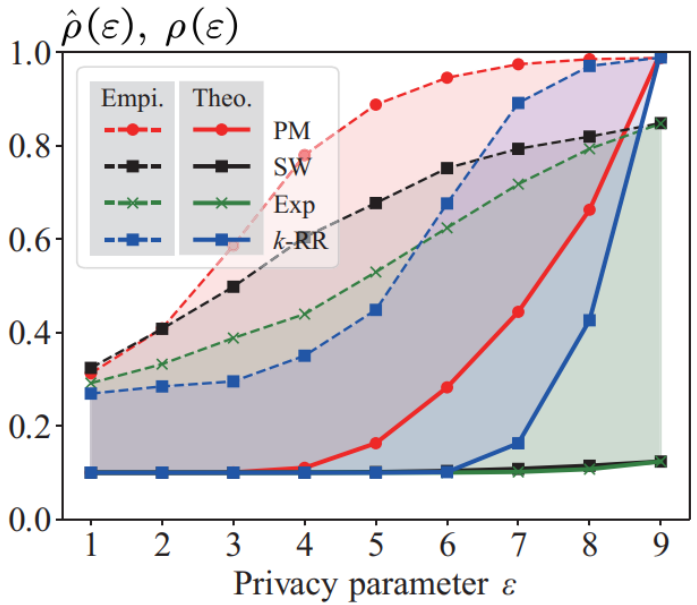
←Negligible time cost

[a] Time of 2000 samples + inference.

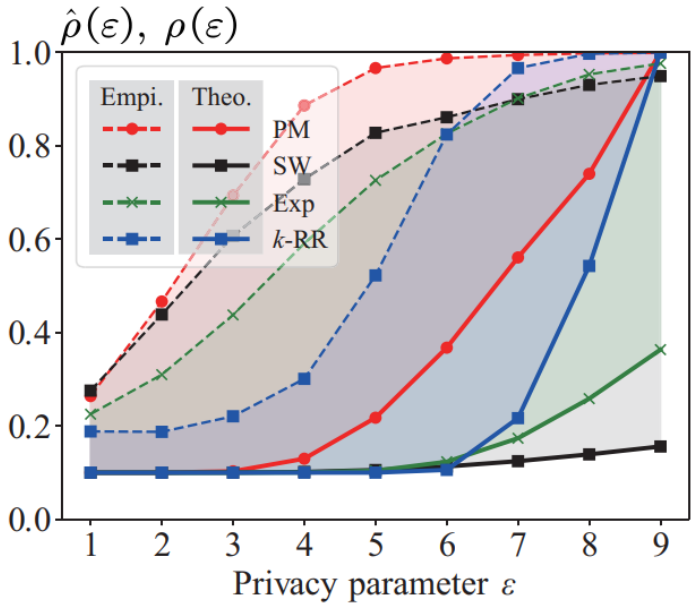
[b] Time to compute $\rho(\varepsilon, S)$ only; computing S takes 5.80 ms but is a one-time cost amortized across all ε values.

Case Studies – Neural Networks

- Conservative analytical utility for high-dim classifiers (49-dim)
  - but high analytical utility → high empirical utility



1st image



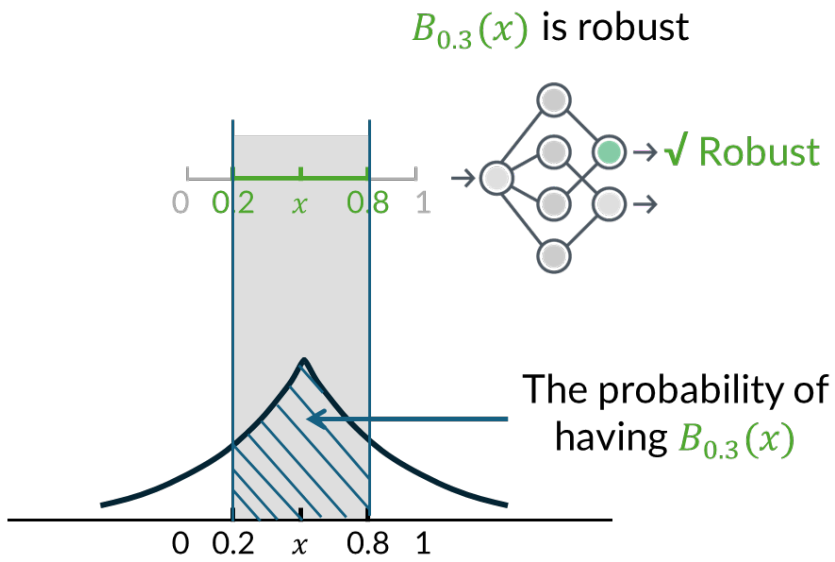
2nd image

Summary & Takeaway

RIT

- **RQ: Classifier utility under LDP-perturbed inputs**

- Our approach:*
  - **connects LDP with robustness**
  - **provides analytical classifier utility**
  - the analytical utility is accurate for low-dim classifiers



* Quantifying Classifier Utility under Local Differential Privacy, PETS'26
- ***Andreas Pfitzmann Best Student Paper Award***

Improving and Quantifying Utility under LDP

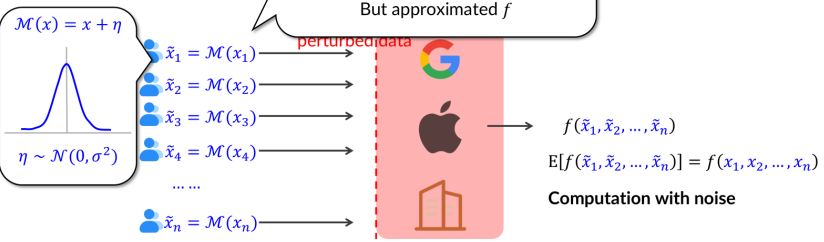
Privacy-Preserving Computation - **LDP**

RIT

- Local differential privacy (L
  - hard to differentiate the
  - each user locally perturb

Advantages:
Negligible computational complexity
No communication between users

But approximated f



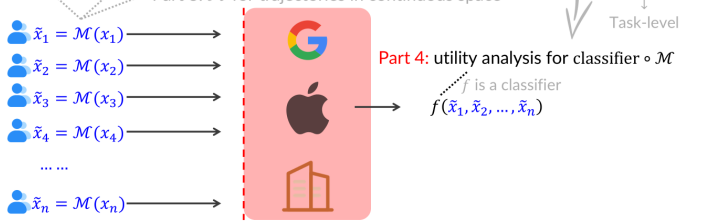
$\mathrm{E}[f(\tilde{x}_1, \tilde{x}_2, \dots, \tilde{x}_n)] = f(x_1, x_2, \dots, x_n)$

Computation with noise

This Proposal: LDP Theory

RIT

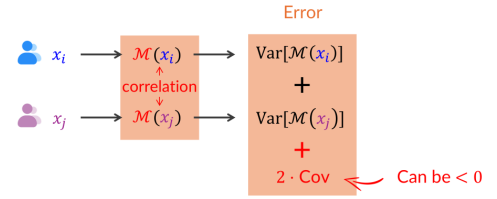
- Advancing LDP's mechanism design and utility analysis
  - Part 1: correlated $\mathcal{M}$ (binary x → numerical x)
  - Part 2: optimal piecewise-based $\mathcal{M}$ (1D x → 2D x)
  - Part 3: $\mathcal{M}$ for trajectories in continuous space
  - Part 4: utility analysis for classifier $\circ\, \mathcal{M}$ (Mechanism-level → Task-level)



This Paper: Joint/Correlated RR (JRR)

RIT

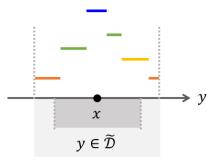
- Existing LDP mechanisms: Each user perturbs their data independently
- Correlated LDP mechanisms: Users' data are perturbed by correlated $\mathcal{M}$



Optimal Piecewise-based Mechanism

RIT

- **Most generalized version:** m-piecewise distribution



$$pdf[\mathcal{M}(x) = y] = \begin{cases} p_{1,\varepsilon} & \text{if } y \in [l_{1,x,\varepsilon}, r_{1,x,\varepsilon}] \\ p_{2,\varepsilon} & \text{if } y \in [l_{2,x,\varepsilon}, r_{2,x,\varepsilon}] \\ \dots & \\ p_{m,\varepsilon} & \text{if } y \in [l_{m,x,\varepsilon}, r_{m,x,\varepsilon}] \end{cases}$$

$\frac{p_{i,\varepsilon}}{p_{j,\varepsilon}} \le e^{\varepsilon}$ (LDP constraint)

Solved $\mathcal{M}$ is the optimal piecewise-based mechanism

Mathematically ≡ to find the **optimal** piecewise distribution under the LDP constraint

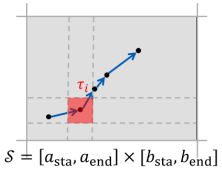
$$\min_{\mathcal{M}:\, p_i, l_i, r_i} \max_{x \in \mathcal{D}} \int_{\tilde{\mathcal{D}}} \mathcal{L}(y, x) \cdot pdf[\mathcal{M}(x) = y] \mathrm{d}y$$

Find $\mathcal{M}$ to minimize the worst-case error

Continuous Spaces: Better & Universal

RIT

- Operate on a continuous space (covering the discrete space)



$\mathcal{S} = [a_{\text{sta}}, a_{\text{end}}] \times [b_{\text{sta}}, b_{\text{end}}]$

Simple sampling: (satisfying LDP for $\mathcal{S}$)

$\Pr[\mathcal{M}(x_i) = \tilde{x} \in \blacksquare] = p_{\text{high}}$

$\Pr[\mathcal{M}(x_i) = \tilde{x} \in \blacksquare] = p_{\text{low}}$

- Benefits:
  1. **Efficiency:** $\mathcal{O}(1)$ sampling complexity
  2. **Trajectory utility:** "n-independent"
  3. **Applicability:** Both continuous spaces & discrete spaces

One-Dimension Example

RIT

- Classifier $h: [0,1] \to \{1, 2\}$ under Laplace mechanism $\mathcal{M}_{\text{Lap}}(x)$

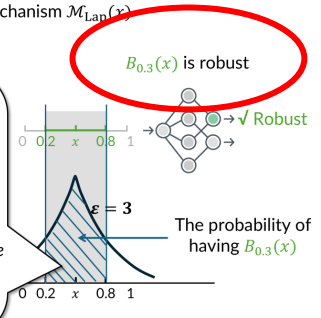
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$= 1 - e^{-\theta\varepsilon}$

For any ε:

With probability at least $p(\varepsilon, \theta)$, the classifier preserves its correct classification under input $\mathcal{M}_{\varepsilon}(x)$.



Thank you!

Q&A

To everyone who helped me arrive here...

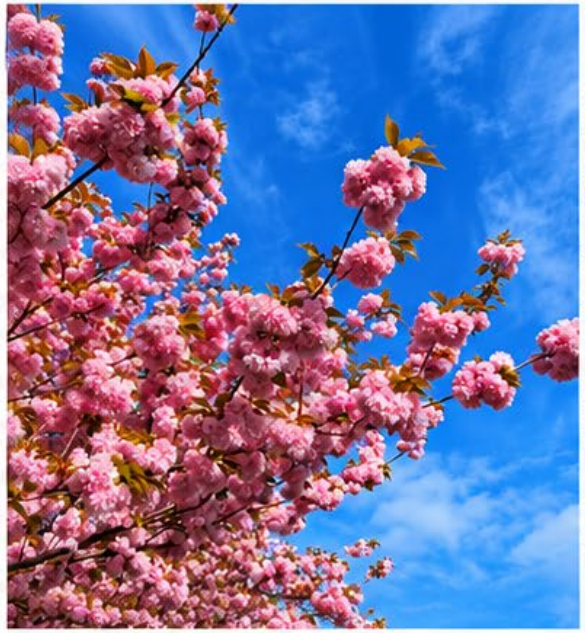






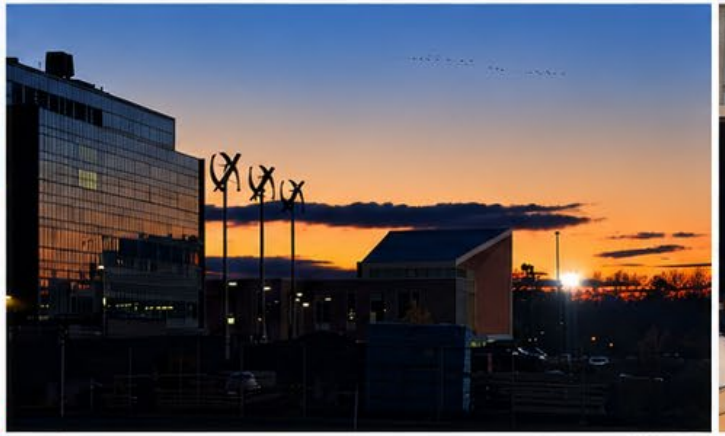





















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